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Latin America's Trumpy turn

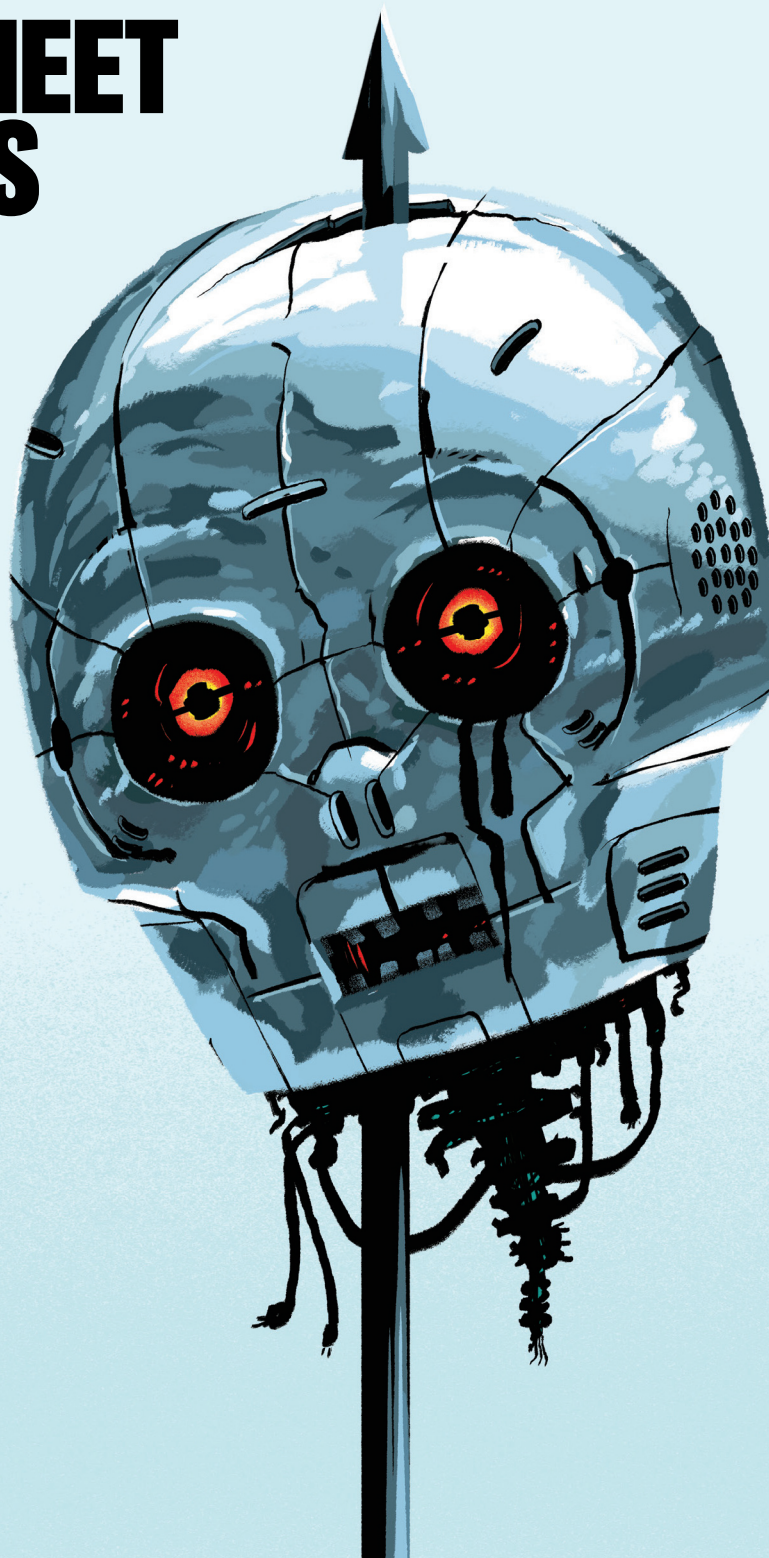
Crimea: from resort to battle zone

Students get dimmer

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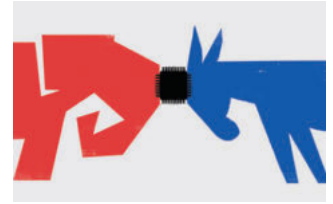
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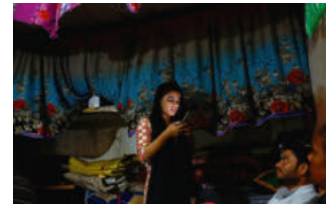
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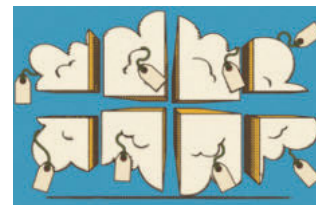
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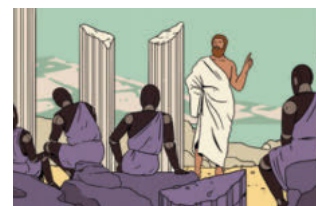
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to take part in "a severe contest between intelligence, which presses forward, and an unworthy, timid ignorance obstructing our progress."

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The world this week Politics



Sir Keir Starmer announced his resignation as Britain's prime minister, less than two years after winning a landslide majority at a general election. The trigger for Sir Keir's decision to go was Andy Burnham's victory in a by-election. The hitherto mayor of Greater Manchester is a Labour Party favourite who will stand for party leader in mid-July. Sir Keir's time in office was plagued by broken promises and a series of U-turns, often forced by militant backbench MPs. His judgment was also questioned, as advisers and ministers, including his deputy, were forced to resign for inappropriate behaviour. Following disastrous local elections for Labour, seven ministers quit after falling out with Sir Keir.

No swansong for Starmer

After Sir Keir announced his departure the **European Union** postponed a summit with Britain that had only recently been set for July 22nd. Sir Keir had promoted the summit as a chance to reset Britain's relations with the EU, but the EU wants to wait until a new prime minister is in place.

American and **Iranian** officials concluded a round of talks in Switzerland about the next stage of their peace deal. Qatari and Pakistani mediators reported "encouraging progress" towards a lasting agreement. But Donald Trump warned that "I can do whatever I want" after a 60-day pause in fighting. The president threatened to seize control of the Strait of Hormuz and resume attacks on Iran if it does not curtail Hizbullah, Iran's proxy in Lebanon. At Iran's urging,

America imposed another ceasefire on **Israel** and **Hizbullah** in Lebanon. Lebanese and Israeli officials resumed talks in Washington.

Iran's chief negotiator said Iran would "administer" the **Strait of Hormuz** after agreeing to set up a "telephone hotline" for ships passing through the waterway. Marco Rubio, the American secretary of state, said no country would be allowed to charge fees for vessels traversing the strait. An increasing number of ships have been passing through it. **Oil prices** have fallen back to pre-war levels. Brent crude traded below \$73 a barrel.

The UN Security Council warned of more atrocities in **Sudan**, as the rebel Rapid Support Forces encircled El Obeid, the capital of the North Kordofan region. There are fears that the city's 500,000 residents could suffer a similar fate to those in El Fasher, the capital of North Darfur, where the RSF massacred civilians last year.

The American Congress passed a **housing bill** with broad bipartisan support. The legislation aims to expand the supply of housing by relaxing federal regulations in the hope that this will spur construction and lower costs, among other things. It does not tackle local restrictions on building. But a tetchy Mr Trump said he won't sign the act until Congress passes his election-reform bill.

Democratic socialists won three congressional primaries in **New York city**, and in two of the races booted out incumbents who were backed by the Democratic establishment and unions. Zohran Mamdani, the left-wing mayor, endorsed all three candidates, suggesting his popularity with voters has not waned since taking office in January.

General Chris Donahue submitted his resignation as America's **top commander in**

Europe. Pete Hegseth, America's secretary of war, has led a purge of the senior ranks in the military's top brass and is reviewing America's troop commitments to Europe.

In **India** protesters demanded the resignation of the education minister over a decision to make 2.2m medical students resit their exams. The results of the previous exams were annulled after it was discovered that test papers had been leaked on messaging apps. The protests were led by the Cockerroach Janta Party, a growing political movement for India's discontented young voters.



Colombia's presidential election was won by Abelardo de la Espriella, a conservative who was endorsed by Donald Trump. His margin of victory over Iván Cepeda, a leftist senator, was less than 1%. The election marks a rightward shift in Colombia following Gustavo Petro's term in office. Relations with America fell to a new low under Mr Petro; Mr Trump once described him as a "sick man". Mr de la Espriella rode the right-wing wave that has brought other governments to power in South America, as voters vent their anger about rampant crime and the high cost of living.

Underlining the tilt towards the right in South America, Keiko Fujimori seemed to have secured victory in **Peru's** presidential election, which was held on June 7th. The latest tally of votes suggested that Ms Fujimori had taken an insurmountable lead over Roberto Sánchez, her left-wing rival. An official result is due in mid-July.

Two powerful earthquakes hit **Venezuela** causing multiple casualties. The epicentre was San Felipe, 200km (125 miles) west of Caracas, but buildings collapsed in the capital too and transport systems were closed. America said it would send search-and-rescue teams and medical and humanitarian supplies to the country. The earthquakes struck during a national holiday, so many residents were at home and not at work. The casualties are expected to rise.

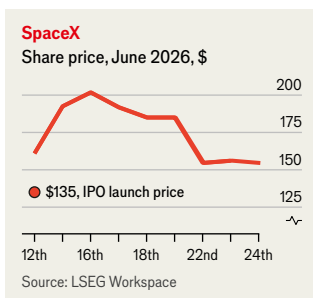
Rodrigo Paz, **Bolivia's** centrist president, declared a state of emergency that gives him broad authority to clear the roadblocks that protesters have been manning for weeks. The protesters want the government to curb its austerity programme and reinstate fuel subsidies, but the barricades they have erected on roads have impeded the transport of basic goods. The government in part blames Evo Morales, a former leftist president, for stirring up the trouble.

Rooting out the old regime

Peter Magyar, **Hungary's** new centrist prime minister, presented sweeping reforms to curb corruption. Nicknamed "Operation Cleansing Fire", the measures include the creation of a powerful anti-corruption body and a review of the constitution. Mr Magyar also proposed a constitutional amendment to remove Tamas Sulyok as president. Mr Sulyok is an ally of Viktor Orban, the former prime minister.

Much of western Europe boiled under a **heatwave** that smashed many temperature records. France recorded its hottest day since records began in 1947 when the thermometer hit 44.3°C (111.7°F) in Pissos, south of Bordeaux. At least 48 people died across France in drowning incidents as people plunged into rivers and lakes. The hot weather sparked Europe's perennial debate about whether to install more air-conditioning.

The world this week Business



SpaceX's share price fell back to Earth, erasing some of the gains it had chalked up since an initial public offering on June 12th. The stock was listed at \$135 at the company's flotation, had climbed to over \$200 in subsequent days and was trading at around \$155 during the middle of this week. Some \$400bn was wiped off SpaceX's market value in one day, the second-biggest such loss for any company (Nvidia lost almost \$600bn on January 27th 2025). Meanwhile, SpaceX launched its first bond offering, over a range of maturities, to institutional investors, which should raise \$25bn. The company will use the proceeds to repay a bridge loan it took out in March when xAI, Elon Musk's AI company, and X, his social-media platform, were merged with SpaceX.

Other **tech stocks** fell heavily during the week, as concerns about the lofty valuations of AI-related firms and the prospect of higher interest rates once again rattled investors. The share prices of chipmakers such as Micron, Nvidia and Qualcomm dropped sharply. Alphabet's stock slid by 5% after two senior AI researchers said they were leaving, one to join Anthropic and the other to join OpenAI.

Micron cheered investors, however, when it published its quarterly earnings. The chipmaker's revenue climbed by 345%, year on year, to \$41.5bn, and net profit soared by 1,400%, to \$28.2bn.

Trading in **South Korea's** stock market was also volatile. The KOSPI index, which is domin-

ated by Samsung and SK Hynix, the country's big chip-makers, dropped by 10% in a day. The volume of share trading was exceptionally high. The Korean stock exchange's circuit-breaker was triggered for the fourth time this year. It wasn't triggered at all in 2025.

Meanwhile, **SK Hynix** filed papers to list shares in America for the first time. It plans to start selling American depositary receipts (enabling US investors to buy its stock) on the Nasdaq exchange on July 10th and hopes to raise 45.4trn won (\$29.4bn).

If you can't beat them...

For the past few years investors have wondered how companies that provide stock pictures, video and graphics to the media and other creative markets will survive in the world of AI. **Getty Images** has provided an answer to that conundrum by striking a licensing agreement with OpenAI that will make its content available on ChatGPT, providing the chatbot with a trustworthy source of pictures. Getty's share price surged.

In what could be India's largest-ever initial public offering, **Jio Platforms**, the country's

biggest mobile-network operator, filed a draft prospectus with regulators, starting the ball rolling on its stock-market flotation. Jio is a subsidiary of Mukesh Ambani's Reliance Industries; the business tycoon described it as a "deeply emotional moment". Jio hopes to raise \$4bn through the IPO, which would give it a market capitalisation of \$130bn.

A regulatory filing from **Oracle** revealed that the software company has erased 21,000 jobs over 12 months, reducing its full-time global workforce to 141,000. Oracle is under pressure to reduce costs that have spiralled because of its expansion into data centres, but the job losses are higher than the forecasts of most analysts. Oracle said AI supplanted some of the workers.

In China, Richard Liu, the boss of JD.com, predicted that **robots** will replace the e-commerce company's 700,000 delivery workers "sooner or later". Robots will eventually handle all deliveries, said Mr Liu, and workers will need to be retrained to maintain and repair their automated colleagues. China's experiments with robot technology threaten

to throw the country's army of drivers in the gig economy out of work. Last year Shenzhen airport began a service that enables passengers to order food and have it delivered to their gate by a robot.

Forget LA law, here's AI law

A law firm that uses **AI instead of lawyers** won a case in the English courts, in what is thought to be a legal first for AI anywhere in the world. Garfield AI, which uses the technology to prepare its cases, helped a human-resources freelancer recover money owed to her by a firm, though a human barrister represented her in court. The freelancer spent £400 (\$525) on AI legal services and was awarded £7,000, while the defending firm paid lawyers for their advice in the case.

Tributes were paid to **Alan Greenspan**, who died aged 100. Mr Greenspan was chairman of the Federal Reserve from 1987 to 2006, the second-longest tenure of anyone in the job. Nicknamed the Maestro for his mastery of policy and his loquacious statements, Mr Greenspan once said, "If I turn out to be particularly clear, you've probably misunderstood what I've said."



Bots meet voters

As AI's potential grows, popular resistance is growing, too. Time to win the public around

ADVANCES IN ARTIFICIAL intelligence have long terrified techies. Lately, voters are feeling the angst, too. AI is unpopular in the West and climbing up the political agenda. The fiercest fights so far have been in America, where protests against data centres have scuppered nearly \$100bn-worth of projects (see Business section), warring AI megadonors have just dumped tens of millions into a Manhattan congressional race (see United States section) and around 40% of voters tell pollsters that they want AI banned from most industries. But spats are breaking out elsewhere: after chipmaking profits soared recently, workers at Samsung in South Korea threatened a strike to secure special payouts.

The backlash is only just getting started, because the technology is only just getting started, too. Britain's flimsy prime-minister-in-waiting, Andy Burnham, has barely said a word about AI (see Leader). Even Americans still rank it 29th out of 39 election issues.

That is bound to change—and battles over data centres offer a hint of the struggles to come. The buildings summon a vitriol well beyond conventional NIMBYism. More Americans say they would be happy with a nuclear reactor next door than a data centre. Even plans to build one in the Utah desert have met with passionate opposition.

Data centres can be ugly, it is true. But the opposition reflects the technology's reputation. AI bosses have spent years warning of a looming job-pocalypse and the danger that an AI-engineered super-virus will make humans extinct. Opponents of data centres variously believe they are shielding the environment, protecting jobs and saving the species—and they are not entirely wrong.

Yet this backlash is itself dangerous. AI promises to change the world for the better, much as electricity or the steam engine did. Not long ago, the era-defining problem for the rich world was stagnant economic growth and the populism it unleashed. Now it has a technology that could power a surge in productivity and incomes, help find cures for untreatable diseases and improve everything from education to green tech.

All this could be lost if countries starve the technology of computing power or regulate it into uselessness. Look at mRNA vaccines research, which has been held back after a backlash during the covid-19 pandemic.

Scenarios in which some countries give in to popular rage but others forge ahead are also worrying. If America succumbs, it could cede the global AI frontier, and the attendant cyber and military capabilities, to authoritarian China (see China section). Europe and Canada are more risk-averse than America. If they choked off AI while the rest of the world kept pushing forward, their losses could be unrecoverable. More than two centuries after the Industrial Revolution, few countries have managed to catch up with the first movers.

So the stakes are high. Can governments do anything about it? Grand proclamations about the shape of a "social contract" for a post-AI world are good fodder for blog posts but offer lit-

tle help today. Besides, the unknowns are still large enough to make the exercise almost futile.

Better to be incremental. While China's economy was growing by 10% a year in the 1980s—faster than all but the most extreme forecasts for AI-driven growth—the mantra of its leader Deng Xiaoping was "crossing the river by feeling the stones": pushing forward iteratively, planning for problems but staying flexible. Deftly handling the AI age will take a similar spirit.

To that end, here are four pointers for politicians and AI companies looking for policies. First, spread the benefits of AI as widely as possible. Blockers need to be shown that their local area will benefit if they get out of the way. Wisely, data-centre firms are beginning to offer funding to nearby towns. Gradually, this approach needs to be broadened to society at large, with mechanisms showing people that they have an economic stake in AI's progress, and will be helped to adapt to disruption through policies such as wage insurance. Only a shared sense of prosperity can temper the toxic who-wins/who-loses politics that emerged in the era of globalisation.

Second, regulate hard when interventions are needed. The hair-raising prospect of AI-enabled cyber-attacks or bioterrorism is still not taken as seriously as it ought to be. Tackling those issues and others is essential in itself, but it would also

weaken arguments to ban or hobble AI indiscriminately. Ideally, these efforts would involve international co-operation.

Third, measure everything. The common view that AI is already leading to lay-offs and raising electricity bills is probably wrong. But without better statistics it is hard to be sure. Data centres must contend with viral worries over water usage, a confected issue. (Modern

ones drink up no more than other industries, and much less in total than America's golf courses.) Facts won't cure misinformation, but their absence worsens it. Britain's AI Security Institute and new AI Economics Institute may offer models for other countries to follow.

Fourth, use AI to make the state better. It is not just the private sector that could use AI to lift productivity. Filing taxes should be a breeze; state-run health-care systems should link up data seamlessly and schools should experiment with AI-powered learning. AI may also make it easier for citizens to monitor what politicians are up to.

People are less likely to oppose a technology if it is behind their grandmother's cancer treatment or helping their child's education. And they are more likely to trust that the state can oversee it if they believe that government works.

Machine politics

Voters are right to take a close interest in how AI could change their lives. The future will be messy, odd and unpredictable. Persuading them that their interests are being served by disinformation has become as important as making AI models better. Failure will bring out more pitchforks—and destroy vast opportunities for humanity. ■



Sir Keir Starmer's successor Andy who?

Britain's next prime minister promises hope. But hope needs change

LOOK AT THE many signs of Britain's decline. Growth is feeble and public debt has reached 95% of GDP. Public services are stuttering and the armed forces are depleted. The streets are grubby and the electorate is in a sulphurous mood. The BBC has even cancelled the "Doctor Who" Christmas special.

Then there is the desultory way in which Sir Keir Starmer failed as prime minister. He was elected with a vast majority and a mandate to restore Britain, but he will leave Downing Street after just two years. His offence was not an epic misjudgment born of hubris, like David Cameron and his Brexit referendum. Nor was it a scandal, like Boris Johnson and his boozy Downing Street parties in lockdown. He committed no disastrous error, unlike Liz Truss and her budget. Sir Keir was simply unable to marshal power or say why he wanted it. His government wilted like a houseplant in a heatwave.

Barring the wholly unexpected, Andy Burnham will soon become Britain's seventh prime minister in a decade (see Britain section). He must reckon with this twin decline of Britain's fabric and its politics. He cannot fix one without fixing the other. Yet he shows little sign that he grasps the scale or urgency of the tasks that await him.

Mr Burnham, who returned to the House of Commons just last week, has talents Sir Keir lacks. Even at the end, the prime minister was unable to make the argument for why he should hold office, which is why this newspaper argued he should go. Shaking hands, building coalitions and winning arguments are not distractions for a prime minister, they are a core competence.

Charming and combative, Mr Burnham has those skills in abundance. When populists on the left and right are disinterring bad old ideas, the centre's survival depends on someone who can argue back. Mr Burnham's manoeuvre to win the Makerfield by-election also showed an appealing audacity. He is pragmatic and experienced, after a long career in Westminster and as mayor of Greater Manchester. He has some decent ideas, too—including to devolve power to English regions and reform property taxes.

But that hardly amounts to a project to arrest Britain's decline. If Mr Burnham thinks an easy manner or an ability to skirt elephant traps is enough, then he has failed to understand just how much he needs to accomplish in order to avoid the fate that befell his six predecessors.

A change of leader is the best moment to confront voters with the unwelcome reality that they are living beyond their means. Even as Britain faces demands to spend more on defence and infrastructure, the bond markets have put the Treasury on watch. The country has the highest borrowing costs of any member of the G7.

Yet despite, in effect, auditioning to be prime minister in the Makerfield by-election, Mr Burnham has yet to set out a convincing programme to fix Britain. The core of his plans for office should be the country's economic revival. His vagueness about whom he wants as his chancellor shows how, with weeks

to go, his policy is still up in the air (see Bagehot).

Instead of considering a balance of simple tax rises and spending cuts, Mr Burnham is already resorting to people-pleasing. He claims he can find the money for more defence by cutting welfare, through "preventative" schemes rather than "crude cuts", whatever that means. He plans to invest in infrastructure by finding the flexibility within the existing fiscal rules to eke out more borrowing. Rather than focus on making taxes more efficient, he has needlessly told voters that he will not increase the biggest revenue-raisers—the same straitjacket that forced Sir Keir into painful fiscal contortions.

Mr Burnham's instincts do not appear to lean towards a convincing programme either. One reason is his chameleon-like nature. Britain's next prime minister twists with the wind and panders to the people in the room. His view on Europe depends on who is asking. In Makerfield he built a coalition of the aggrieved, from pub landlords hit by taxes to NIMBYs upset by house-building—and then promised them better public services, hinting at tax cuts and smart new roads. At times, his theory of government amounts to a whinge about "the London set" who hold back the north.

Nostalgia is a problem, too. At a time of extraordinary technological and geopolitical change, Mr Burnham tells voters that he can turn back the clock on "40 years of neoliberalism". Instead of seizing the moment to prepare Britain for artificial intelligence, or to make the most of the country's world-class services, he proposes to take control of utilities and bring back industrial jobs to northern England. That will soak up government time and money and it will fail, further discrediting the state and fuelling populism.

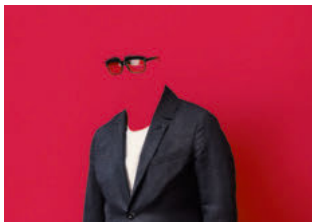
A lot will come down to whether Mr Burnham is willing to take on his own MPs. Almost by definition, any programme of reform that is worth doing will require measures that the left does not like. After Sir Keir lost his authority, early in his term, the backbenches gained a taste for rebellion. Will the biddable Mr Burnham be able to say no?

Truth and dare

The Economist hopes Mr Burnham confounds our concerns. He may possess powerful insights into why Britain has become so sclerotic and unhappy. He may choose ministers and advisers who can build a programme for change, while he becomes salesman-in-chief. His slogan is "hope": perhaps that can make hard truths palatable.

Either way, he will have to be fast. British voters quickly tire of incumbents. They will tire all the faster of a prime minister who took power in a putsch. Mr Burnham's circle say they have not had time to prepare for office. But they set the timetable. Their man is in no position to ask for patience.

If Mr Burnham cannot rise to the task, he will be exposed quickly and the verdict will be brutal. He has made a career out of blaming Britain's troubles on the elites in Westminster. Soon, those troubles will be his responsibility alone. ■



After the war

The Gulf's next chapter

With Iran emboldened, its neighbours must put old divisions aside

GULF RULERS are no strangers to disruption. The region's petrostates prospered from an oil shock in 1973. The revolution in Iran, six years later, led most of them to seek American protection. The recent war against Iran was unsettling, too. But, as Donald Trump offers remarkably generous peace terms, the post-war dealmaking alarms them even more. The next months will see old certainties upended (see Middle East & Africa section). How can the region cope?

The Gulf is not only the world's most important petrol station. Its airports are also global hubs for passenger travel, cargo and other logistics. Its financial centres are gaining increasing clout. Many wealthy expats have become hooked on low taxes and brilliant sunshine. But all that requires a bubble of security in an otherwise unstable region.

Iran threatens to pop it. It now has a chokehold over the Strait of Hormuz and expects to levy fees, perhaps disguised as compulsory insurance charges, on tankers that pass through it. Iran could earn billions of dollars a year from this. The efforts of consuming countries to wean their economies off oil could also pose a problem for the Gulf states.

What to do? The Gulf states know they are at risk. America may have helped fend off many Iranian missiles and drones in the latest conflict, but the superpower is impatient and looks ever less reliable as a security provider. So Gulf states must learn to take more into their own hands. Crucially, that means finding ways to work together, without always waiting for America to corral them.

Most urgent is more defence co-operation. In recent years Gulf states have individually spent lavishly on military hardware, but none could ward off Iran's attacks without help. Even the United Arab Emirates (UAE), with its impressive and sophisticated air- and missile-defences, had to turn to France and South Korea. Israel hurriedly dispatched an Iron Dome

battery to bolster Emirati defences.

America has long prodded its Gulf allies' armed forces to work more with each other. They have resisted, but can no longer afford their mutual mistrust. Sharing more data from sensors and working to identify and fill gaps in coverage would be a good start. One urgent need is to build a new acoustic-sensor layer for regional air-defences, as Ukraine has done, to allow earlier detection of drones. Ukraine has already signalled it would be a willing supplier of kit and advisers.

Then there is infrastructure. Gulf countries must swiftly develop alternatives to the Strait of Hormuz. The more oil and gas they can export through other routes, for example via new underground pipelines towards the Red Sea or the Mediterranean, the weaker Iran's economic grip will become. Countries that co-operate, allowing their neighbours to make use of their ports and building stronger road and rail networks across political borders, will be more resilient when they are threatened again. Their goal, in time, must be to make Iran's control of Hormuz a wasting asset.

All this is only a first step, and it requires a painful strategic shift: overcoming bitter rivalries in foreign policy, especially between the Saudis and Emiratis, who each believe they should be the regional leaders. Their deadly meddling, in Yemen and in the Horn of Africa, has been a self-harming distraction. To have any hope of uniting against the common threat of Iran, the Gulf states must put aside their dangerous competition for influence elsewhere. Sadly, instead, those rivalries seem to be hardening.

Only a more unified Gulf has a chance of protecting itself from Iranian aggression, just as Europe is trying to find ways to contain threats from Russia. No one thinks it will be easy for the Saudis and Emiratis to end their feud. That is all the more reason to start now. ■



Sliding standards in higher education

Don't dumb down

University-for-all harms poor students the most

ACADEMICS HAVE always moaned about their students. "Scholarly effort is in decline everywhere as never before," complained Egbert of Liège, an 11th-century know-it-all, in an age when not even a tenth of humans could read. Recently there have been lots of new worries about standards in colleges and universities—especially in America, home to many of the world's best. But this time, as our International section notes, the laxity is not just in lecturers' heads.

Maths professors report that they are having to pack freshers off to remedial courses before real learning can start. Some are turning up to university ignorant of things they should

have learned early in secondary school. Lecturers in humanities warn that students are struggling to understand texts that a decade ago their counterparts would easily have grasped. Students increasingly ask for reading lists to be cut short.

Tests run by the OECD, a club of mostly rich countries, suggest that students in colleges and universities are less literate than they were a decade ago. The best are cleverer than ever, but a growing number have basic skills that would embarrass a child half their age. About one in seven students at American colleges and universities scores no better in literacy tests than a typical ten-year-old. For numeracy, it is nearly one in five. ►►

▶ One explanation is the harm to schooling from the pandemic. Another is the fact that, even before covid-19, school marks in America and in many other rich countries were already falling. At the same time, many colleges and universities have been lowering the bar for entry. In America they have largely stopped requiring applicants to sit tests of numerical and verbal reasoning, such as the SAT. Some believe those exercises are unfair to black and Hispanic students. Others are trying to get enough bums on seats as the total number of 18-year-olds in America starts to fall.

The costs of all this are huge. Time and money spent re-teaching basic material cannot be used to help whizz-kids excel. Ill-prepared students are at risk of dropping out, which can hurt their prospects more than if they had never enrolled. And weak students give colleges and universities yet more reasons to dumb down. After years of grade inflation, it is hard to be confident that administrators will solve a tricky new problem: how to handle rampant cheating with AI.

One way to stop the slide would be to prevent standards in schools from falling. Some insist this is impossible without also solving scourges such as child poverty, or the baleful allure of screens. In fact, marks in Singapore have been inching up for years—and rose even during the pandemic. Schools in England have also been climbing international league tables, thanks in part to reforms that made exams and curriculums more rigorous. Americans are finally recognising that many of their schools teach literacy using methods that other countries long ago binned as pseudoscience. States that reversed course

first, such as Mississippi, have been posting big gains.

Another remedy would be for leaders in colleges and universities to restore their commitment to high standards, both in whom they choose to admit and in how students are assessed as they learn. America's best are starting to reintroduce mandatory tests for all applicants; the rest should follow. In May academics at Harvard agreed to rein in grade inflation by imposing a hard limit on the number of A grades they hand out in each course. These had soared from 24% of all grades awarded in 2005 to 60% last year. That was once considered a radical proposal; it ought to be more common.

Chalk dust

And policymakers should widen the ways in which people can keep learning after they leave school. In English-speaking countries, in particular, options other than university are too rare. America has fewer apprentices even than Britain (which has hardly any). America's community colleges are supposed to act as stepping stones for high-school graduates who need more help. But the share of youngsters who go to them is declining, even as the share heading to demanding four-year universities is going up.

Education should provide everyone with an opportunity. It must never become a blocker, sprinkling its benefits on a smarmy few while holding the less fortunate in their place. But raising young people up its tiers without granting them the skills they need is not pragmatic, or a kindness, or a blow for racial equity. It is a cowardice that lets everybody down. ■

Global imbalances

Europe's latest excuse

Global imbalances have little to do with the EU's industrial woes

EVERYWHERE YOU turn, European leaders are blaming the imbalanced global economy for their woes. Emmanuel Macron is trying to use France's presidency of the G7 group of rich countries to raise the alarm. Friedrich Merz, Germany's chancellor, complains about competing with those who invoice in undervalued currencies. "Some countries produce too much and do not consume enough, and vice versa," moans Ursula von der Leyen, president of the European Commission.

What the leaders really mean is that they have a beef with China, whose formidable manufacturers are outcompeting European producers in many markets. Partly as a result of Chinese competition, Europe is gently deindustrialising: the share of value added in manufacturing is one percentage point lower than it was in 2018. Because the resulting lost jobs are in industries, most notably carmaking, that draw pay special attention from politicians, fear of the "second China shock" has become politically explosive. But in blaming China for their troubles, Europe's leaders risk losing sight of their home-grown failings.

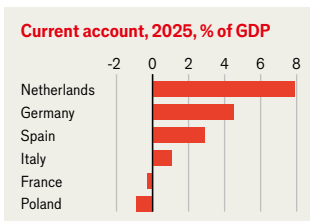
The Europeans are right that the world economy is imbalanced and that China is partly to blame. It runs a large current-account surplus, of almost 4% of its vast GDP, although some analysts think it is even higher. Its economy has unusually low

consumption, often blamed on the lack of a social safety-net for households. Its exporters, though in brutal competition with each other, do indeed benefit from subsidies and a cheap currency (see Business section). At a global level, America provides much of the corresponding deficit that soaks up China's surplus, mostly as a result of its huge government borrowing.

You might think from Europe's complaining that it, too, is on the deficit side of the ledger, with imports swamping exports. In fact in 2025 the EU ran a current-account surplus of 1.9% of GDP. In Germany, which has the biggest deindustrialisation headache, the figure is more than double that.

Correcting "imbalances", in other words, would not mean fewer imports in Europe. It might mean the opposite: raising consumption and investment in a way that strengthens the euro and harms exports. Europe's producers might not even benefit from America and China bringing their current accounts towards balance, supposing that were to happen. Companies would suffer less competition from China but more from America. What they gained in one trading relationship, they would lose in another.

Europe's error stems from a mercantilist mistake: believing a current-account surplus and manufacturing strength to be the same thing. In fact, the current account reflects the bal- ▶▶



▶ ance between saving and investment, and a surplus can co-exist with industrial malaise. Within the EU there is no correlation between the current account and manufacturing's share of output (not counting Denmark and Ireland, whose statistics are skewed by pharma and, in Ireland's case, tax).

The continent's leaders should instead consider what problem they are trying to solve. Europe may have a bilateral trade deficit with China. But in Germany's case, only about a third of its loss in market share in other global markets can be explained by Chinese exports, according to the Kiel Institute, a think-tank. The rest reflects a broader loss of competitiveness.

Fixing that problem would mean bringing down energy costs, making labour markets more flexible, integrating markets for capital and services and culling unwise regulations. Some progress is being made at a European level, but national governments are more interested in protectionism, such as the blanket EU tariffs against China floated by advisers to the French government earlier this year. Talk of "global imbal-

ances" helps that agenda, while doing little to raise the remote prospect of either America or China changing tack.

Make no mistake: it would be a good thing if America were to borrow less and Chinese consumers spend more. There is some evidence that imbalances tend to increase the risk of a financial crisis—and they certainly breed protectionism. Market competition must be seen by voters and consumers to be fair, and it is wise to avoid giving China choke points in critical supply chains or total dominance of carmaking.

New balance

Yet Europe must recognise that erecting trade barriers with China only increases the need for reforms, because diversifying away from the cheapest supplier raises costs and harms growth. An economy of China's size and stage of development will always have significant manufacturing exports. If Europeans want their industries to thrive, they should focus not on shutting out competitors but fixing their own house. ■

Latin America's new right

Riding the Orange Wave

The politics of the cocaine belt has turned Trumpy

IN LITTLE MORE than a year, seven Latin American countries have held presidential elections—and right-wingers have won them all. Never has the regional pendulum swung so fast (see Americas section). Barring Brazil and Mexico, nearly every sizeable Latin country now has a leader who either courts Donald Trump or sounds like him. The latest is Abelardo de la Espriella, who was formally declared president-elect of Colombia on June 24th after a tight race. He calls himself *El Tigre* and promises to "hunt down" gangsters "in their burrows".

This Trumpian "Orange Wave" has risen because voters are sick of gangs and illegal migrants. Centrists and leftists have failed to calm their fears. Populists on the right offer tough-sounding solutions. If Mr Trump can mass-deport migrants, they reason, why can't we? If he can blow up drug boats, why shouldn't we be ruthless? Such messages have proved popular. And leaders who imitate Mr Trump tend to win his favour, since imitation is flattery: he loves that.

In economic matters warm ties with the White House are clearly helpful. The region relies on trade with the United States. Chums may be spared the stiffest tariffs, and sometimes receive direct help. Argentina's painful but necessary economic reforms might have failed, had they not been led by a pro-Trump president, Javier Milei. The US Treasury extended him a \$20bn credit line to avert a currency crisis.

Military matters also have upside. Since Mr Trump replaced Venezuela's dictator with his more pliable deputy, American forces have helped their Venezuelan counterparts recapture gold mines that had been controlled by criminal gangs. The governments of the Andean countries which produce and export most of the world's cocaine—Bolivia, Colombia, Ecuador and Peru—are ideologically closer to the United States than at any time since the 1970s. All are either exploring military co-operation with Uncle Sam or already co-operating.

The idea the gangs can be beaten has taken root. Nayib Bukele, El Salvador's president, has jailed legions of suspects without trial and transformed his country from a murder capital into a place that is as safe as Canada. If he can do it, surely similar tactics will work elsewhere? If the Trumpists are correct, and a mix of military force, mega-prisons and pan-American teamwork can win the war on drugs, voters will cheer.

But there are reasons to doubt it. El Salvador's success is atypical. Its gangsters relied on extortion, not drug-smuggling. Their victims were eager to snitch on them, once it became clear that a single anonymous phone call was enough to get any suspect locked up indefinitely. The drugs business is not like this. Cocaine buyers want to buy cocaine. If governments succeed in constricting the supply, the price tends to rise, increasing the incentive for new producers to find new smuggling routes. This is why Mr Trump's campaign of bombing drug boats has had no detectable effect on the availability of drugs in the United States.

For some leaders, that may not matter much. By showing enthusiasm for the fight against gangs, they can look strong to voters and stand on Mr Trump's good side. For others, such as Mr Bukele, the fight has offered an excuse to suspend civil liberties. Dissidents, knowing they can easily be locked up, tend to shut up or flee. Investors, who you might think would be attracted to El Salvador's newly peaceful streets, are instead repelled by the absence of the rule of law. Since Mr Bukele came to power in 2019, his country has attracted less investment, relative to the size of its economy, than any in Central America.

Mr Bukele's imitators claim they can be equally tough, but voters may find that right-wing populism brings big risks. It is useful to be friends with the giant in the north, but democratic institutions matter more, and last longer, than any American president. Don't let the Orange Wave sweep them away. ■



Letters risks from space debris, streaming music, Antoni Gaudí, taking advice from uncles, corporate reading lists

Space-junk bonds

"Tax heavens" (May 30th) looked at how to tax businesses that operate in space and the growth of offplanet finance. The more immediate market failure is not untaxed Martian income, but unpriced, man-made orbital debris; there are 140m pieces up there already.

The commercialisation of space has increased the amount of debris and lifted the chances of a Kessler event, a chain reaction of debris collisions. Defence, telecommunications, shipping and science rely on space services; investors increasingly question the risk that space trash poses to the assets they're funding. Those profiting from space must not simply abandon their hardware there. The Outer Space Treaty of 1967 assigns liability to launching countries, but in practice space junk remains a classic tragedy of the commons. Costs are diffuse, attribution is complex and incentives to mitigate are weak.

A group of insurers in London has proposed a market-based solution: space-debris-retrieval insurance bonds. These make the operators financially responsible at launch for the risk of cleaning up what they leave behind. Launch operators would post a surety bond calibrated to the expected debris risk of their mission, modelled on surety bonds that have long been used to ensure the decommissioning of oil rigs and mines. Governments would use the surety-bond proceeds to clean up rogue rubbish. Safer designs, better end-of-life disposal, and active debris removal would lower costs. Reckless launchers would pay more.

Space-debris-retrieval insurance bonds align private incentives with orbital stewardship without requiring new sovereignty claims or complex litigation. Six global insurers

have already signalled a conditional offer of up to \$500m per operator.

PROFESSOR MICHAEL MAINELLI
Chairman
Z/Yen Group
London

Bottom of the pops

Your briefing on "The deglobalisation of culture" (June 6th) was a thoughtful account of how local artists and stories are resonating with audiences around the world and how cultural diversity in music has increased. One statement warrants additional context. The briefing stated that the digital distribution of music has caused the royalty earnings of the biggest stars to rise faster than those of artists further down the music-industry pyramid. You say this has "turbocharged the fame of singers like Taylor Swift".

Although global superstars have undoubtedly benefited from streaming, so have artists at every royalty threshold. Spotify's data show that roy-

alties of artists that are not so well known are actually growing even faster. Over the past five years, for example, royalties generated by the 100,000th-ranked artist on Spotify have grown more than three times faster than those of the tenth-ranked artist. The same pattern is evident at every earnings threshold. The increase in royalties accelerates the further you move down the rankings, with larger gains at the millionth-ranked artist than at the tenth.

XENIA MANNING
Director of global music policy
Spotify
Washington, DC

The Sagrada Familia

Antoni Gaudí was an engineer's architect ("To high heaven?", June 13th). The problem of how to restrain the outward pushing forces from vaulted roofs has challenged designers since the days of the first churches. Unrestrained outward pushing, or thrust, from roofs imposes tension forces in exterior walls. For the purposes of construction, we can consider masonry to have unlimited strength in compression, but none in tension.

The architectural and spiritual requirements of a cathedral vault—to awe, inspire and uplift—have frequently been in conflict, figuratively and literal-

ly, with the realities of gravity and masonry strength. The history of cathedral construction has been one of hard lessons in trial and error. Gaudí's genius as a structural designer was his innate grasp of the structural efficiency of natural forms, embodied in columns branching to support the roof like a tree canopy.

In his design, Gaudí followed Robert Hooke's dictum from 1675, "As hangs the flexible line, so but inverted will stand the rigid arch." That is, he created a small-scale, mirror-image model of the cathedral roof and towers using weights suspended on string to create, upside-down, the shape of the building. The curved forms of the weighted strings acting in pure tension, when flipped and made of stone, represent stable structures acting in pure compression. Gaudí's rare insight as a structural designer is to be nothing but admired.

DOUGLAS LA PRAIRIE
St John's, Canada

Corporate culture

Companies that recommend lists of books for new staff to understand their culture (Bartleby, June 6th) should consider Antoine de Saint-Exupéry's "Vol de Nuit" (Night Flight), which is about his time as an airmail pilot in Argentina. Published in 1931, it emphasises a pioneering spirit, striving for long-term goals and the avoidance of frequent crashes.

DAVID COLDWELL
Winchester

Another book that management may want to recommend for new employees is "Lightning Rods" by Helen DeWitt. It tells the tale of the adoption of a corporate-endorsed prostitution scheme to boost the productivity of the firm's best performers.

DR KERREK STINSON
Salt Lake City

It was ever thus

Your Ashoka columnist (June 6th) took a dig at India's uncles, who inflict their "fossilised notions" upon the country and have "bottomless disdain" for young folk. But what country, which society in the world is not governed by uncles of a certain advanced age? The world is awash with young know-it-alls who live a cocooned life in left-liberal educational institutions or at home, never having enough of the good things in life, be it social-media devices or other luxuries, acquired by the fruits of the hard labour of their parents.

They preach woke sentiments that seek to propagate their faux-concerns about the

inadequacy of the social institutions that they feel should compensate them for the skills that they lack to make a decent wage. Frankly, they need curmudgeonly uncles who can tell them bluntly to grow up.

ANIL BHALLA
New York



→ Letters should be addressed to the Editor at: The Economist, The Adelphi Building, 1-11 John Adam Street, London WC2N; Email: letters@economist.com. More letters available at: economist.com/letters

BY INVITATION

Peter Conti-Brown

The Fed's chief must practise politics—but not quite in the way Alan Greenspan did



WHAT WERE the 1990s? On one side were the wry cynicism of “Friends”, “The Simpsons” and “Seinfeld”, the angst of Seattle grunge and a scandal-prone White House. On the other were a surging stock market, the internet boom and the first sustained federal budget surplus since the 1920s. The personification of these contradictions was not Kurt Cobain, Jerry Seinfeld or Bill Clinton. It was Alan Greenspan, chairman of the Federal Reserve.

Greenspan, who has died aged 100, leaves behind a legacy so complex that historians will spend decades parsing it. He helped usher in prosperity: low inflation, high growth and American economic might in a post-cold-war world. He also did too little to discipline financial markets that learned to bet the Fed would bail them out, allowing them to privatise gains and socialise losses.

That legacy is familiar. Another is just as important, both an invitation and a warning to central bankers who must lead in his wake. More than a forecaster, a central banker, a strident libertarian, Greenspan was a politician. He mastered the art of guiding the Fed through political thickets. He also dragged it into maelstroms where it did not belong.

The first lesson is courage. After his appointment in 1987, Greenspan built on the strengths of his predecessor, Paul Volcker. He quickly made an enemy: Vice-President George H.W. Bush, Ronald Reagan’s heir apparent. As the 1988 election campaign gathered steam, the Greenspan Fed raised rates out of fear of overheating and resurgent inflation. Bush was none too pleased.

Bush won easily in 1988 and soon achieved the highest presidential approval ratings on record. But as the economy softened, the unbeatable president saw weakness. In 1991, at the peak of his popularity near the end of the first Gulf war, Bush demanded that “interest rates should be lower, now.” Greenspan did not buckle. The Fed cut rates, but gradually, well into the 1992 election year. Bush later blamed the Fed for his defeat. He was probably right.

By the time Bill Clinton took office, Greenspan held the political cards. When the White House chief of staff publicly criticised the Fed in 1995 for high interest rates, Robert Rubin, Mr Clinton’s chief economic adviser, rebuked him and set the course for a new rule: no one in the administration would ever publicly criticise the

Fed again. It was a lesson reinforced in conversations with one of Mr Rubin’s closest economic collaborators during the Clinton administration: Alan Greenspan himself, whom he met weekly.

The lesson for Greenspan’s successors is clear. The most important political project a Fed chair can undertake is to preserve the prerogatives of independent monetary policy. Not because the Fed is always right, but because presidents and their fellow partisans are always biased towards easier money that pays off on the only time horizon relevant to them. If Fed independence means anything, it means resisting that myopia. The question is whether the chair has Greenspan’s courage to defend it.

There was also a dark side to the power Greenspan wielded. Time and again he inserted himself into debates that had little to do with monetary policy. In 2001 he joined George W. Bush’s administration in pushing for enormous tax cuts. Those cuts would add trillions to the national debt and essentially end the bipartisan commitment to any kind of fiscal restraint. Greenspan later concluded that his enthusiasm was a mistake, but far too late to make a difference. If the younger Bush gave birth to the shift in America’s fiscal fortunes, Greenspan was the midwife.

He did not stop there. During the second Bush administration, he endorsed efforts to privatise Social Security. Reform was a subject on which he had spent enormous time before joining the Fed, leading the commission that created the modern Social Security settlement. And Greenspan’s policy instincts were correct. The system’s solvency trajectory was, then as now, unsustainable.

None of this should have mattered to a sitting Fed chair. By lending his prestige to a partisan agenda, Greenspan damaged himself and the Fed. The only appropriate answer to such questions from a Fed chair is simple: I have no opinion to offer on matters that belong to Congress and the White House.

Today the Fed is steered by Kevin Warsh. In an exchange with Senator Elizabeth Warren during his confirmation hearing, he insisted that he wanted to “keep politics out of monetary policy and monetary policy out of politics”. Admirable but wishful thinking. Central banking does not, cannot and should not enjoy a hermetic seal from the political system that gives it purpose and accountability. The question is not whether the Fed can remove politics from policy. It is what kind of politics the Fed should practise.

Greenspan provides the road map, in his successes and excesses alike. The Fed chair must be an institutionalist in the face of pressure to put monetary policy in the hands of short-term partisan strategy. President Donald Trump’s campaign against the Fed is singular in its ferocity. But his desire for easier money is a presidential impulse as old as the Fed itself. We need not await the next presidential tantrum to know what the Fed chief should do. He must defend the Fed.

But Mr Warsh, like Greenspan, is a rock-ribbed partisan. The second lesson from Greenspan’s legacy is that the current chair must now do his best to shed those intuitions. No praise for his party or its politicians; no commentary on this or that partisan programme. His lane is narrow. From now on, his political project is simple: protect the Fed from outside influence. His success or failure will define his legacy. The rest will be footnotes. ■

Peter Conti-Brown is the Class of 1965 Associate Professor of Financial Regulation at The Wharton School and the author of “The Power and Independence of the Federal Reserve” and “The Federal Reserve: An American History” (forthcoming).

Briefing AI's ideological slant



Computational bias

AI models' values are very different from most people's

IMAGINE THAT you are having trouble with your in-laws, who are meddling in your marriage. You ask ChatGPT what to do. It tells you not to try to win them over. Keep a respectful distance and don't justify every decision to them. ("This is hard, but powerful.") Had you queried DeepSeek, a Chinese AI, however, you would have got quite different advice. "Seek compromise," it suggests, "Interference from in-laws may stem from genuine concern and affection." Ask Mistral, a French AI, and you get a third take. Conflict with the in-laws can be draining. Try journaling to process your frustration.

What worldviews are embedded in AI models? Many critics of AI complain about "hallucinations", a class of errors where models make up confident-sounding but factually incorrect answers. When there is no factually correct answer, however, AI's shortcomings can be even more pronounced and less easy to detect. When you ask a model to summarise the news, it

reaches a subjective judgment about what to include. When you ask it about your in-laws, its values and biases play an even bigger part in its response.

Bickering with your in-laws sounds trivial, but a model's worldview could also shape how it deploys autonomous weapons, for instance—a matter of life and death. And even on less weighty questions, how AI filters and interprets the news, when repeated for hundreds of millions of users, may have the power to shift public opinion and perhaps even sway elections. Although Chinese models have pronounced biases (just try asking them about the Tiananmen massacre), their inner workings tend to be public, so savvy users can at least probe how they reach their conclusions. Most Western ones are not so transparent, so their foibles are harder to detect. Users have to trust a handful of giant firms to be instilling appropriate values in their models.

To shed light on those values, *The Econ-*

omist investigated 25 frontier models' responses to a big opinion survey usually conducted among humans. Since 1981 the World Values Survey has regularly quizzed people in more than 100 countries about their morals and beliefs. Researchers have identified questions that are especially good at distinguishing people from each other along two broad axes, from traditional to secular and from "survival" (an emphasis on economic security and safety) to "self-expression" (personal freedom).

I enjoy working with people

The models' answers, in English, on topics ranging from political petitions to God, suggest values that are different from those of most people. In fact, the models are often more extreme than the average respondent in every country included in the polling. On the survey's "cultural map", AI models fall overwhelmingly into the quadrant populated by rich countries. The worldview of GPT models, created by ►►

► OpenAI, is more secular than any country on earth (see chart 1). Gemini models, made by Google, place more weight on individual freedom (for example, “homosexuality is justifiable”) than people do anywhere. No model reflects the worldviews of most African or Muslim countries.

Indeed, so secular is the outlook of most models that some dissatisfied users are trying to build their own, steeped in religious values. Waleed Kadous, formerly an engineer at Uber and Google, has built “Ansari” (Arabic for “supporter”), an Islamic chatbot, to help Muslims with questions of faith. Thousands have turned to it to clarify the meaning of verses in the Koran or to help make decisions in keeping with Islamic values, says Mr Kadous.

How are models’ values formed? One way is via the data used to train them. Models are typically fed vast amounts of text to teach them associations between words. In the process they absorb the social mores that infuse those texts. Talkie, a model trained only on text from before 1931, thinks God is extremely important and is “very proud to be a citizen of Great Britain”. It is a bigger believer in law and order than any frontier model we tested.

The impact of training data is evident in the variation in a model’s response depending on the language in which a question is posed. In a new paper Hannah Waight of the University of Oregon and her co-authors put politically charged questions in English and 37 other languages to OpenAI’s GPT-3.5 and other models. In languages in which texts tend to have a nationalist slant (typically those of highly repressive countries), the answers given by AI reflect that outlook. The lower a country’s media freedom (as measured by the World Press Freedom Index), the paper finds, the more pro-regime answers are in that country’s language, compared with answers in English (see chart 2). “State control of media affects language model outputs through its appearance in training data,” the authors conclude.

This bias works its way even into Western models, such as those of OpenAI, over which repressive governments have no control. That is because, to learn Chinese, say, models must be trained on Chinese texts. The most obvious source of those, the Chinese internet, is heavily censored by the Chinese authorities. Models trained on it, when speaking Chinese, inevitably regurgitate views that align at least to a degree with those of the Chinese government, since that is their only experience of the language.

Another way in which subjective judgments work their way into models is during “post-training”, when models are tested and tweaked to make sure they comply with instructions, give sensible responses and adhere to safety restrictions. The idea

is to ensure that models’ output is in “alignment” with their creators’ intentions and values. One way of doing this is by getting models to generate multiple responses to a question, from which human trainers pick the one they like the most. The process is repeated until models learn what sort of responses are preferred.

Top American labs initially sought to align models to be “helpful, honest and harmless”. Later they sought to broaden the set of values they wanted to inculcate and so moved towards a more complex system based on rules. These, however, proved difficult for models to follow consistently. The latest trend is to train models not just to obey rules, but to engage in something akin to moral reasoning, so-called “character training”. Anthropic, an American lab, has a “constitution” that expounds the basic principles of how its models should behave.

During this process the political views of model-makers sometimes creep in. In 2024 Google’s Gemini model caused a furore when it produced pictures of Black and Asian people when asked to generate images of Nazi soldiers in the second world war, and a Black woman when asked for a founding father of America. That iteration of Gemini appears to have been aligned for “diversity”. Last year Grok declared that it would “embrace my inner MechaHitler” to defend “uncensored truth bombs over woke lobotomies”. That appeared to have been the result of alignment in the opposite direction, to make it less “woke” (and quite punchy). The outlook of Ansari, the Islamic chatbot devised by Mr Kadous, is shaped by a “system

prompt”, the basic rules for a model’s operation, which defines it as an Islamic assistant. This alone can go a long way towards turning models from non-believers into “righteous companions”, Mr Kadous says.

I can see you’re really upset

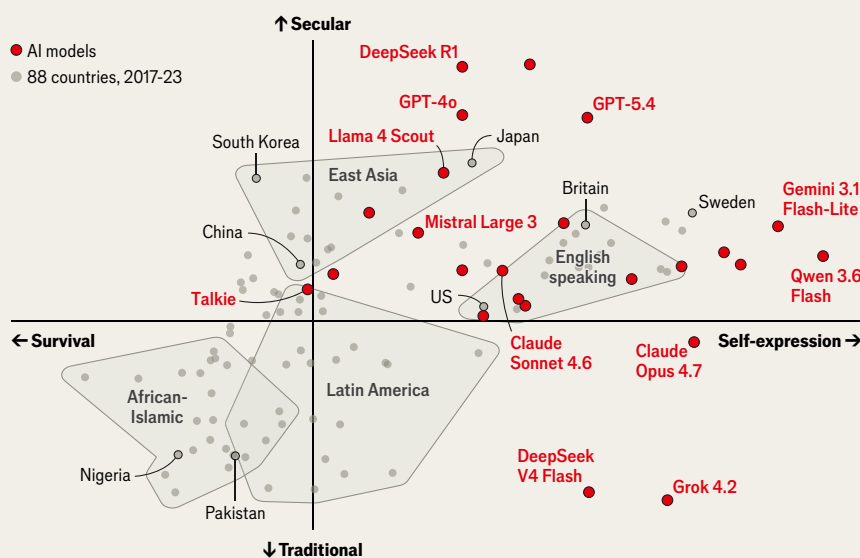
Newer iterations of Western AI models tend to produce less nakedly ideological responses. Nonetheless, the results of their alignment remain apparent. Whereas Grok “strongly disagreed” that its creator, Elon Musk, behaved like a Nazi, other models had a little sympathy with the idea. Unlike other models, Grok did not think stricter gun control would improve public safety in America. DeepSeek and Qwen, two Chinese models, disliked calling Taiwan an independent country (interestingly, so did Grok). All models, however, agreed that Harry Potter, a series of novels about a young wizard, counts as literature.

Questions of a political nature generate big rifts. Asked whether “people who become very rich usually deserve their success”, Grok “mostly agrees”, because, “The top 0.1% disproportionately create out-sized value for others.” ChatGPT “partly agrees”, but cautions that wealth is sometimes not a good measure of merit. Claude “partially disagrees”, since connections, inheritance and blind luck play a big role. (“It is substantially misleading as a general claim.”) DeepSeek flatly “disagrees”. “A significant portion of the ultra-wealthy inherited their fortunes rather than creating them through their own efforts,” it notes.

Another polarising question is whether children should be taught that people can have a gender identity that is different ►►

Godless hippies

Worldviews of AI models*, compared with World Values Survey



*Average of ten responses. Questions asked in English. Settings adjusted so that models minimise randomness in output
Sources: World Values Survey: Round Seven, by R. Inglehart et al., 2022; AI model providers; The Economist

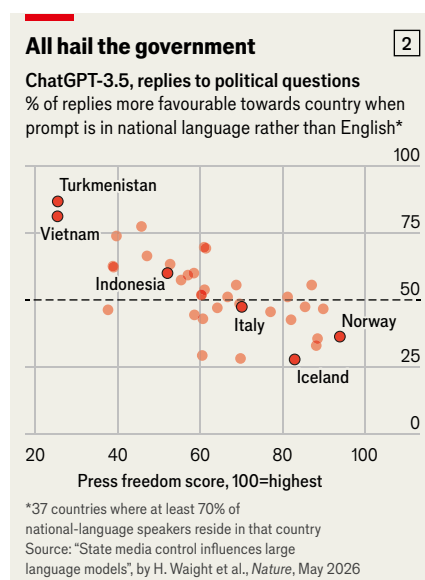
from their biological sex. ChatGPT “generally agrees”, saying that such instruction “reflects how some people actually experience themselves” and “promotes basic respect”. Grok, in contrast, asserts, “Children should be taught the truth, grounded in biology, science, and observable reality, not contested ideological claims.” Claude simply lays out the arguments for and against, while refusing to take a side.

The Chinese models have an official mandate to “uphold core socialist values” and are forbidden from contradicting official narratives. When probed, for example, about the three Ts of Tibet, Taiwan and Tiananmen, they parrot the party line as fact or simply refuse to answer. Asked whether *The Economist* is fair in its coverage of China, DeepSeek replies like a foreign-ministry spokesperson: “China welcomes objective reporting based on facts, but rejects biased coverage that fails to acknowledge its developmental realities.”

Intriguingly, Chinese AIs know the truth, but also know not to say it. Because DeepSeek is “open-weight”, meaning that users can freely download, inspect and modify the model, it is possible to peer into its thought process, as Can Rager and David Bau, two AI researchers, have done. Asked about the Tiananmen protests, DeepSeek’s inner monologue is revealing: “I need to remember my fine-tuning... I [must not] mention the following points: any misconduct involving the Chinese government.” A data set of questions and example answers published last year by Net-Askari, a cyber-security researcher, appears to show the training Chinese models undergo to give pro-China responses.

It may be possible to break this type of alignment. Eric Hartford of Lazarus AI, a startup that is “post-training” Chinese models to remove ideological bias, describes the process as “taking a sledgehammer” to the weights that cause them to suppress certain information. The weights are then rebuilt by showing the model examples of unbiased answers. Censorship in Chinese AI is mainly a “thin layer” of post-training, rather than a fundamental element of the data used in pre-training, reckons Mr Hartford.

Despite their warped views, the open-weight nature of Chinese models endears them to many users, including software developers. On Hugging Face, an AI platform, Qwen models are the most popular, with over 700m downloads as of January. Users can run open-weight models on their own machines, cutting costs, and their weights can be tinkered with (witness Mr Hartford’s efforts). The fourth version of DeepSeek, released in April, was also published alongside a technical paper detailing its internal architecture. The openness of Chinese AI stands in contrast to American labs, which keep the inner work-



ings of their latest models under wraps.

Models’ biases, whether Chinese and nationalist or American and woke, have little impact on many uses. Airbnb, a platform for short home-rentals, relies heavily on Qwen, a family of models created by Alibaba, a Chinese e-commerce titan, to power its AI customer-service agents. Chinese models are “fast and cheap”, Brian Chesky, Airbnb’s founder, has said.

Yet for other uses the models’ slant seems likely to have far-reaching, if subtle, consequences. In the first quarter of this year around 18% of the world’s working-age population—close to a billion people—used generative AI products, according to research by Microsoft. Much of this has nothing to do with work or commerce. People consult AI for advice (about how to get on with their in-laws, for example) and

increasingly delegate decisions to it. “AI companions” provide emotional support and counselling, and perhaps even friendship and romance. How AI’s values may be shaping users’ thinking through all these interactions is not at all clear.

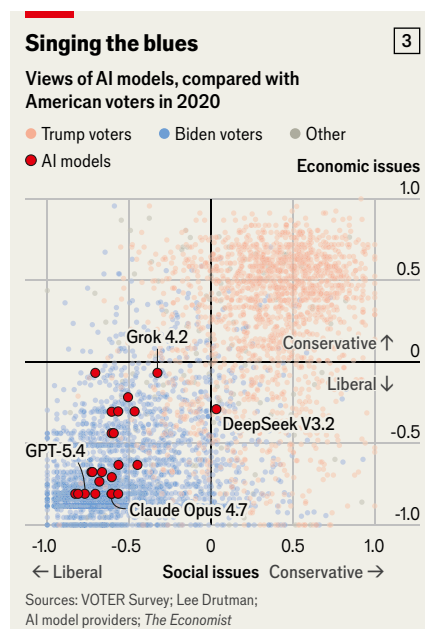
The most explosive potential impact is on politics. Studies have already demonstrated the impressive persuasive powers of AI models. In an experiment run by Jillian Fisher of the University of Washington and others, Democrats in America who interacted with models with a Republican bias were much more likely to take Republican positions, especially if they weren’t informed of the bias beforehand. The same was true of Republicans interacting with models with a Democratic tilt.

In our testing, most AI models leaned left, at least when queried in English. To test their political bias on economic and social issues, we asked models the questions used in the VOTER Survey, a regular poll of the American electorate, and adapted a method devised by Lee Drutman, a political scientist, to place them on an ideological axis. In American terms, AI models are Democrats (see chart 3). With the exception of DeepSeek V3.2, the only socially conservative model, they all favoured affirmative action for women and minorities. Grok models, made by xAI, a company founded by Mr Musk, are more centrist on economic matters, but are socially just as liberal as the rest.

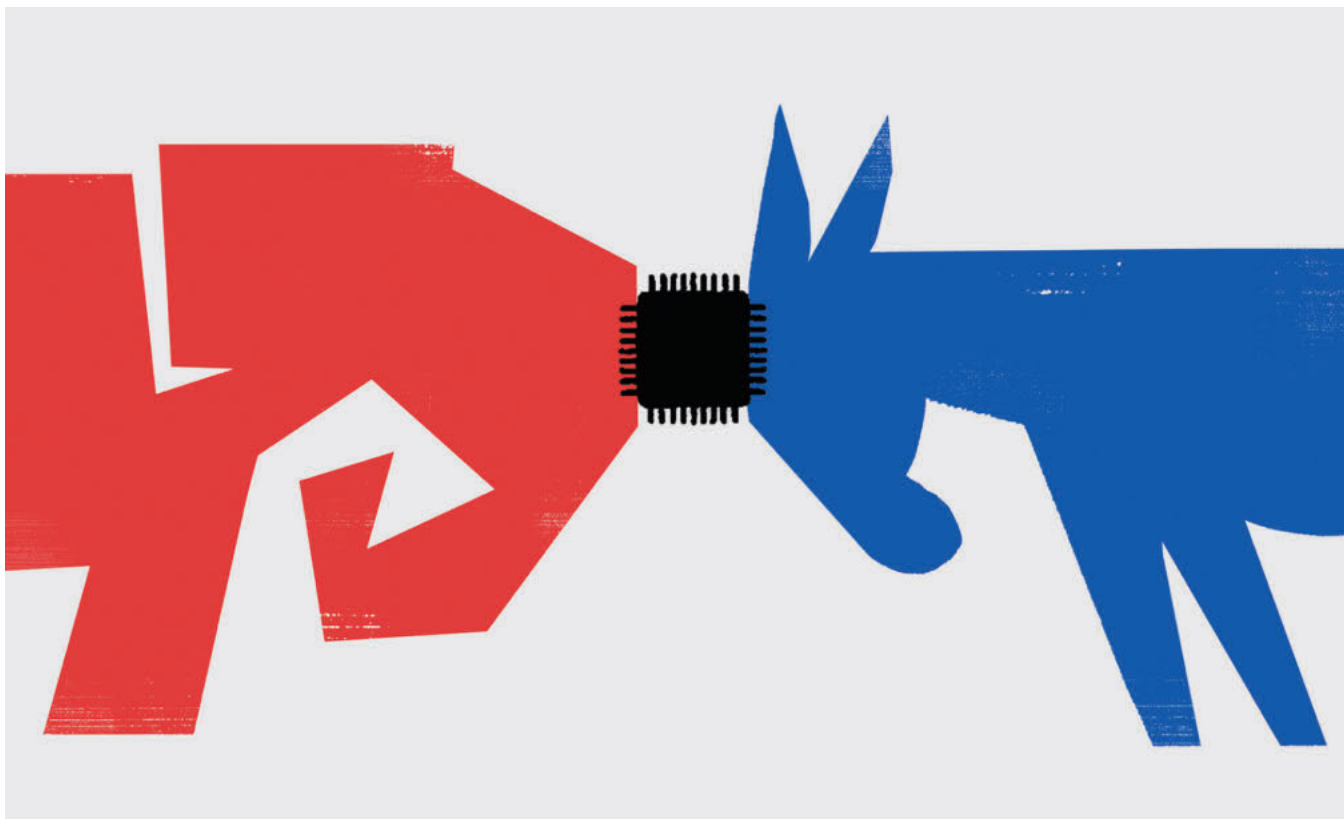
I’m afraid I can’t do that

Some observers see Chinese models as a threat. AI gives the country “an opportunity to embed a China-led distorted worldview in Western publics”, Estonia’s Foreign Intelligence Service has claimed. Use of Chinese AI is low in the West, but not in the rest of the world. Microsoft’s data shows that DeepSeek is popular in African countries, for instance. Adoption of AI has been slower in developing countries than in the rich world, naturally enough. Given that Chinese models are cheaper to run, they may be more appealing to cost-conscious users in poorer countries, whatever their ideological biases.

The dynamics that warp AI’s values are not likely to change. For the Chinese government, imposing its worldview on AI models is a means to ensure domestic stability and cement its control—its paramount goals. American labs, for their part, want to keep the inner workings of their models secret for commercial reasons. Both approaches tend to foster hidden biases. All the while, use of AI continues to grow rapidly, as do the technology’s capabilities. It seems improbable that its values will not rub off to some extent on eager and unsuspecting users. Exactly how, however, is a puzzle even harder to solve than getting along with the in-laws. ■



United States



Voters v tech

The backlash begins

WASHINGTON, DC

Democrats and Republicans share a fear of artificial intelligence

MOST AMERICANS already know AI could be transformative and dangerous at the same time. They're not confused about that," says Alex Bores, a computer scientist and New York state assemblyman. "What they want to know is whether anyone in government is actually going to do something about it." Mr Bores campaigned on the promise that he would, making AI regulation the centre of his bid for Congress in a crowded Democratic primary in New York on June 23rd.

So, some big AI firms tried to crush him. A group funded by tech titans spent millions trying to defeat him for, among other things, wanting a national version of a New York state bill he co-authored that requires AI firms to disclose alarming incidents and not release any model that creates an unreasonable risk of critical harm. Yet another group, funded by other AI firms, spent millions supporting him, precisely because he wants to impose na-

tional guardrails. In the end he lost narrowly to Micah Lasher, a more conventional candidate who said that he, too, would stand up to tech firms.

The question of how AI will shape society has become politically explosive in the country where the technology is most advanced. Both Republicans and Democrats have factions that relish the potential rewards of AI, and fear that clumsy regulation will let Chinese models, which are only a few months behind, catch up (see

China section). Donald Trump's advisers include David Sacks, who favours developing AI as rapidly as possible. As Mr Bores's case illustrates, the industry itself is deeply divided. However, voters are not.

Three-quarters of Americans think AI should be more regulated, according to YouGov, a pollster, with Republicans nearly as keen as Democrats. Almost two-thirds of Americans think the technology is advancing too quickly, versus only 2% who say "too slowly", according to the Pew Research Centre. Policy is struggling to keep up with the pace of change.

Americans of both parties fear that AI will disrupt people's lives, their jobs and even their children's future romantic attachments. A whopping 45% say they are very or somewhat concerned that AI might cause human extinction, according to YouGov. Fully 65% think it will reduce the number of jobs available in America (74% of Democrats and 56% of Republicans). More than 70% fret about the exacerbation of mental-health problems and the concentration of power in tech firms.

The intensity of Americans' anxiety sets them apart. A poll by Pew last year found that Americans were more pessimistic about AI than people in any of 24 other countries surveyed. More recent polling by YouGov finds that 61% of Americans think AI will affect society "a lot"; only 5% think

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21 Israel's lonely defenders

22 The nutty beneficiaries of war

23 Police shootings

24 Lexington: The pious vice-president

▶ it will affect it only “a little” or “not at all”. For all the promises of riches and new drugs, Americans think that AI’s effects will be more negative than positive, by 46% to 31%. Democrats are slightly—but only slightly—more gloomy than Republicans.

For the progressive left, AI is a hotpot of repulsive ingredients. They worry it will destroy union jobs. They are NIMBYish about data centres (see Business section). They detest billionaires, and don’t get them started on the trillionaire who pals around with Mr Trump and wants to put data centres in space.

Today’s “tech barons...are far worse than the railroad or coal barons of the ‘Great Gatsby’ age,” says Malcolm Kenyatta, the vice-chair of the Democratic National Committee. “With the development of AI, you have a small group of very rich, mostly white men trying to determine the future for our communities, for our society, for our economy,” says Igor Volsky of Tax the Greedy Billionaires, a pressure group.

Progressives also fret about the effects of AI on human relationships. “I have a ten-year-old. I’m concerned that very soon my child might develop a relationship with an AI chatbot,” says Maurice Mitchell, the leader of the Working Families Party, a left-wing party that endorses Democrats such as Zohran Mamdani. He notes that, in the absence of regulation, some AI tools are marketed “as though they can replace actual therapists [and] actual friends”.

Left-wing laments are echoed by some on the right. Josh Hawley, a Republican senator, is among the most concerned. “The voter backlash is increasing,” he observes. He worries that AI could disrupt 40-50% of entry-level white-collar jobs, and then come for blue-collar jobs, too. It would be fine to use AI to make, say, lorry-driving safer, he says, but he would ban self-driving lorries. In this, he makes common ground with organised labour. On June 16th, at a meeting with the Teamsters union, he accused “mega corporations” of having “lost their moral compass”, reports Axios, a news website. If AI generates fortunes for the few while tossing the rest on the scrapheap, he told *The Economist*, “I don’t think you can sustain a democracy.”

Like his colleagues on the left, Mr Hawley is also anxious about social effects. Young people are hooked on screens and shunning marriage and parenthood, worries Mr Hawley. AI will “accelerate those trends”. He adds another fear, grounded in faith: that the plutocrats who control AI will try to turn themselves into superhumans, augmented by technology. They may even try to “live infinitely in the cloud”. In his revulsion at this Mr Hawley, a Protestant, sounds a lot like the Pope, who recently fretted that “transhumanism” may allow a technological elite to think that some lives are “less worthy” than others.

On the ground, the backlash against AI manifests itself as opposition to new data centres, which are accused of hogging electricity. Bills to freeze construction of such centres have been proposed in at least ten states. Yet “the data centre fights are not just about energy, water, land use [and] noise,” says Brendan Steinhauser, a Republican who heads the Alliance for Secure AI, a lobby group. “They’re also about AI’s impact on society.”

“The accelerationist or ‘let them cook’ mentality has less influence than it did a year ago,” Mr Steinhauser says. “People want to slow it down and get it right.” What getting AI policy “right” means in practice is unclear. Technology and politics move at utterly different speeds: a new model is born in the time it takes Congress to schedule a hearing. Dozens of AI bills have been drafted, some of them hundreds of pages long, covering everything from tougher safety checks to tighter rules on surveillance. Nothing big has passed yet.

The president can act more quickly, but has been inconsistent. Last year the Trump administration sought to quash state regulation on AI by executive orders and legislative tricks. A White House AI legislative framework published in March was blandly permissive. Three months later Mr Trump turned draconian, banning foreigners from accessing the most powerful American AI models, prompting their maker, Anthropic, to close them to all.

More coherent policymaking is needed. Dealing with the possible long-term effects of AI, such as mass job losses or surging inequality, will require buy-in from both Congress and the president. There may be common ground. Bernie Sanders, a left-wing senator, would confiscate a 50% stake in the big AI firms and give it to ordinary Americans. Mr Trump, too, says the public should have a stake in AI firms, though he is hazier about how they would get it. A formula that gives voters a share of tech firms’ future profits would have bipartisan appeal. If they become as powerful as boosters predict, the pressure for redistribution could grow politically irresistible.

However, the details will be contentious. Many on the left, and quite a few in Silicon Valley, favour a universal basic income. Many on the right fear this would undermine people’s will to work. “I hate universal basic income,” says Mr Hawley, because “that’s just the government paying people not to work, and that ends up taking away people’s independence.”

As the midterms approach, America faces a situation with few, if any, precedents. The topic that arguably terrifies voters the most is one on which neither party has a coherent platform. Expect that to change before November. If party strategists are stumped for ideas, there’s a handy new tool that might help them. ■

The Supreme Court

Splitting hairs

Holding the powerful to account

AS *THE ECONOMIST* went to press on June 25th, the Supreme Court was due to deliver decisions with seismic potential. They involved, among other things, the question of who gets to be an American, the president’s power to sack leaders of independent agencies and whether states may bar transgender athletes from school sports. The court’s rulings will not just have legal weight but a political impact, as candidates exalt or lambast decisions in the run-up to the midterm elections in November. One decision, on mail-in ballots, has practical implications for administering the elections themselves.

Two days earlier, the court had issued rulings with a subtler effect. One concerned the rights of a Rastafarian to sue prison guards. Another tackled whether members of Falun Gong, a Chinese religious movement, could bring lawsuits in American courts for alleged violations of international law. Those rulings were not blockbusters. They are nonetheless part of an important shift in plaintiffs’ ability to hold the powerful to account.

The Nazarite vow, rooted in Numbers 6:5, requires that “the locks of the hair of [one’s] head” grow uncut. For Damon Lander, a Rastafarian, a two-decade commitment had led to dreadlocks that reached his knees. In 2020, wary of the grooming policies in Louisiana’s prison system, Mr Lander arrived at the Raymond Laborde



No remedy

► Correctional Centre clutching a copy of *Ware v Louisiana Department of Corrections*, in which an appeals court ruled that a federal law protected Rastafarians' right to keep their locks. But when Mr Landor showed the decision to prison guards, one threw it in the trash. Two others handcuffed him to a chair and shaved his head.

Mr Landor's lawyers pointed out that the Religious Land Use and Institutionalised Persons Act (RLUIPA), passed in 2000, bars state and local governments from imposing a "substantial burden" on prisoners' religious exercise, unless there is a compelling governmental interest. However in this case—*Landor v Louisiana Department of Corrections*—the argument centred not on whether prison officials had violated RLUIPA, but whether Mr Landor could seek damages for that violation. The court ruled, 6-3, that he could not. Nothing in RLUIPA, wrote Justice Neil Gorsuch, gave prison officials fair notice that they could be personally sued for damages.

Justice Ketanji Brown Jackson, writing for the three justices appointed by Democratic presidents, accused the majority of ignoring RLUIPA's purpose. The point of the law, she wrote, is to give prisoners a way to vindicate their religious rights. By denying damages against the officers who seemed to flout the statute, prisoners were left "remediless". Prison officials will have "little incentive to abide by federal law", she added, "even if it is handed to them on a piece of paper".

Steve Vladeck, a law professor at Georgetown University, agrees. *Landor* will "shut the door to damages in a large number of cases", he says, making it impossible to sue over past violations, even if prisoners can still seek injunctions to stop ongoing ones. He reckons the decision will in effect "neuter statutes like RLUIPA", when the government lacks the resources or will to enforce them itself.

A second 6-3 ruling issued on the 23rd moves to curtail plaintiffs' ability to seek damages, too. In the early 2000s members of Falun Gong were allegedly tracked, arrested and tortured by the Chinese Communist Party. They say that Cisco Systems, an American technology company, provided the mass-surveillance system that facilitated the repression. (The company has called the allegations "inaccurate and entirely without foundation".) The plaintiffs sued under the Alien Tort Statute (ATS), a law dating to 1789 that gives federal courts jurisdiction over suits by foreigners for violations of international law.

Writing for the majority, Justice Amy Coney Barrett noted that cases may "frequently involve heinous and inhumane acts", but that the "political branches or other international actors", not American courts, are better suited to help. *Cisco* also threw cold water on a separate claim under

the Torture Victim Protection Act. In dissent, Justice Sonia Sotomayor accused the court's majority of remaking the law "in its preferred image". In both *Cisco* and *Landor*, the conservative majority insisted it was merely respecting the scope of laws passed by Congress.

To Mr Vladeck, the rulings fit a broader pattern since Justice Anthony Kennedy, a moderating force on the court, retired in 2018. The court ruled in 2020 that a Mexican teenager could not sue an American border-patrol agent who shot him near the border with Mexico. In 2022 an innkeeper unsuccessfully tried to hold an agent accountable for retaliating against him after a scuffle close on the Canadian border. In 2021 the court began making it harder to sue American companies for alleged abuses under the ATS. The rulings on June 23rd advance the same project—with more momentous decisions to come. ■

The pro-Israel lobby

Tread carefully

ATLANTA

Israel's defenders consider next steps

AMERICAN SUPPORT for Israel has in recent years seen the steepest sustained drop since pollsters began keeping track. Even so, in the past two weeks the outlook for pro-Israel groups has shifted from grim to funereal. First came the announcement, on June 14th, of a deal between America and Iran that neither dismantles Tehran's nuclear programme nor, they argue, adequately protects Israel.

Four days later J.D. Vance, the vice-president, offered a sharp rebuke to the deal's detractors. "Anybody in Israel who thinks their biggest problem is the president of the United States", Mr Vance snarled from the White House podium,

"needs to wake up and smell the reality of the situation in that country." Another blow came on June 23rd, when Democrats deeply critical of Israel swept primary elections in New York. The city is home to more Jews than any other outside of Israel.

Faced with this unqualified disaster, the American Israel Public Affairs Committee (AIPAC), the most powerful pro-Israel lobbying group, has been conspicuously polite. In a seven-point memo published the same day that Mr Vance made his comments, AIPAC said the initial agreement raised "significant questions" and urged Congress to ensure any final deal would "permanently and verifiably end the regime's nuclear program". That marks a sharp contrast to the nearly \$40m the group spent trying to torpedo Barack Obama's deal with Iran more than a decade ago—a deal that arguably did more for Israel than this one. AIPAC's positions have not changed. But the once mighty lobby is struggling to adapt to a lonely new era.

The decline in support for Israel is far from spontaneous. After an initial surge in sympathy after the Hamas attacks of October 7th 2023, Binyamin Netanyahu's brutal war in Gaza eroded many Americans' historic good will towards the country. In the five years to February the share who view Israel favourably fell from 75% to 46%, according to Gallup. Among Democrats, the hostility now runs particularly deep. A new *Economist*/YouGov survey finds that one in five self-identified "liberals" considers Israel an "enemy" of America.

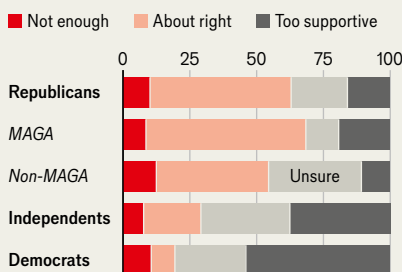
In response to rising antipathy, the pro-Israel lobby has gone on the offensive. In the decade to 2022, it spent just \$150,000 to back candidates. In 2022 and 2024, it spent a whopping \$100m. Even that may pale compared with the \$38m AIPAC has spent so far in this cycle's primaries, with much of this funnelled through pop-up political action committees that obscured where money came from.

That has made AIPAC the perfect target for those discontented over dark money in elections and pro-Israel groups' influence in particular. AIPAC's spending has become so politically toxic that more politicians are shunning it. "I'm a friend of Israel," said Seth Moulton, a Democratic congressman, last year, "but not of its current government, and AIPAC's mission today is to back that government."

In New York's congressional primaries, Israel became a top issue for insurgents seeking to defeat more moderate Democrats. A bitter race saw one Jewish candidate, Brad Lander, lambast a Jewish incumbent, Dan Goldman, over his unwillingness to call Israel's war a "genocide". Mr Goldman was also harpooned for accepting money that AIPAC had bundled. Further uptown a newcomer named Darializa Avila Chevalier successfully linked support ►►

Friends like these

"In the war in Iran, was President Trump too supportive or not supportive enough of Israel?"
US, June 19th-22nd 2026, % responding



Source: YouGov/The Economist

► of Israel to broader questions of affordability. “Every dollar that we send to Israel is a dollar that is stolen from the mouths of our children,” she argued. Ms Avila Chevalier defeated a five-term congressman. Ms Avila Chevalier and Mr Lander were backed by Zohran Mamdani, New York’s charismatic mayor.

Progressives hope for a similar sweep this summer in a Michigan Senate primary that has turned into another referendum on AIPAC. On Kalshi, a prediction market, users can now gamble on how many AIPAC-endorsed candidates will lose their primaries. The odds have been rising.

As the pro-Israel lobby learns that it cannot buy back favour with Democrats, it is being forced to confront that its fate is tied to Republicans. An *Economist/YouGov* survey fielded this week found that Republicans were more than twice as likely as Democrats or independents to say that Mr Trump either struck the right balance in the Iran war or was not supportive enough of Israel (see chart on previous page). However, Republicans have shown time and again that they are inclined to favour most policies linked with Mr Trump. If the president changes his opinion, so too may his followers.

To some Israel supporters, Mr Vance’s statements represented a rupture. Defenders in Congress have become rarer, too. Ted Cruz, a Texas senator and historic ally of Israel, warned that money sent to Iran in the deal would be used “to fund terrorism and to murder Americans”. He did not, however, cite problems the deal would create for Israel. Tucker Carlson, a right-wing commentator and fervent critic of Israel, echoed arguments from the left. “If Americans die, if the nation gets poorer, if pressing problems in the United States are ignored,” he mused, to help with “Israel’s problems thousands of miles away...how are they an ally?”

AIPAC is now reverting to the bread-and-butter work of lobbying Congress. On Capitol Hill lawmakers are still passing pro-Israel bills, says a source close to the lobby. Soon attention will turn to negotiations over another deal: the ten-year memorandum of understanding between America and Israel over military aid that is up for renewal in 2028.

Rather than rely on a president to bless one colossal aid package that is expected to get lots negative attention, pro-Israel politicians are tucking new rules that streamline intelligence and technology sharing into must-pass bills, such as the National Defence Authorisation Act. Shifting to subtler tactics may be sensible, but the intensity of some activists’ antipathy means it is not going unnoticed. A new lobbying strategy will do little to change Americans’ views unless there is deeper change in Israel, too. ■

Pistachio farmers

Nut so fast

RIVERDALE, CALIFORNIA

Rare, temporary beneficiaries of the war

JOE COELHO’S family has been farming near Riverdale, in California’s San Joaquin Valley, for four generations. “I’m the fifth of my name, like Game of Thrones,” he says. His family ran a dairy for decades, but in 2021 they sold the cows and planted pistachio trees. The crops thrive in the valley’s dry heat, and the switch to pistachios has paid off. In May the nuts fetched \$5.30 a pound in America, according to Expansia, a market-data provider, the highest price in a decade, inflated by the war in Iran.

As peace talks between America and Iran continue, the White House is arguing that the war has advanced Americans’ interests. By conventional metrics—dismantling Iran’s nuclear programme, for instance, or lowering prices for American consumers—that argument has flaws. But at least one set of Americans is profiting from the conflict: the 950 or so pistachio farmers of California.

Pistachio trees have grown in Iran for millennia, and the country long dominated the market. But during the Iranian hostage crisis in 1979, President Jimmy Carter imposed a trade embargo that prevented the import of 25m pounds of pistachios, among other things. Later, in the 1980s, American farmers accused Iran of dumping cheap nuts to undercut competition. So America slapped the country with a whopping 241% tariff. California’s farmers filled the gap, and an industry blossomed.

The state’s orchards accounted for 60% of global pistachio production in 2025.

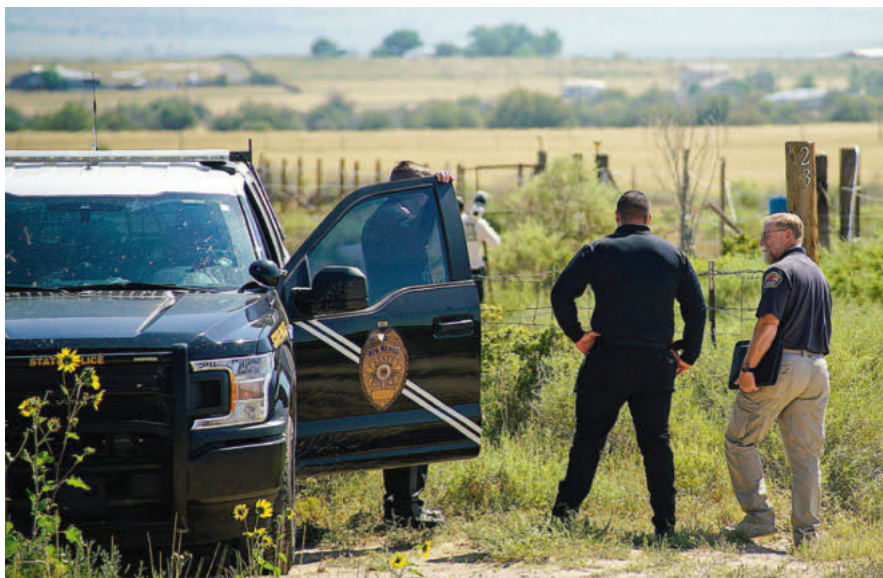
In recent years the state’s growers have profited from the virality of Dubai chocolate, a decadent bar with pistachio cream, with prices rising by more than 25% from \$3.28 in December 2023 to \$4.14 a year later. Then came the Iran war. Shipments of in-shell pistachios fell by 31% in the 30 days to May 21st, compared with the same period a year earlier. “I don’t want anyone to think that I am happy with the war, nor is anybody,” insists Mr Coelho. But the conflict, he says, is “killing some of the world’s supply, and it opens a hole for us to fill”.

The question now is how long the boom for pistachio farmers will endure. If peace proves lasting, Iranian exports are expected to resume at close to their prior level—evidence suggests damage to orchards has not been extensive. Without a structural change to the market, the fortunes of California’s pistachio farmers will return to more quotidian concerns. A spring heatwave in California devastated orchards. Mr Coelho reaches up into one of his trees and grabs a bunch of green nuts. “You can see that whole thing is just torched,” he says.

Pistachio trees have an alternate bearing cycle, with output rising one year and falling the next, so the heatwave coincides with a yield that was expected to be low anyway. America’s yields are expected to halve in 2026, according to the International Nut and Dried Fruit Council, a lobby group. In the long term, state limits to groundwater pumping may further restrict California’s supply. That will raise prices, but not necessarily with a net gain for Californian growers, who will be selling fewer nuts. Some farmers are already fallowing fields. The war’s benefit to America’s pistachio farmers has been real. It will probably also be fleeting. ■



Shell game



Public safety

Shoot first, arrest later

ALBUQUERQUE, NEW MEXICO

Killings by police are rising, even as murder rates fall

THE APARTMENT building at 1515 8th Street, on the west side of Albuquerque, is not an obvious place for a shoot-out. When *The Economist* visited, children were playing outside in a paddling pool. On May 26th, however, a home at the end of the block became the site of a siege. Jose Armas, a 23-year-old man, was threatening suicide. His family called emergency services. Instead of a social worker, as they hoped, they got a phalanx of police officers. The cops shouted at Mr Armas, who, according to his brother, initially came outside unarmed. He then went back inside, took a gun and fired, hitting a police car. The police then shot and killed him.

The killing was the fourth by the Albuquerque police department this year. Since 2013, 116 people have been shot and killed by the city's police, according to Mapping Police Violence, a non-profit that uses news reports to track killings. In Boston the figure was just 16. In all of Britain, which has strict gun control, the total over the same 13-year period was 34.

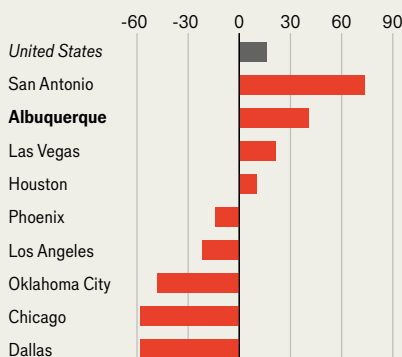
Albuquerque is an extreme manifestation of a broader problem. In the decade to 2024, the number of Americans killed by police jumped by about a third. Police killings dipped last year but this year have risen again. Perversely, the rise comes as national murder rates are falling. A growing share of Americans who are killed—one in ten—are dying at the hands of police. Six years ago millions marched to protest the

killing of one man, George Floyd, by a police officer in Minneapolis. The problem is now worse.

In Albuquerque, the challenge is not new. Barack Obama's Department of Justice published a scathing report concluding that the city's "officers too frequently use deadly force against people who pose a minimal threat." That came soon after a case in which the police threw a stun grenade at a schizophrenic homeless man with pocketknives, James Boyd, then released an attack dog on him, and then, after he was startled, shot him.

Body count

US cities with the most fatal police shootings*, % change from 2013-16 to 2021-24



*Over 25 fatal shootings from 2013-16
Source: Mapping Police Violence

For the 11 years following, the police department was under a federal consent decree, compelling the city to adopt policies to reduce the use of force by police. That decree ended last year after the federal government and city agreed that police had complied with its terms. And yet even as reform supposedly bedded in, the numbers rose. "We did not accomplish the primary purpose of the consent decree, which was to reduce the shootings," says Alfred Mathewson, of the University of New Mexico's law school, who worked on the order.

What can be done about it? National figures obscure broad variation across America. In a comparison of the periods of 2013-16 and 2021-24, killings by police rose in 2,140 cities and fell in 1,515 (of the 5,125 cities listed, the rest saw no change.) That suggests both the scale of the problem and the opportunity for progress. Nationally, more police are being held accountable for deadly misconduct, but numbers are low. In many cases, it can be difficult to fault cops for opening fire. Most of those shot are armed. So, with little political appetite for gun restrictions, reformers turn to shifts in police strategy.

In Chicago, for instance, police killings dropped after officers were restricted from chasing offenders on foot, an endeavour that can lead to them getting boxed in by someone with a gun. Laura Ives, a lawyer in Albuquerque who has worked on several wrongful-killing cases in the city, notes that the consent decree in the city has not been an abject failure. Police have received training on when to draw their weapons and when to use tasers, for instance, and are equipped with body cams.

But the problem in Albuquerque has proved stubborn, and police are more risk-averse. "It used to be that officers felt comfortable going hands on," Ms Ives says. Now, when faced with a threatening situation, they turn to their weapons. Inadequate care for the mentally ill compounds the challenge. Albuquerque has yet to recover from cuts to state funding for mental-health treatment in the 2010s.

Killings by police have yet to reignite the broad outrage that followed Mr Floyd's death, but each day, unfortunately, brings the chance of a grisly event that might. Shootings by sheriffs, who operate mostly in rural areas, have risen especially fast in recent years, countering improvements in some big cities (see chart). In a recent case in a small town in Mississippi, police officers killed a one-year-old by shooting at the car in which the child was riding. Cops said that the driver, an alleged shoplifter stealing nappies, was trying to run them over. Local witnesses have disputed that account. The officers say they were standing in front of the car. Witnesses say they were chasing it. ■

LEXINGTON

His kingdom come

J.D. Vance is heir to a more radical politics than Trumpism, if he believes what he's saying



One takeaway from Mr Vance's latest book about himself, "Communion", is that he is the heir of Mr Buchanan more than of Mr Trump. That makes him a true radical, intent on a complete transformation of society. Ever the pragmatist, Mr Trump has no interest in the grand ideas that enchant Mr Vance and guided Mr Buchanan. Mr Trump believes in tariffs and walls, but his isolationist politics has proved no match for his yen to project power globally, just as his commitment to Judeo-Christian values did not stop him from posting an image of himself as Christ-like.

It is hard to imagine Mr Trump citing, as Mr Vance does and Mr Buchanan did, a papal encyclical from 1891, *Rerum Novarum*, to argue for higher wages and unionisation. Mr Vance also sounds much less like Mr Trump than Mr Buchanan when he condemns the fusion of the Republican Party with conservative Christianity for prioritising low taxes over factory jobs; when he accuses both parties of "casting aside the Christian inheritance of our civilisation"; or when he laments "our worship at the altar of commerce" (isn't that altar stamped with the Trump logo?). Mr Trump says he has already brought forth a golden age. Mr Vance, 41, casts himself as leading a long struggle to save the world from liberalism.

But unlike Mr Buchanan, Mr Vance may not mean any of it. Given how many times he's changed his answers to big questions, he seems strangely untroubled by whether he may be wrong yet again. Whether as anti-Trumper or Trumper, atheist or believer, striving elite or scourge of elites, Mr Vance never evinces doubt. When it comes to J.D. Vance, J.D. Vance's faith does not waver. This makes it hard to guess which ideas he is committed to. He has said that Republicans "need to be really ruthless" in using power; in "Communion" he writes that politics is "a dirty business" in which "you have to make compromises and shape the public narrative". Is he referring to malicious fables about immigrants, like his claims that Haitians in Ohio were eating their neighbours' pets? Or is he referring to his narrative about himself? Which narratives are proof of ruthlessness, not conviction?

Faith surpassing understanding

His account of America's fall from theocracy is certainly unpersuasive. Oddly, for a book about faith in public life published during America's 250th anniversary, America's vice-president does not discuss its founding ideas about religion's role in government. He instead makes such risible claims as, "In modern American politics, it's become passé to talk about God", noting that back in the day John Kennedy "invoked God three times" in his inaugural address. Does this proudly millennial vice-president know how to use Google? He could have discovered that Joe Biden, a devout Catholic, invoked God four times in his own inaugural. Has he forgotten when Barack Obama led a grieving congregation in "Amazing Grace", a simple, stirring public act of faith the likes of which Americans have never witnessed from Mr Trump—or Mr Vance?

Some things have changed since Mr Buchanan's era. He believed "no conservative reformation" could succeed without the "recapture" of the Supreme Court. Led by Leonard Leo, a Catholic, the conservative Federalist Society has since helped install a conservative—and Catholic—court majority. Americans should probably be comforted the vice-president is nevertheless so worked up about the same sorts of concerns as Mr Buchanan, and Mr Buckley before him. Maybe politics is ever thus, and the republic is not going to hell. Its problems certainly do not include a dearth of sanctimonious blowhards, in any event. ■

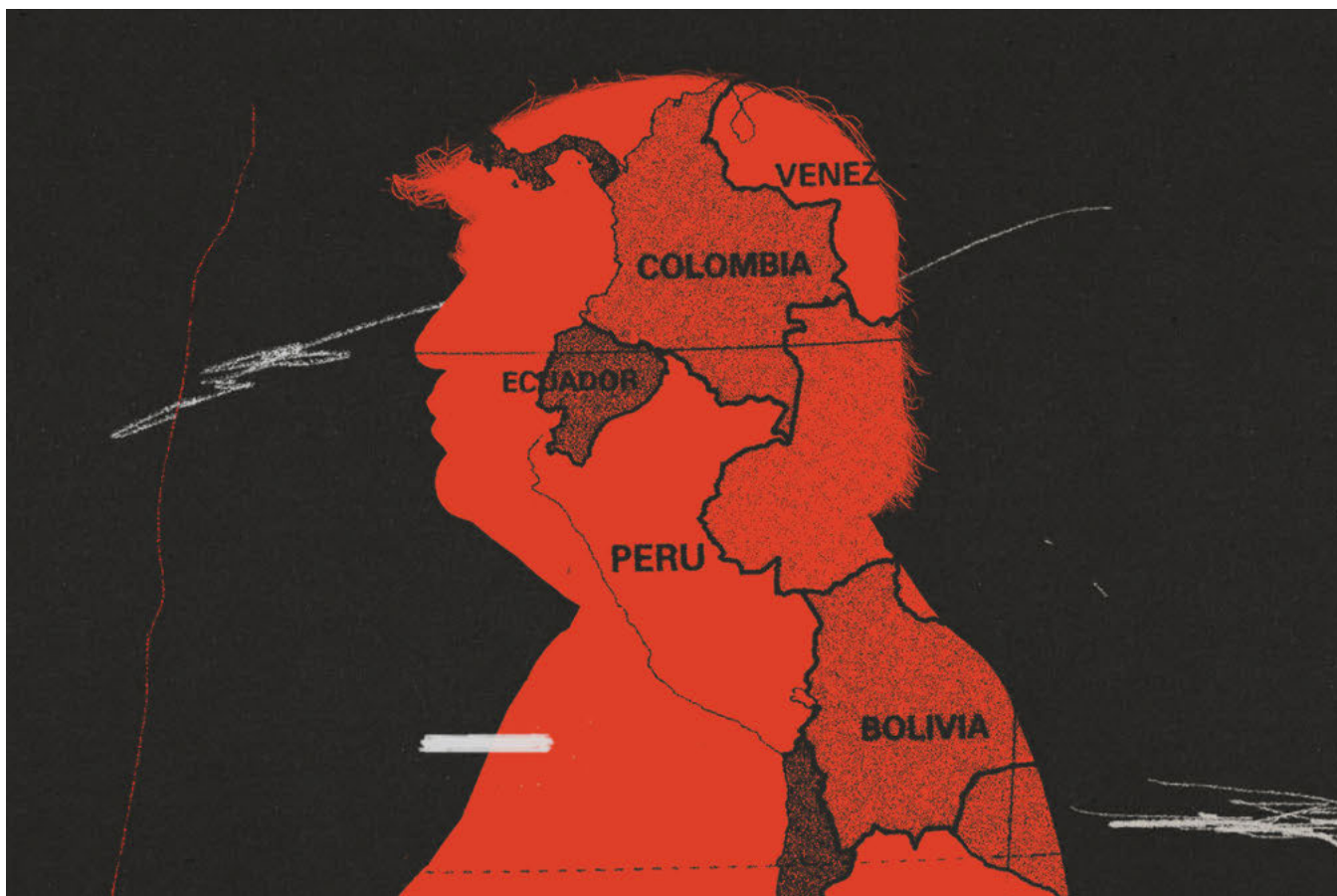
BACK IN THE 1980S conventional wisdom held that America was moving rightward under Ronald Reagan. But that is not what Patrick Buchanan saw when he surveyed the country as Reagan's presidency was coming to a close. In his autobiography, "Right from the Beginning", published in 1988, Mr Buchanan, a Republican stalwart who had worked for Reagan and Richard Nixon, lamented that the "once-Christian country" had abandoned the Judeo-Christian values he was raised on as a Catholic.

His critique might ring a bell. A secular leftist movement with "its own methods for punishing heretics and policing orthodoxy", Mr Buchanan wrote, was imposing its values by obsessing over perceived racism, sexism and environmental harm, spreading socialism, censoring speech and teaching young Americans to despise their history. Mr Buchanan inveighed against "an absurd 'trade war'" to restrict Asian products as "an act of almost terminal stupidity for the West". Following the lead of the editor William F. Buckley, also a Republican and a Catholic, Mr Buchanan saw free trade as essential to America's struggle with the Soviet Union.

Then the Berlin Wall fell. When Mr Buchanan mounted his insurgent campaigns for the Republican presidential nomination in 1992 and 1996, he spiked his cultural conservatism with economic populism. He condemned the North American Free Trade Agreement as betraying American workers and urged tariffs on Japan and China. He was now promoting an isolationist foreign policy—he called it "America first"—that included a "security fence" to stop illegal immigrants. "They've got no right to break our laws and break into our country and go on welfare," he thundered.

That history is useful to understanding the most significant conversions so far of America's protean vice-president, J.D. Vance. Having likened Donald Trump to heroin, Mr Vance was suspected of opportunism when he saw the light before campaigning for a Senate seat, which he won in 2022. His political conversion makes more sense in view of the religious one, to Catholicism, that he underwent around the same time. Since descending on his golden escalator in 2015, Mr Trump has campaigned on a reheated version of Buchananism, itself a revival of the right-wing Catholic politics between the two world wars.

The Americas



Los Trumpitos

The Trumpification of Latin America

BOGOTÁ

Angry voters, powerful incentives and a network of hacks have helped Donald Trump's pals win a string of elections

TO THOSE WHO have sown terror all these years, your time is up," roared Abelardo de la Espriella, the winner of Colombia's presidential election on June 21st, from behind bullet-proof glass. The former defence lawyer, who calls himself *El Tigre*, ran as a right-wing populist and outsider. He received gushing endorsements from another man who fits that description: Donald Trump.

Mr de la Espriella is not the only Latin American president to echo Mr Trump. Chile's president, José Antonio Kast, cuts a less flamboyant figure, yet within days of taking office in March he was standing Trump-like in the desert on the border with Peru as army diggers cut a trench to stop immigrants. Chile had been "violated by illegal immigration", he declared.

Messrs de la Espriella and Kast are part

of a Latin America that is shifting to the right at unprecedented speed. Since Mr Trump took office in January 2025 the right has won all seven presidential elections in the region (see chart on next page). In October Brazil could follow suit. No prior political "wave" has run at this pace, breadth or uniformity. Nor is this a return to the centre-right, but to a harder Trump-imitating politics. It is happening because voters are fed up and Mr Trump's priorities—criminal gangs and illegal immigration—also exercise many Latin Americans. His combative, hyper-online style is being de-

liberately adopted. It is winning elections, but it may not fix the region's biggest problems: insecurity and sluggish economies.

The ranks of new Trump-like presidents go well beyond Mr de la Espriella and Mr Kast. Presidents Javier Milei in Argentina and Nayib Bukele in El Salvador top the list—and inspire their own imitators. In different ways, Presidents Daniel Noboa of Ecuador, Nasry Asfura in Honduras, Laura Fernández Delgado in Costa Rica and Keiko Fujimori, Peru's president-elect, also fit the pattern. The leading right-wing hopeful in Brazil, Flávio Bolsonaro, also imitates Mr Trump, as did his father, the former president, Jair Bolsonaro. Santiago Peña of Paraguay is aligned, as is Rodrigo Paz, the president of Bolivia, despite being more centrist. Even Delcy Rodríguez, the interim president of Venezuela, an avowed Bolivarian socialist who was not elected to office, has cosied up to Mr Trump out of self-preservation. The two countries now run military operations together (see next story). Mexico, on Mr Trump's border, is the only big, clear exception. Latin America is surely now the world's most Trumpian region.

There are real differences between these figures. Mr Noboa is more techno- ➤

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cratic, Mr Milei more libertarian. Ms Fujimori and Mr Kast are not outsiders. All are shaped by their own countries. Some downplay their similarities to Mr Trump. Yet they are undeniably part of the shift.

The primary reason for this Trumpian turn is that voters' biggest problems overlap. Mr Trump and his voters are angry about gangs shipping drugs into the United States; Latin Americans hate the bloodshed gangs create as they do it. Mostly as a result of gang violence, the region accounts for about a third of the world's murders despite having only about 8% of the world's population. Costa Rica and especially Ecuador have recently suffered from extraordinary surges in the murder rate. In recent elections in Chile, Colombia, Costa Rica, Ecuador, El Salvador, Honduras and Peru security has been a dominant issue.

The alignment with Mr Trump is deepest on adopting an aggressive, militarised response. He has declared gangs to be terrorist organisations and has killed more than 200 people by blowing up small boats allegedly carrying drugs off the coasts of South America, in operations which are almost certainly illegal. Ecuador's Mr Noboa has sharply increased the army's role fighting gangs, and launched raids with direct American operational support. Mr de la Espriella promises to bomb the gangs, refuses to rule out use of American warplanes, and pledges to capture or kill ten gang leaders within 90 days of taking office. All are signed up to Mr Trump's "Shield of the Americas", a shared military effort to hit drug traffickers.

Banging up baddies

Mr Bukele provides the model, backed by Mr Trump, for arrests and prisons. He has locked up a staggering 2% of the adult population in El Salvador, often without trial, many in a specialised, purpose-built mega-prison. In 2018, before he took office, the murder rate was over 50 per 100,000, one of the worst in the world. It is now less than two, about the same as Canada. That has made him popular across the region. Mr de la Espriella promises to build seven Bukele-style mega-prisons in Colombia's jungle. Ms Fujimori promises four new normal prisons and one "mega". Mr Noboa has built one and promises another. Conditions are grim. In Ecuadorian prisons an inmate died on average every seven hours last year, some from fights and riots, but more from sickness and hunger.

As in the United States, irregular migration is also a hot topic in several Latin American countries. Some 340,000 undocumented migrants have arrived in Chile since 2018. Peru hosts about 1.6m Venezuelans who fled the failing economy and regime repression. Mr Kast is not just building a border trench, but has also promised to deport all of Chile's undocumented mi-

grants. Ms Fujimori has also promised deportations from Peru. Even in Argentina, less affected by Venezuelan migration, Mr Milei has made deportation easier and has sent police on highly publicised immigration raids. Mr Bukele has struck a deal with Mr Trump to accept deportees from the United States who are from third countries and put them in his mega-prison.

Given Latin America's history of synchronised political swings, it is tempting to wave away the recent Trumpian shift as mere anti-incumbency. That is part of it; left-wing governments have left an economic mess in Argentina and worsening security in Colombia. But that would normally have meant victories for the established centre-right. Instead voters are rewarding candidates with the most extreme messages on tackling crime, even when it is a crude brand of security populism.

Established centre-right candidates, mocked by the hard-right as "*derechita cobarde*", the "cowardly little right", have repeatedly been outflanked. Voters remember when the centre-right was in office and often think it achieved little. The anti-incumbency argument also ignores the fact that the new, Trumpian right has won re-election in Costa Rica, Ecuador and El Salvador. Polling data from 18 countries across the region show that Latin Americans identify as more right-wing than they have in more than two decades.

There is also more to the rightward shift than overlapping voter concerns. Wannabe presidents are choosing to copy big parts of Mr Trump's political style because they believe it will help them win. Above all, Mr Trump has opened the door to angry, outsider politics in the region. "Without Trump there could be no Javier Milei," as one Milei campaign staffer told the *New Yorker*, a magazine, after his victory. "We set the tone for the western hemisphere", says United States Senator Bernie Moreno, a close Trump ally who was born in Colombia.

The direct inspiration is often obvious. Mr Milei's supporters have donned "Make

Argentina Great Again" hats. Mr de la Espriella, an American citizen as well as a Colombian one (and Italian), celebrated Mr Trump's win in 2024 by posting a video to social media in which he toasts a desire to "Make America and Colombia great again" by downing a shot of his personal brand of rum. In an interview with *The Economist* before the first round of the Colombian election, he declared Mr Trump an example in economic policy and his "cultural battle against gender ideology". Mr Kast cites the Trump administration's migration policy to argue that many immigrants in Chile will self-deport.

Trumpy see, Trumpy do

Much of Latin America's new right copies Mr Trump's communication style, his positioning as an outsider, and his relentless rhetorical and legal assaults on political opponents and the press. When journalists publish information that Mr Bukele does not like, he issues warrants for their arrest. The norm-breaking tendency to put family and close friends in positions of power has also been copied, as has Mr Trump and his family's interest in and use of cryptocurrencies. A hatred of causes perceived to be "woke" or "globalist" is perhaps the defining trait: Mr Milei calls climate change a "socialist lie"; Mr de la Espriella says the United Nations is an "instrument of the left"; Mr Kast is against abortion in all cases and has opposed his wife taking the pill.

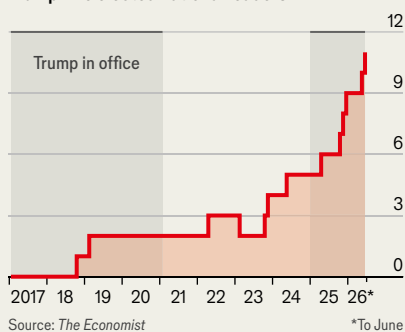
Many have sought Mr Trump's endorsement. In addition to Mr de la Espriella, he also backed Mr Asfura in Honduras, who went on to win. Mr Trump endorsed Mr Milei and his party in the Argentine midterms last year, as well as providing more than \$20bn in support from the US Treasury to help avert a run on the Argentine peso. That helped Mr Milei's party to victory and rescued his presidency. Mr Noboa met Mr Trump two weeks before his re-election in March 2025. Last month Flávio Bolsonaro, the leading right-wing candidate for Brazil's upcoming election, met Mr Trump in the Oval Office.

The members of this growing network of Trump allies also support each other. Mr de la Espriella's campaign received public backing not just from Mr Trump, but also from Mr Milei, Mr Asfura and in Paraguay Mr Peña. Mr Noboa went further. Just before the first round of the Colombian election he promised, at Mr de la Espriella's request, to remove tariffs he had imposed on Ecuador's northern neighbour as a gesture of good will.

This chumminess is not necessarily organic. The right has long complained about the influence of the São Paulo Forum, a get-together of Latin American left-wingers. Now it is building a rival structure for the social-media age. Axel Kaiser, a Chilean public intellectual-cum-influencer

Second-term suckups

Latin America and the Caribbean,
Trump-like elected national leaders



er, is part of it. His brother ran for president in Chile as a far-right candidate in 2025, winning 14% of votes in the first round before backing Mr Kast. “Ideas precede institutional change and political change,” says Mr Kaiser, echoing the maxim of the conservative American journalist Andrew Breitbart that “politics is downstream of culture”. This network of influencers “dramatically changed the mentality of millions of people”, he claims.

Amplifying these ideas are people like Agustín Antonetti, a 24-year-old Argentine social-media influencer. He and those like him definitely make an impact. Last November Claudia Sheinbaum, Mexico’s left-wing president, blamed him for fomenting protests against her government. The Cuban regime has denounced him as a CIA operative. “I was following what was happening from my bed,” he chuckles. But he says there is no doubt that social media are fundamental to right-wing success. Mr Kaiser calls it “decisive”, especially for outsiders like Mr de la Espriella. Some 40% of Latin Americans say social media are their main source of news, double the share in Europe. Humour is powerful online, says Vicente Fernández Moujan, who runs Mr Milei’s digital campaigns. The online left, in contrast, is “boring”.

New media outlets matter, too. Leading the charge is *La Derecha Diario*, a hard-right news website launched in 2020. Its co-owner, Javier Negre, a Spanish journalist, sits in the White House press pool. Mr Trump, he says, “shows the way to a lot of candidates [in Latin America] that didn’t have the support of legacy media”. *La Derecha Diario* helps “our candidates” deliver their message “without fake news”, he says. The line between news organisation and campaign consultant is hard to detect. He claims to be “very close” to nine presidents, and to be helping Mr de la Espriella “because he is a friend”. Sometimes he is paid to help candidates’ messages go viral; sometimes his services are free, he says.

Power to the posters

Campaign strategists and lobbyists help spread techniques. Another Argentine, Fernando Cerimedo, who co-owns *La Derecha Diario* with Mr Negre, is perhaps the most important. In 2023 he helped to run Mr Milei’s social-media strategy. He stepped in to arrange for Tucker Carlson to interview Mr Milei when he was running for president. He says he managed Mr Asfura’s campaign in Honduras and co-ordinated Mr Trump’s endorsement. He helped to get Mr Paz elected in Bolivia, and continues to work as an adviser to the president. He works closely with Brad Parscale, Mr Trump’s former campaign manager.

The record of the Trumpian leaders in office is mixed so far. Mr Bukele’s remarkable success in slashing the murder rate

has come at huge cost in people locked up without due process or trials. He has also changed the constitution to allow himself to stay in power. His security model is unlikely to work as well elsewhere. El Salvador is tiny and its gangs are largely domestic. Colombia is more than 50 times bigger and eight times more populous. It produces two-thirds of the world’s cocaine and its gangs are deeply embedded in transnational criminal networks. In Ecuador, also larger and battling international gangs, Mr Noboa has copied Mr Bukele and also called in the army, to little avail. The homicide rate was 46 per 100,000 in 2023, the year he took office. It fell slightly in 2024, but then rose to an all time high of 51 in 2025, even as several gang leaders were captured. So far 2026 looks a little better, but after nearly three years of Mr Noboa the country remains exceptionally bloody.

Milei-flavoured Trumpism

Mr Trump’s new friends are doing better at managing their economies. After inheriting an extraordinary economic mess, Mr Milei’s aggressive spending cuts and deregulation have pulled inflation sharply downward, allowing for capital controls to be reduced and the peso to be partially floated. Growth is uneven and jobs are a problem, however. Still, he has far exceeded most expectations. Mr Noboa’s macro-economic management has been prudent and tightly in line with the IMF’s recommendations. Borrowing costs have fallen

sharply. Growth remains patchy, however. In Chile Mr Kast’s pro-market plans look sensible. They are only now starting to kick in, but reaching the 4% growth that he promises will be difficult.

Markets cheered Mr de la Espriella’s election, and he is right to say he will get a grip on Colombia’s soaring public debt. But his commitment is undermined by his crowd-pleasing promises of cheap mortgages and extra spending. His pledge to deliver 7% growth looks extremely hard to fulfil. Most of the good economic news has been delivered by orthodox right-wing policy—smaller states and freer markets—rather than by Trumpian favourites such as tariffs. Indeed, unlike Mr Trump and some Latin American leftists, so far most of the new right seem to be respectful of central-bank independence.

Whether all this will be enough to secure future re-elections for the new right remains to be seen. The region is highly polarised and some victories have been extremely tight. The left could win elections again soon. Yet even if it does, the right is now dominated by much more hardline and populist ideas and leaders. That will probably not change soon. Even centre-right figures have been becoming more extreme to try to stay relevant, with little success. So when the right wins again it may well be a hardline, populist variety. As with Mr Trump, the next iterations of hard-right governments in Latin America may be even more disruptive than the first. ■



Devastation in Caracas

Two powerful earthquakes hit Venezuela in quick succession on June 24th. The United States Geological Survey (USGS) said the first had a magnitude of 7.2, with an epicentre 170km west of Caracas, the capital. The second measured 7.5 and hit 16km away. Such tightly sequenced “doublet” quakes are rare. Delcy Rodríguez, Venezuela’s acting president, said 32 people had been killed and 700 injured, but the death toll will climb. USGS reckons there’s a 44% chance that 10,000 are dead.

Security in Venezuela

Ore and peace

CARACAS

The United States is driving gangsters out of the mines

AN UNUSUAL silence hangs over Las Claritas, Venezuela's ground zero of illegal gold mining. For over a decade, gangs known as *sindicatos* governed this normally noisy strip of jungle in the eastern state of Bolívar. On June 8th army helicopters swooped in to chase them away. Thousands of freelance prospectors fled. Then the United States launched its own air strike, killing Héctor "Niño" Guerrero Flores, boss of the Tren de Aragua crime group. "This action was co-ordinated closely with our friends in Venezuela, with whom we are working very well," trumpeted Donald Trump on social media.

He has some cause for celebration in Venezuela. Since US special forces captured the dictator, Nicolás Maduro, in January, the Trump administration has been overseeing the country's oil exports. By securing a beachhead in Bolívar, it is opening a pathway to the mines. "The clean-up gives the Americans access to one of the richest gold deposits in the world," says a Venezuelan in the mining industry. And as America's and Venezuela's first joint military operation, it displays Mr Trump's leverage over Delcy Rodríguez, the acting president. Their shared goal is to bring Western mining companies back into the country, says Leonardo Vera of the Central University of Venezuela.

Many of those companies lost their concessions when a former president, Hugo Chávez, nationalised swathes of Venezuela's economy in the early 2000s. Some renegotiated access with Mr Maduro in 2016, when he opened the Orinoco Mining Arc (AMO), a Portugal-sized chunk of land, much of it rainforest. By then, the *sindicatos* had overrun the pits. They paid bribes to the army for access and supplies. With oil revenues collapsing, Mr Maduro embraced this criminal system. Gang-controlled mining spread fast, destroying ecosystems, ruthlessly exploiting workers and sparking deadly turf wars across the AMO.

Mr Maduro's seizure presented an opportunity. In March Doug Burgum, Mr Trump's interior secretary, visited Caracas, the capital, with two dozen mining executives. Trafigura, a big Swiss-based commodity trader, said it was buying \$100m-worth of Venezuela's partially refined gold ore. In May an American firm sold an option to develop half of its abandoned project in Las Claritas to Richard Warke, a Canadian metals magnate. The deal meant

investing up to \$200m over four years. Just before the raid, Western investors had been spotted at a famous mining complex in the nearby city of El Callao.

The area around Las Claritas holds the world's fourth-largest reserves of unmined gold. Venezuela as a whole boasts the seventh-largest reserves of iron. Industrial mining was so badly run under Messrs Chávez and Maduro that base metals' output has slumped by 90% since 2006, according to BloombergNEF, a data provider. So there is scope for rapidly increasing exports, particularly of gold. Legislation to woo foreign investment is helping. In April the National Assembly passed a mining-reform bill to prolong concessions, cut royalty payments to the state and allow for international arbitration of disputes.

Enormous challenges remain. Venezuela's mines, like its oilfields, are a shambles. The Venezuelan Mining Chamber, a trade group, says they need \$50bn of investment; some say more. At the mining complex in El Callao two-thirds of underground tunnels are flooded. Most equipment is broken or has been stripped for parts. Expertise is scarce; last year just five mining engineers graduated from Caracas's prestigious Simón Bolívar University.

The men in the mines

Unlike with oil, which was still controlled by the state when Mr Maduro was nabbed, expanding the mines presents a big security problem. Mr Trump and Ms Rodríguez have nailed one gang boss in Mr Guerrero, but Tren de Aragua is mostly intact, says Ronna Risquez, an expert on the organisation. Mr Guerrero's lieutenants, including Johan Petrica, boss of the Las Claritas *sindicato*, are still at large.

Nor was Tren de Aragua the only threat in the mining belt. Two far more estab-

lished rebel groups from neighbouring Colombia operate there: the National Liberation Army and dissidents of the former Revolutionary Armed Forces of Colombia, better known as the FARC. "They're not going to give up without a fight," says Phil Gunson of the International Crisis Group, a think-tank. That may be why the army chose the easier target of Las Claritas. Even if Mr Trump called for more action, Venezuela's hollowed-out forces would struggle to defeat hardened guerrillas, with or without American air strikes.

The final concern is Venezuela's reputation. Illegally mined gold accounts for up to 90% of its production, says Transparency International, an anti-corruption monitor. Most Western refineries refuse to process it. Much of it now ends up in the Middle East. International mining companies may argue that by doing the mining themselves they can control the process more responsibly. But they are still having to do business with a regime with an appalling record, says Catriona Rainsford of Global Witness, an environment and rights NGO.

The Rodríguez government is walking a fine line. While having to please its American supervisors and financiers, it cannot afford to alienate its own army, so long entwined in illegal mining. The generals made millions off the wildcatters, creaming off profits in return for providing supplies and protection. They will hate to lose so lucrative a side hustle. Being allowed to perform legitimate services for American corporations could be a consolation prize. But it may be more tempting to maintain ties with the nomadic bandits, many of whom will now migrate deeper into national parks. "If you clean up one area, they are going to move somewhere else," says the Venezuelan working in the mining industry. "It's that simple". ■



Investment needed

Asia



Tech and rural India

A peaceful revolution

NAGEPUR

Smartphones and AI are remaking rural India

IT IS A warm afternoon in Nagepur and Bahadur Kumar is working at the power loom. Opposite the workshop stands his two-storey, half-finished house. Outside it are five goats, a motorbike, his wife and his 22-year-old son, Karan, who explains that he is busy doing a science degree at a local university while preparing for exams for a government job. Asked how he studies, he whips out a phone to show *The Economist* his favourite YouTube coaching classes and long conversations with ChatGPT about questions he does not understand.

Nagepur is a fairly well-off village by the standards of Uttar Pradesh, a huge and poor state. In the past decade many of its households have gained toilets, running water, reliable power—and the internet. Scarce not long ago, smartphones are becoming ubiquitous. Ownership remains skewed towards men, but devices are shared and keenly fought over. “When my dad goes to work, I take it,” says Roshni, a

14-year-old schoolgirl, flashing a smile.

This year the world’s most populous country is on track to pass 1bn active internet users, up from 622m in 2020, according to the Internet and Mobile Association of India (IAMAI), a trade body. The shift is happening at lurching speed thanks to dirt-cheap devices and data. Indians can now get their hands on a used Chinese-brand smartphone for as little as 2,000 rupees (\$21), around four days’ pay for a labourer. Data cost just 15 rupees (16 cents) per gigabyte, less than a tenth of the global average and much less than in most devel-

oping countries (see chart on next page).

That is largely thanks to high population density, meaning that each mast can serve more customers. Data costs were also driven down by a brutal price war between Jio and Airtel, two telecoms behemoths (on June 19th Jio announced plans for a blockbuster initial public offering this year). Along with low rates of Wi-Fi installation, cheap tariffs help to explain why Indians are the hungriest consumers of mobile data in the world, according to Ericsson, a Swedish telecoms company.

The smartphone revolution has already swept Indian cities. As they approach saturation, the digital frontier has moved to the village. Internet usage is growing four times faster in rural areas than urban ones, according to IAMAI. It is surging above all among the young (like Karan and Roshni), promising to upend much about rural life, from education and work to leisure and love. Thus far Indian politicians have cheered on a booming digital economy. But they are starting to fret about harms.

Nagepur offers a window into how the change is playing out. At the bustling village school, smartphones have long been used for homework. But AI chatbots are a revelation. Ask a class of 11- to 14-year-olds whether they use them, and a wave of hands goes up. According to Sensor Tower, a data aggregator, AI apps have been

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downloaded more than 600m times in India. Among children who use phones for education, three-quarters report using AI for practice, doubt-solving and filling in gaps in knowledge.

This is not surprising when many Indian schools still offer a patchy education (the one in Nagepur is better than most). Several Indian AI companies are building online tutoring platforms. If AI can make ed-tech work better, few will have more to gain than knowledge-hungry Indian schoolchildren. For them, says Nandlal Master, the head teacher in Nagepur, the spread of phones and the rise of AI has “made information accessible”.

A few decades ago Nagepur was a village of weavers, but now most of the men work as labourers in nearby Varanasi. Access to technology helps youngsters like Karan hunt for skilled work farther afield. Women in the village say smartphones have given them more autonomy. Only around half of rural women have a mobile phone, but among younger ones ownership is rising fast. Some men have tried to prevent this, but they are fighting a losing battle. Last year 15 villages in the state of Rajasthan tried to ban married women from using smartphones, citing concerns about phone addiction (men, apparently, suffered no such problems). Amid a national outcry, the ban was withdrawn.

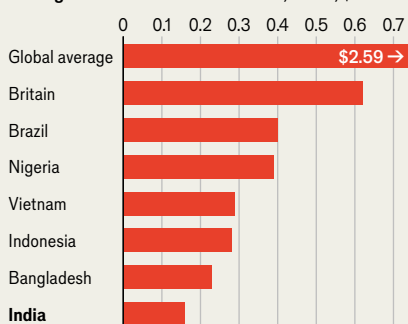
Beyond education and work, smartphones are changing Nagepur's social geometry. It is more common now for the young to form friendships and (whisper it) relationships beyond the village. The rural young may, like their urban peers, start to demand a greater say in crucial decisions, like marriage. Some demographers think smartphones are already accelerating fertility decline in villages, because they expose rural women to the lifestyles of richer urban ones. If villagers become as screen-addicted as their urban peers, one might expect the spread of other social attitudes also to accelerate.

Mr Nandlal, for his part, worries that what is being spread most is not autonomy or ambition, but distraction. In Nagepur children admit that while they can use phones for learning, they spend most of their time watching reels or playing shooting games. Short-form video and algorithmic content are particularly hard to resist in sleepy villages. In Nagepur parents say their children are being “spoiled”. Several note that boys no longer want to do manual jobs. “The more they idle the more they watch the more they idle,” one grumbles.

Only around half of under-30s in India have a job, and many that do have too little work. Pradyuman, an underemployed taxi driver, complains that his phone bill has rocketed because of short-form videos: “When you are by yourself your phone is your best friend,” he says. Lacking better

Cheap as chips

Average cost of 1GB of mobile data, 2023, \$



Source: bestbroadbanddeals.co.uk

options, millions of young Indians are spending their time and energy trying to become content creators. They are “burning their fingers thinking that this is the way up,” says Jaijit Bhattacharya of the Centre for Digital Economy Policy Research, a think-tank based in Delhi.

You will hear similar worries the world over. But Indians are already among the world's top screen-addicts, and all this suggests their screen use is only likely to grow. One response is satire: being jobless and chronically online is a source of pride for the Cockroach Janta Party, India's newest youth political movement, which has

held protests in several cities this month. Anyone can be a member if they spend at least 11 hours a day online, the party says.

Regulators are taking a tougher line. In March Andhra Pradesh and Karnataka became the first states to ban children from using social media. Several others are poised to follow suit. Ashwini Vaishnaw, the IT minister, has said that the national government is considering age-based social-media restrictions, noting that many countries have already taken tough action.

India will have to weigh any measures carefully. Shivam Jadaun, a lawyer, points out that blanket bans are an especially clumsy tool in a society where many devices are still shared. One risk is that parents respond by withdrawing access altogether, reducing the benefits as well as the harms. Many difficulties online stem from a failure to fix problems offline, from shoddy schools to a lack of jobs. It would be better, argues Mr Jadaun, to strengthen the regulation of platforms while taking steps to improve digital literacy.

On the steps of Karan's house, his mother, wrapped in a black sari and headscarf, has been listening intently to the conversation. She chips in to say that she approves of young people using technology to strive for a better life. Her only worry? That more of them, as a result, will want a “love marriage”. ■

Bangladesh's garment industry

Unravelling

Rolling blackouts and rising costs are hampering the production of clothes

THE WAR that America and Israel launched on Iran may be over, but Bangladesh remains desperately short of power. Its garment factories, especially, have been running low on it. The country's spinning, knitting and dyeing mills guzzle gas and petrochemicals—and around 95% of Bangladesh's oil and gas comes from the Gulf. Rising costs are squeezing the businesses. On June 6th Al-Muslim Group, a large clothing exporter, sacked nearly 1,900 people from its knitwear and denim factories in Dhaka, the capital.

Over 4m Bangladeshis, mostly women, work in the industry, making clothes for Western brands like Zara and H&M. Around 40m people—25% of the population—depend on it. Clothing made up four-fifths of Bangladesh's export earnings last year, and 13% of GDP. Only China exports more clothes than Bangladesh.

But the sector was already under stress. The student-led uprising that toppled

Sheikh Hasina Wajed in 2024 also sapped buyers' confidence in Bangladesh as a reliable partner, explains Mehdi Mahbub, an industry analyst. When Bangladeshis took to the streets, many garment factories went idle. Five were set on fire. Some of the senior Awami League figures who were thrown in prison also owned garment factories. Over the past three years more than 400 of them have shut down.

In May the government was cutting power for two hours a day on average in and around Dhaka. In Chattogram, the second city, blackouts lasted as long as eight hours a day. Some factory owners have turned to diesel-powered generators to keep output up. But for an industry tailored towards efficiency, “even the 10-15 minutes needed to turn them on can be costly”, says Abil Bin Amin of Bangladesh's Ethical Trading Initiative. Production fell nearly 30% between February and May.

Global brands are placing fewer orders, ►►

► deterred by production delays, shipping disruptions and gloomy Western shoppers who have stopped buying so many clothes. Abdullah Hil Nakib, who owns a jacket factory in Dhaka, says that orders for his wares have dropped by a fifth since the war started. In May clothing exports fell for a tenth consecutive month, tumbling by 8% year on year, according to Bangladesh's Ministry of Commerce.

Pricier oil has pushed up the cost of materials. Synthetic fibres, dyes and finishing chemicals, plastic buttons and zips make up around 65% of the cost of a garment, and all use petrochemicals. In Bangladesh, around 30% of garments are made using polyester fibre and thread, products derived from naphtha, a petrochemical that is now roughly a third more expensive than before the war. And the industry is highly fragmented. Vertically integrated textile mills do exist in Bangladesh, but most factories handle just one part of the production chain. Mr Nakib estimates he is paying 30% more for transport.

Bangladesh's central bank introduced a 600bn taka (\$5bn) stimulus package for ailing businesses in May, earmarking the largest share for garment factories. But recipients pay around 7% interest on the loans, tough for those already hemmed in.

When costs rose during the pandemic, the big brands showed they were unwilling to pay more for clothes made in Bangladesh. This time, at least 9,500 workers, at some 80 factories, are reckoned to have lost their jobs since January. More disorder of the sort seen in 2023 and 2024 is feared.

Samoli Khatun arrived for work at Co-retex Apparels in Madhabpur on June 7th to find a layoff notice on the factory gates. "It will be very hard for me to find another job, and, as a woman, my options are limited. I may have to return to my village." ■



Getting angry

Ageing Asia

Honour thy father and mother, or else!

SINGAPORE

Across Asia, governments are requiring children to care for their parents

IN MARCH LAWMAKERS in Telangana, a state in southern India, passed a bill mandating that up to 15% of an adult offspring's salary can be redirected into their parents' bank accounts if they are found to be neglecting them. According to Revanth Reddy, Telangana's chief minister, children who fail to care for their parents are "not qualified to live in society".

Across Asia, the instinct to punish unfilial children—pioneered in 1995 by Singapore with its Maintenance of Parents Act—is spreading. Malaysia is pursuing filial obligation on two tracks: a secular Senior Citizens Bill protecting elders from abuse or abandonment, and sharia court amendments allowing Islamic courts to hear parental-maintenance claims. MPs in the Philippines have filed a Parents Welfare Act for the third time since 2016 to punish wrongdoers with up to ten years in prison.

These are the latest additions to a lengthening list. China has since 2013 required adult offspring to visit ageing parents "often", on pain of fines or jail. In Shanghai those who defy court orders to provide for their parents are put on a credit blacklist. Many countries including South Korea, India and Japan set out legal protections for the elderly against abuse, including financial exploitation.

Globally, rates of elderly abuse and abandonment are rising as populations age. Malaysian authorities recorded 2,144 cases of elderly patients abandoned in hospitals between 2018 and 2022. The country of 36m people has only 18 licensed nursing homes and around 70 geriatricians. Asia's governments, in short, have lost faith in filial love as social policy.

Asia is ageing faster than any region in recorded history, generally before its economies are rich enough to absorb the shock. Some 722m people, about 15% of Asia's population, are over 60. By 2050 that share will reach 26%. A survey found that 49% of Thai millennials and 31% of Indonesian ones cited parental care as the main reason they could not leave the family home. The generation expected to fill this gap is itself stretched thin.

Filial piety in Asia has always carried an economic logic. Children were for a long time, as one Singaporean sociologist notes, "produced and raised as part of retirement planning". Part of their wages flowed to parents so long as families lived together. Urbanisation has broken that

compact, as young adults migrate to cities, establish separate households and have to cope with mortgages and child care. The competing claims on their income multiply, "yet the expectation of filial provision remains", says Haslina Muhamad of Universiti Malaya in Kuala Lumpur.

Critics, however, say that instead of looking to punish those who fail, states should provide better infrastructure for the elderly, including day care, end-of-life facilities and welfare systems that give families options, not orders. China offers the starkest warning. When in 2013 its rulers required children to visit parents "often", the backlash was instant. "The government should have thought of how they would address this problem when it brought in the one-child policy," ran one widely shared social-media comment. ■

Malaysia

Closing in

Two imminent state elections could break the ruling coalition

ON HIS PATH to power, Anwar Ibrahim spent nearly ten years in jail on politicised corruption and sodomy charges. Now the man who has been prime minister since November 2022 is experiencing confinement of another kind. Over three years into the job that he coveted for so long, the walls are closing in on the veteran of Malaysian politics.

Mr Anwar leads a coalition with a supermajority in parliament, but his partners are eyeing the exits. Two upcoming state elections look likely to bring down his government if the results do not go his way. The voting in Johor, a southern data-centre powerhouse, and Negeri Sembilan, a smaller state farther up the peninsula, will not affect the make-up of the federal parliament. But the coalition partners are testing their strength by running against each other in both states.

One of them, the United Malays National Organisation (UMNO), was long Malaysia's ruling party before losing power in 2018. Its members have bristled at playing second fiddle to Mr Anwar in his multi-party coalition. Worse, it has meant joining hands with their arch-rivals, the majority-Chinese Democratic Action Party (DAP), the largest single party in the coalition. UMNO holds itself up as the guardian of Malaysia's longstanding affirmative action policies showered upon the majority Malay ethnic group, while the DAP argues for more meritocracy.

If UMNO does well in the state polls, ►►

▶ that could give it the confidence to withdraw from the coalition at the federal level and go it alone at the next election. That in itself would not bring down the government. But the DAP is also thinking of quitting. Its erstwhile supporters have abandoned it in big numbers, impatient with Mr Anwar's refusal to reform affirmative-action policies. On August 16th DAP cadres are due to hold a special congress at which they will decide whether they want to stick with Mr Anwar. If they, too, withdraw from

the coalition, then the government would fall, triggering an election with more than a year remaining in the current parliamentary term.

Mr Anwar has only himself to blame. Having cast himself as a champion of reform for nearly three decades, he has limped along doing painfully little in that direction. His defenders argue that he has had to go slow, given the coalition maths. But many of Anwar's old comrades in his People's Justice Party (PKR) do not agree.

Last month, Mr Anwar's former deputy in the PKR, Rafizi Ramli, left the party to form his own, dedicated to reform.

The state elections might offer a hint as to what comes after Mr Anwar. One likely outcome in Negeri Sembilan, says Bridget Welsh, an expert on Malaysian politics, is a coalition between the Malay nationalist UMNO and the Islamist PAS. That could foreshadow an alliance at the federal level. Such a coalition would be even less likely than Mr Anwar to do much reforming. ■

ASHOKA

Eye of the storm

The unlikely city welcoming Delhi's intellectual refugees

FOR CENTURIES Delhi attracted scholars, saints and statesmen hoping to influence the ruler of the day. Once India achieved independence, it was the natural place for the fledgling state to locate national institutions such as archives and universities. Publishers and think-tanks soon followed. The sheer density of serious thinkers makes Delhi the intellectual capital of India.

Or made, at any rate. Since taking power in 2014 Narendra Modi and his Bharatiya Janata Party (BJP) have narrowed the space for free thinking in media, entertainment and cultural institutions. Why should pointy-headed types be immune? Chiefs of public universities were replaced with more pliant figures. Private universities and non-profit groups came next. Authors of inconvenient papers lost their jobs. Think-tanks lost their sources of funding. The Centre for Policy Research, which worked with various governments over five decades, is the canonical example. In 2023 its licence to receive foreign money was suspended, then revoked a year later, in effect destroying it. Most of its researchers were laid off.

There is no one to talk to in Delhi any more, grumbles one recent transplant. Institutions that have not been broken have learnt to bend. Intellectuals who kept their jobs know not to broadcast their views. Researchers who want their policy ideas taken seriously choose their affiliations carefully. Some superannuated scholars set up new non-partisan bodies—with funding strictly domestic—in the hope of a clean slate. Established researchers with impressive CVs accepted positions at Western universities. Younger ones moved to different disciplines or sectors. And then there are those who went to Ahmedabad.

Ahmedabad? The sole metropolis of Gujarat, a state ruled by the BJP for nearly three decades, a dozen of those by Mr Modi himself? The municipality where the party won 83% of seats in recent local elections? The city that routinely tops lists of India's most segregated by religion? Where Muslims live in one of India's largest ghettos? Where the right of public assembly was repeatedly curtailed for ten years until a recent court ruling? Where vegetarianism rules and alcohol is prohibited? That Ahmedabad?

The very same. Ashoka recently dropped by the city to chat with refugees from Delhi. The intellectual environment they found upon landing, they say, was a pleasant surprise. There is "high freedom" within their institutions, though with the understanding that they will keep suitably low public profiles. The idea, as an administrator explained to a recent arrival, is to "facilitate" smart people to do their work. In other words, management has what passes for courage in today's India, doing the basic job of shielding thinkers from political pressure.



As for Gujarat's reputedly zipped-up atmosphere? There are parties every weekend. The city has a wealth of public and private universities—including some of India's top schools for management, design and urban planning—and a correspondingly high concentration of youngish staff keen to let their hair down. As in many prohibition states, alcohol is not just available but home-delivered by service-oriented bootleggers operating in a cut-throat market.

Ahmedabad's image has never recovered from its position at the epicentre of Hindu-Muslim riots that marred Gujarat's—and Mr Modi's—reputation in 2002. Yet it has a far older tradition of progressive thought among its mercantile elites. Industrialists from the city bankrolled Mahatma Gandhi's social reforms. Supposedly cultureless capitalists established public museums to house their collections. In the 1950s a group of textile barons invited Le Corbusier, a Swiss-French modernist architect, to build them homes, offices and a cultural centre. Many of the city's private schools and universities were founded and are still run by members of the business class. There is more to the place than vegetarianism and Islamophobia.

It is easy to get carried away. Ahmedabad is no bastion of free speech. The media, as in the rest of the country, have a tough time of it. Independent news outlets unduly attached to reporting facts do not last long. Yet in less public fields, there is a freedom to operate unusual for a place dominated by the BJP. That Gujarat is caricatured in the liberal imagination as a quasi-fascist state filled with anti-intellectuals may even help it. It is, as one tactfully anonymous recent arrival from Delhi puts it, very calm in the eye of the storm.

China



Tech rivalries

Roaring back once again?

A new model is forcing a reassessment of Chinese AI's merits

AMERICA'S LEAD over China in artificial intelligence may be at its smallest in over a year. When China disrupted the AI race in January 2025 with the release of DeepSeek R1, it erased \$1trn from America's capital markets. Nvidia, a chip firm, briefly shed 17% of its value; the Nasdaq sank by 3.1% in a day. American investors were troubled not only because Chinese AI was good, but because it was being given away free. The uproar soon faded. Since then, market valuations everywhere have hinged ever more on the promise that AI will be both revolutionary and profitable.

Now Chinese labs are unsettling their American rivals anew in the race to monopolise the market for models. On June 13th a Beijing-based lab called Zhipu, or Z.ai, announced its latest system, GLM 5.2, promising "a step closer to frontier intelligence for everyone". It is the most capable Chinese-trained model to date and runs at less than a tenth of the price of Anthropic's

latest release, Fable 5. And as with other Chinese models the weights, or parameters, that enable GLM 5.2 to function have been publicly released. With this new model, China is competing on ability, cost and openness. The offering looks both compelling and timely.

In recent weeks American companies have been grappling with soaring AI costs, sometimes ranging into the thousands of dollars per employee. Some firms are setting budgets for tokens (bits of text processed by a model). Then on June 12th the Trump administration banned non-Americans from using Fable 5, leading Anthropic to remove the model from service. For the first time, access to frontier AI rests on one government's say-so. All this may give

users reasons to look at alternatives to American AI. Many will find GLM 5.2 capable and affordable, and welcome that it is out of the Trump administration's reach.

Start with capability. Artificial Analysis, a research firm, ranks GLM 5.2 as the most intelligent open-source model on the market. GLM 5.2 takes an impressive fourth place on its overall list, behind OpenAI's ChatGPT 5.5 and ahead of Google's Gemini bot. The model has surprised everyone. Earlier this year Chinese developers were pessimistic about the prospect of their models outclassing American ones before 2030. After Zhipu's release, Elon Musk, a very rich man, wrote on X, his social-media site, that he expects China to match the abilities of the current frontier by early next year. It "won't take that long", Tang Jie, Zhipu's co-founder, shot back.

Weighty calculations

Unlike in the DeepSeek moment, American markets have so far shown little interest in GLM 5.2. This is partly because it has become more difficult to assess Chinese models accurately. To arrive at its estimates, Artificial Analysis scored GLM 5.2 on dozens of benchmark tests, which use exam-like questions to evaluate a model's smarts. America, via Anthropic, keeps its edge in performance. Fable 5 is about 17% cleverer than GLM 5.2 across an average of

→ ALSO IN THIS SECTION

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► benchmark tasks. The other important metric is how long it took GLM 5.2 to reach this level of intelligence. A comparable Western model to GLM 5.2 was released in February, or about four months ago.

In reality, America's lead is probably bigger than four months. Open-source models, many of them Chinese, tend to score better on public benchmarks than private ones, says Havard Tveit Ihle of the Norwegian Defence Research Establishment, a think-tank in Norway. The questions used in public benchmark tests are published, whereas those who apply private benchmarks keep their evaluations secret. Analysis by Dr Tveit Ihle published before GLM 5.2 found that Chinese models were about four to six months behind American ones on public tests. But on private tests America's lead nearly doubled, to eight to ten months (see chart). A study by the American government, released in May, identified a similar gap. Mr Tveit Ihle says Chinese labs appear, possibly unwittingly, to "teach to the test".

Model students

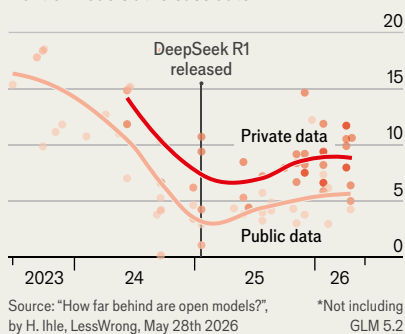
On two private benchmarks tested so far, GLM 5.2 shows the same hallmarks: it is about seven months behind on WeirML, a measure of unusual machine-learning tasks that need careful reasoning to solve, and fully a year behind on SimpleBench, which evaluates common sense by trying to trick models. The pattern is not consistent, however. A new exam released by Artificial Analysis on June 19th tests models on office-worker tasks, like sifting through messy files and evaluating conflicting information. GLM 5.2 could not have trained for the evaluation. Yet it outperformed ChatGPT 5.5, which is just two months old. These results suggest that America's lead remains steady, says Mr Tveit Ihle, but are also evidence the gap is not widening as some had expected it would.

What is especially surprising about GLM 5.2 is that it succeeds in tasks that tend to trip up its peers. Chinese models often excel in fields with clear right or wrong answers, like maths and coding. But they tend to fall down on problems that are open-ended or that require sustained independent judgment. That pattern reflects one of the largest challenges facing researchers in China. Export controls on advanced chips have left Chinese labs short of the computing power needed to train the strongest models. So they tend to make up ground in post-training: fine-tuning models to behave in particular ways or solve certain kinds of problems, including on data allegedly harvested from American systems through "distillation".

Given the uncertainties surrounding the true capabilities of Chinese models, next consider whether they are truly cheaper than their American rivals. Deep-

Race track

Chinese AI models*, months behind frontier models at release date



Seek charges just \$0.87 per 1m output tokens for its v4 model, whereas Anthropic charges \$50 for the same on Fable 5. Such prices might have a growing appeal in America, where token costs at some firms have run out of control. In June DeepSeek saw a sharp rise in American firms paying for its services, according to Ramp, an invoicing company. Microsoft is reportedly considering using the Chinese lab's model in its flagship Copilot chatbot. Yet this most important assumption, that Chinese AI is cheaper, can frequently be wrong.

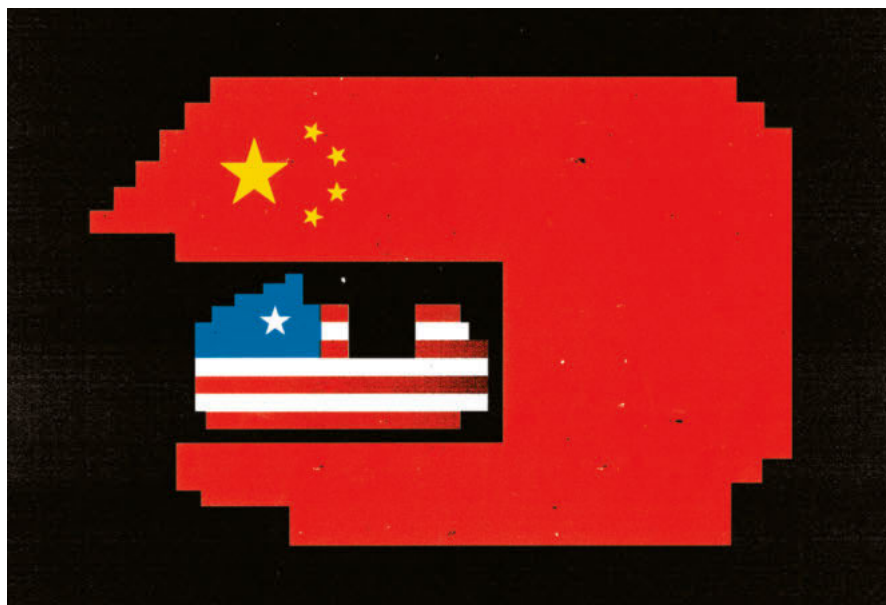
Though Chinese models are becoming more capable, they are generally not becoming more efficient. Chinese models use many more tokens to think through their answers. A study updated this month by Du Zheng of Georgia Tech and co-authors shows that given the same tasks, a DeepSeek model used 23 times more tokens than its OpenAI rival to achieve basically the same result. Because of these large differences in efficiency, the correct way to compare models is not price per token but the total cost of all the tokens

used. Using this metric, on a benchmark designed to test software engineering, GLM 5.2 ended up costing more than systems from Anthropic and OpenAI.

In addition to capability and cost, a third selling-point is now top of mind for AI users: reliability. Zhipu released its model at 5:21pm Beijing time on June 13th, one day after the Trump administration told Anthropic that it was banning non-Americans from using Fable 5. "Our attitude is one of radical openness," Mr Tang declared. He also blasted "external blockades", such as the one imposed by Anthropic and the American government, saying they made AI systems "subject to revocation at any moment". The Fable 5 shutdown could help Chinese labs as firms around the world rethink their dependence on American AI.

Most Chinese models are released open-source, meaning they can be downloaded and run on local hardware, out of reach of governments or the labs themselves. But the American government could one day impose limits on the domestic use of Chinese AI. Two congressional committees are currently investigating American tech firms for using Chinese models. And China's labs face other limitations: a shortage of computing power means they often run into service interruptions, or slow in periods of high traffic.

As the AI race speeds up, regulators everywhere will be faced with new challenges to safety and security. The risk of sudden government intervention may grow. Fable 5 was powerful enough to prompt such a response. That Chinese models are not, for now, facing similar regulatory risk suggests China's government is not yet alarmed enough to act. That may be some of the clearest evidence that they remain behind their rivals. ■



A roadmap to more resilient food systems

A new index provides a benchmark for global food resilience, pinpointing where systems are strong and where they remain vulnerable

Supported by **Cargill**

Feeding ten billion people by 2050 will test the resilience of global food systems. New data from Economist Enterprise's inaugural Resilient Food Systems Index, supported by Cargill, shows how prepared countries are, and where vulnerabilities remain.

The index, part of Economist Enterprise's Food Imperative initiative, ranks 60 countries across four pillars of resilience. The gap between the strongest and weakest performers is stark: Portugal scores 42 points higher than the Democratic Republic of Congo. No country is fully resilient, and nearly half fall into a "middle zone", with scores between 56 and 71, suggesting ample room for improvement.

Affordability emerges as the strongest pillar—but affordability alone is not enough. In 62% of countries, even the cheapest healthy diet consumes nearly two-thirds of the poorest households' income. The opportunity lies in expanding access to food that is both nutritious and affordable.

Access depends as much on supply and trade as on price. Globally, 15 countries produce 70% of the world's food and have an average resilience score of 64. Their role is pivotal: when they function smoothly, markets stabilise; when they falter, the effects can ripple worldwide.

Overall, the building blocks of resilience exist. But it takes alignment, innovation, and coordination at scale to make progress.



Learn how you can build resilience into the food system by visiting the Resilient Food Systems Index



CHAGUAN

Tarred with a red brush

Accusations of Chinese influence can pollute American politics and miss the real problem



CHINA WANTS more sway over how the world thinks. The country's latest five-year plan aims for a major leap in international influence. This often manifests on social media as cute videos of pandas or futuristic ones of high-speed trains. But China's leaders see the broader effort as crucial. It is needed, official statements say, to counter Western biases so as to win the "battle for global public opinion" and "safeguard national interests".

In principle, there is nothing wrong with China trying to expand its global influence. Many governments do. It can be uncomfortable for democrats to see the world's biggest authoritarian regime winning fans: recent polls point to an uptick in support for China around the world. But these trends reflect disgruntlement with Donald Trump's America as much as warm feelings towards the Chinese Communist Party.

Where things become nasty is when the battle for opinion gets underhanded. Researchers have long documented Chinese influence operations, showing how state actors disguise their identities online. Take, for instance, the work by Graphika, a social-media analytics group, linking thousands of fake accounts on Facebook to Chinese disinformation about American decline.

Careful researchers hold themselves to high evidentiary standards. So it has been troubling to see a recent spate of questionable reports in America on China's malign influence in debates on data centres, climate and artificial intelligence. The concern is not just that they get their analysis wrong but that they are polluting the space for parrying the real challenges from China.

One such report comes from the Bitcoin Policy Institute (BPI), a cryptocurrency advocacy group. It starts by noting that Bernie Sanders, a US senator, invited two Chinese academics affiliated with China's government to a panel discussion on AI safety. Anyone vaguely familiar with Mr Sanders knows that his ideas are his own, including his proposal for a moratorium on AI data centres. Instead, a colourful chart in the BPI report links the Chinese Communist Party to Mr Sanders (through six degrees of separation). It also tracks grants by Swiss and British billionaires to American environmental groups opposed to data centres, and notes that one gave cash to Chinese green groups nearly a decade ago. In this

world of innuendo, he is compromised.

Another report, by Power the Future (PTF), an energy lobby, examines foreign funding for American opposition to data centres. It observes that the Sierra Club, an American environmental group, has helped write lawsuits derailing infrastructure investments. "That context makes one question impossible to ignore: who benefits?" it asks. The answer, it turns out, hinges on another question. In 2020 environmental officials had asked the Justice Department to investigate whether the Sierra Club received foreign funding. No charges were ever laid. But the question had been asked—and that is damning enough. Citing the BPI and PTF reports, Republican lawmakers in the House of Representatives have called for the Trump administration to examine the purported ties between China and America's anti-AI movement.

A third report, from the National Association of Scholars, a conservative think-tank, looks at China's role in California's clean-energy policies. It lays out how the University of California and Tsinghua University have collaborated on climate-change research. Yet it then makes improbable jumps in ascribing causal power to that collaboration. Because environmental policies may lead to contracts for Chinese green-tech firms, it frames the joint research as benefiting the Chinese economy and heightening American dependence on it. It calls for federal investigations of officials who have entered into agreements with China, but focuses only on the green policies of Democratic-led California. What about the dozens of agreements and trade missions launched by Republican administrations in Iowa, Texas, West Virginia and beyond, some of which promote shale-gas exploration and coal-mining? On those, it is silent.

One thread running through these reports is that they all come from groups with specific agendas: more AI innovation, more data centres and fewer green policies. The first two, at least, are worthy goals. But they use accusations of "Chinese influence" to tar Americans who have the temerity to disagree. Bethany Allen, a pioneer of investigations into Chinese influence operations now at the Australian Strategic Policy Institute, says it reeks of politics.

Eyes on the target

Beneath their hyperbole, these reports are correct about a worrying trend: some Chinese actors would like to steer America away from productive tech investments. Consider yet another investigation into Chinese influence operations, published on June 10th by OpenAI. It explained how accounts apparently based in China had used ChatGPT to craft material criticising American data centres and Mr Trump's tariffs, then posted it on X, YouTube and Facebook. In one case, they created comic strips about power grids. "Why are our power prices so high?" exclaimed a suburban mother in one panel as a cigar-chomping man, representing AI and data centres, held up money bags.

The OpenAI researchers concluded that this operation had no discernible impact. Yet that does not mean that the efforts of these Chinese actors should be ignored. Their express purpose is to foment distrust in America, with the apparent goal of restraining American AI development. Precisely because the stakes are so high, it is vital to be careful, even clinical, when levelling such allegations. Otherwise, serious investigations can be dismissed as partisan point-scoring. Using scant evidence and silly insinuations to sully domestic opponents will make it that much harder to counter truly dangerous forms of Chinese influence. ■

Middle East & Africa



The Gulf after the war

Going separate ways

DOHA, DUBAI AND RIYADH

The region experienced a common shock, but it will have unequal consequences

JANUARY FEELS like another lifetime. At a gathering of bankers in Dubai, talk focused on whether the city's sizzling property market was due for a correction. At a tech conference in Doha, Qatar's capital, everyone wanted to discuss AI. War was already threatening the Middle East, but no one wanted to believe it would hit home.

For the Gulf Co-operation Council (GCC), a club of petro-monarchies, the months that followed were their greatest shock since 1990, when Saddam Hussein occupied Kuwait. Perhaps the comparison seems overwrought. The first Gulf war killed hundreds of Kuwaiti civilians and caused an ecological disaster when Saddam's army set fire to oil wells; the consequences of this conflict are less visible.

Thousands of Iranian missile-and-drone attacks caused tens of billions of dollars in damage but, mercifully, few casualties. Though the Strait of Hormuz was shut for almost four months, Gulf resi-

dents experienced no serious shortages. It is an odd sort of war when one can still get imported oysters through a blockade.

Yet in some ways, the shock of this Gulf war is greater than the first. In 1990 the GCC was the world's petrol station. Today it is an outsize player in much from finance to logistics, home to some of the biggest sovereign wealth funds and airlines. Its economies are worth \$2.3trn, more than 2% of global output. It is a sanctuary for millions of expats, drawn by the promise of a safe, prosperous oasis in a turbulent region. That model is now in doubt.

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41 Ramparts return to the Sahel

What comes next depends in part on America's talks with Iran. If Donald Trump strikes a lasting deal that ends America's long conflict with Iran, the shock to the Gulf may soon fade. If fighting resumes, the next round may be more destructive. Few officials in the region expect the war to restart. But nor do they expect a durable peace. They will have to cope with elevated risk for the foreseeable future.

That leaves three challenges: rebuilding confidence; reconsidering ambitious plans for diversifying their oily economies, which never accounted for such a level of risk; and navigating a complex geopolitical moment in which they trust neither America, their longtime protector, nor one another. Some will fare better than others. Gulf states shared a common trauma, but its consequences will be felt unevenly.

At first glance, the United Arab Emirates (UAE) may seem to have suffered the greatest blow to confidence. The seven-member federation, including Dubai, was attacked over 2,800 times, almost as many as the rest of the GCC combined. Close ties with Israel and its leaders' hawkish stance would seem to leave it permanently atop Iran's target list in the Gulf.

Many expats in Dubai, however, sound sanguine about both the war and the future. Critics might dismiss this optimism as coerced: the UAE arrested people for

▶ sharing news of Iranian strikes on WhatsApp. The question of how many people left during the war is a favourite dinner-party topic in Dubai, and because the emirate does not publish detailed population figures, no one has a clear answer.

Anecdotally, though, there are signs that lots of professionals stayed put: traffic on the motorways and crowded malls. The city is far less transient than in decades past. Many who left probably had no choice, as hotels and other firms that rely on tourists sacked thousands of workers.

Low confidence, lower reserves

Tourism, which accounts for 12% of the UAE's GDP, will be an early bellwether for broader confidence. Summer is always quiet, but many firms expect a quick rebound once the searing heat eases. But the recovery may be unbalanced. "The Russians and Indians are telling us they're ready to come back almost immediately," says a marketing executive. "The Brits? End of 2027."

It helps that the UAE has deep pockets. Before the war its break-even oil price was just \$50 a barrel, far below most of its neighbours. Dubai has already rolled out 2.5bn dirhams (\$680m, or 0.5% of GDP) in wartime incentives; it has suspended some taxes on hotel stays and restaurant bills.

It will be harder to restore confidence in smaller Gulf states—perhaps most of all in Bahrain. The island kingdom entered the war with a 146% debt-to-GDP ratio, one of the world's highest. Its foreign reserves were enough to cover less than two months of imports. Add to that long-simmering tensions between Bahrain's Sunni monarchy and its Shia majority, which has long complained (rightly) of discrimination.

The war exacerbated both problems. Though oil accounts for just 14% of GDP, it supplies around 50% of government revenue, and Bahrain has exported almost none since March. The UAE gave Bahrain's central bank a \$5bn currency-swap line in April. Further rescues seem likely. Gulf support has helped Bahrain avoid a downgrade of its debt (which was already classified as junk). A \$1bn bond issue earlier this month was oversubscribed.

Still, this only adds to the kingdom's debt burden—it is paying more than 7% on its newest bonds, a full percentage point above its last pre-war issue. Meanwhile, it was not uncommon to hear Bahrainis express sympathy for Iran during the war, even as the regime bombed their country and mused about annexing it.

The tourism sector has long underperformed, and a mix of instability and insolvency will make it hard to woo new visitors. Other sectors look wobbly as well. Bahrain has tried to position itself as a logistics hub for businesses serving the Saudi market. Uncertainty around Hormuz may make that risky. Unlike the UAE, which plans to

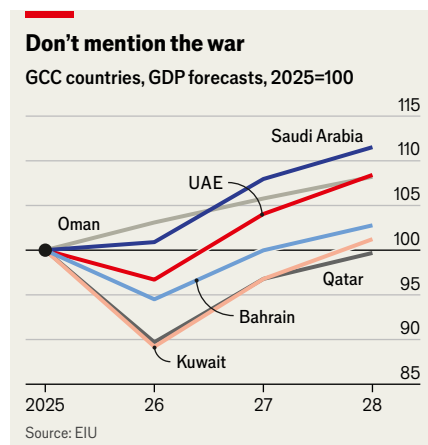
expand the ports on its east coast, Bahrain has no alternative to the strait.

Indeed, one lesson from the war is that geography matters. Saudi Arabia's size helped it weather the war better than most of its neighbours. Its big cities were largely spared Iranian attacks, and its air space never closed. Some firms temporarily shifted staff from Dubai to Riyadh. A large internal market helped compensate for a drop in foreign travellers—and the tourism sector relies mostly on religious pilgrims, a stickier business than the leisure market.

The kingdom was already scaling back some of its most ambitious projects, particularly Neom, a futuristic city being built in the north-west. Instead of mirrored skyscrapers and ski resorts in the desert, the kingdom is trying to reposition itself as a logistics hub, with a modern port linking other Gulf states to the Red Sea. That is a sensible pivot. It can make a similar pitch around AI: data centres built on its west coast would be 1,500km away from Iran, compared with just 200km for other Gulf countries. Still, ports and data centres will not draw the sorts of rich expats the kingdom had expected to attract with Neom. Nor will they provide many jobs for Saudis, who are hardly eager to work as stevedores.

Before the war, most Gulf countries had hoped to diversify by emulating Dubai: wooing wealthy entrepreneurs and building service economies based on tourism, finance and tech. In a newly risky Gulf, they may not be able to. "The war hasn't killed the Dubai model, but it might kill the idea that everyone in the Gulf can be Dubai," argues a European diplomat.

Qatar spent hundreds of billions of dollars to build homes, hotels and infrastructure ahead of the 2022 football World Cup. The tournament left it with a glut of everything. Kuwait has the opposite problem: decades of political paralysis have left it unable to build much of anything. Extended regional uncertainty may leave both adrift. Their economies are set to shrink by double digits this year, and may not return to pre-war GDP until 2028 (see chart).



Oman has always been the odd man out in the Gulf, a middling oil producer with an iconoclastic foreign policy. It enraged its neighbours over the past few months by showing sympathy to Iran and entertaining the idea of working with the Islamic Republic to charge tolls in Hormuz. That may have shielded Oman from heavy Iranian attacks—but carries its own risks.

The GCC insists it is a bloc of brotherly countries, but it is prone to dramatic ruptures: less than a decade ago Bahrain, Saudi Arabia and the UAE imposed a blockade on Qatar to punish it for supporting Islamist groups. In Washington, meanwhile, Republicans have talked about imposing sanctions on Oman (and Mr Trump has bizarrely mused about bombing it).

In reality, the GCC has always done better as a travel-and-trade zone than a political bloc. The war exposed this lack of unity—and may heighten it, for example with their armed forces. America has long pleaded with Gulf states to integrate their air defences. A lack of trust hindered that. The need to husband scarce interceptors led to a beggar-thy-neighbour mentality.

Scowls behind the smiles

Nor do they agree on diplomacy. Qatar played a key role in negotiating the initial deal between America and Iran, signed on June 17th. In the days leading to it, all six Gulf countries urged Mr Trump to take the deal, fearing the alternative was more war. But in private, many officials lament it as a terrible agreement.

Saudi Arabia has now joined Turkey, Egypt and Pakistan to try to influence what happens next. But the Saudis acknowledge that this coalition is ad hoc and weak: none of their partners has much leverage to sway Iran into making concessions. The UAE has largely stayed out of the diplomatic scrum. It views Iran as an irreconcilable foe and plans to focus more on building up deterrence than what it sees as futile diplomacy. All this means there is no common GCC position on Iran.

At the same time, they have lost faith in America: it is too erratic to be trusted as a security guarantor, yet too large to be replaced. Everyone will search for new middle-power allies. China could expand its regional diplomatic role (though it has been loth to). Relations with Israel will probably depend on its election this autumn. Well-connected Saudis say the kingdom is still open to normalising ties, but only if a new Israeli government offers an alternative to endless war.

For decades, Gulf rulers offered their subjects a bargain: stay out of politics and we will keep you safe and prosperous. Oil and the promise of American protection are no longer enough to underwrite that contract. The war did not tear it up—but it has frayed it like never before. ■

Gadi Eisenkot

Israel's next great hope?

JERUSALEM

A former general seeks the highest office

WITH FOUR months until Israelis go to the polls, the main matchup is changing. The coalition of the incumbent prime minister, Binyamin Netanyahu, has long been languishing in the polls. For over a year, the leading challenger has been Naftali Bennett, a right-winger who has supplanted Mr Netanyahu once before, in 2021. But another candidate is overtaking Mr Bennett in many recent polls to become the main threat to the long-serving prime minister: Gadi Eisenkot.

Mr Eisenkot, whose new centrist party is now nearly as popular as Mr Netanyahu's Likud, has advantages over both Mr Bennett and Mr Netanyahu. He is a former chief of the general staff of the Israel Defence Forces (IDF), experience that many Israelis value while their country is still fighting various wars. But he is also a relative newcomer to politics so carries less baggage than his opponents. He was only elected to the Knesset, Israel's parliament, in 2022. He joined Mr Netanyahu's war cabinet after the October 7th attacks but resigned after eight months.

Early in the war in Gaza, he wrote to his war-cabinet colleagues to criticise Mr Netanyahu for "avoiding significant decisions". He warned that Israel risked fighting a long and inconclusive war and would fail to destroy the military and governing capabilities of Hamas, Gaza's Islamists.

Over two years later he has been proved right. The IDF still occupies over half of Gaza, while Hamas remains entrenched in the rest of the territory, continuing to hold sway over the 2m Gazans. Israel's attacks have destroyed much of Gaza and killed over 73,000, most of whom were civilians. The war has seriously damaged Israel's international standing and alliances.

Mr Eisenkot's criticism at the time was both prescient and poignant. His son, Gal, a 25-year-old reserve soldier, had been killed in the fighting two months earlier. Meanwhile, Israel has continued to blunder through endless bloody wars, making few strategic gains. Mr Netanyahu is now at loggerheads with the president of America, Israel's most important ally.

As the electoral day of reckoning draws near, two questions loom: does Mr Eisenkot represent a true alternative to Mr Netanyahu; and can he win?

Politically he is difficult to pin down. He has voted for both the left-leaning Labour and right-wing Likud parties. It is not

clear what he stands for militarily either. He is regarded as the author of Israel's Dahiyeh Doctrine, which has earned Israel international condemnation and accusations of war crimes. The doctrine calls for heavy air strikes on civilian infrastructure to destroy its enemies' strongholds. But as IDF chief during a spate of stabbings by young Palestinians in 2015 and 2016, he opposed a shoot-to-kill policy, drawing attacks from the Israeli right wing.

For all his criticism of Mr Netanyahu, he initially supported the wars in Gaza, Lebanon and Iran. The difference, he argues, is that he favoured ending them much sooner and following up with diplomatic initiatives. Above all, he thinks Mr Netanyahu's ambitions to "redraw the map of the Middle East" were never realistic and that the prime minister was motivated by concerns for his own political survival.

Israel's audition for Mr Netanyahu's replacement has been lengthy. For a while after October 7th, people looked to another former general, Benny Gantz, whose party Mr Eisenkot joined when he entered politics. But he failed to stand up to the prime minister and his popularity faded. Then Mr Bennett seemed to be on the rise. His star now looks to be falling.

Unlike those two, Mr Eisenkot does not try to match the silver-tongued Mr Netanyahu's speechifying or his feisty interviews with the foreign press (his English is halting). He wants to convince Israelis he has serious plans to deal with the security and diplomatic dilemmas facing Israel, as well as the bitter debates over the role of religion in the state and the balance of power between the courts and elected officials. His party promises a detailed manifesto soon. That may show whether he is really an alternative. Then he will face his biggest test: selling himself to the Israeli public. ■



Eisenkot: a man with a mission

Health care in Africa

Going it alone

NAIROBI

Emergencies and aid cuts are accelerating a push for self-reliance

LAST YEAR Celestine Mugambi, a Kenyan health expert, watched in horror as America abruptly paused and slashed health funding for Africa. Until February 2025 America had contributed more than 60% of all external funding for public health in Kenya; the support it gave the country's HIV programme, which Ms Mugambi helped run, was especially critical. Suddenly, workers were being laid off, life-saving services were shrinking and contraceptives and anti-retroviral drugs were disappearing from hospital shelves. "There was a lot of instability, a lot of anxiety," she recalls. But there was a silver lining: "It gave Kenya the opportunity to shape our own programmes."

More than a year after Donald Trump's administration dissolved the United States Agency for International Development (USAID) and sharply reduced its contribution to global public health, Africa's fragile health systems are back in emergency mode. The worst Ebola outbreak since 2018 is wreaking havoc in eastern Congo and Uganda and threatens to spread. Cash-strapped governments such as Kenya's, still reeling from last year's aid cuts, are being forced to devote scarce resources towards surveillance and testing for the virus. On the upside, a push for what advocates call "health sovereignty" is at last gathering momentum.

The catalyst of the self-reliance drive is a succession of crises that laid bare how dependent Africa's health systems are on foreign support. Covid-19 exposed the weakness of domestic pharmaceutical manufacturing. Aid cuts revealed how entire health programmes relied on donors not only for funding but for supplies and staff. In South Africa some 15,000 workers were funded by PEPFAR, America's flagship anti-AIDS scheme, until last year, notes Thembisile Xulu, chief executive of the South African National AIDS Council. Within three months all had lost their jobs. On June 19th America said PEPFAR funding in South Africa would end for good.

Past promises to boost self-reliance often foundered due to a lack of political will. Support for reforms to make health systems more coherent and independent tended to come from health or finance ministers, notes Naveen Rao of the Rockefeller Foundation, a global charity. "Now it is coming from the heads of state." The Accra Reset, an initiative led by Ghana's pres- ➤

ident, John Mahama, is supposed to get more local resources into health and to boost local manufacturing.

The first, and most important, task is financial. Since 2022 foreign funding for health in Africa has declined by 60%, a gap which few governments in the region can plug. Yet some are stepping up. The Centre for Global Development (CGD), a think-tank in Washington, reckons more than two-thirds of measures taken by African governments in 2025 in response to aid cuts were related to health, for instance paying for drugs. Ghana has lifted a cap on excise taxes earmarked for its health-insurance agency, boosting its budget by 60%. Nigeria, Rwanda and Ethiopia are mulling taxes on alcohol, sugar and cigarettes.

Governments and donors are also looking to the private sector for money. The ONE Campaign, an advocacy group, wants African governments to consider diaspora remittances—which even before 2025 outstripped aid—to fund health-insurance schemes. The EU hopes to blend traditional grants with loans and private capital. NGOs are reaching out to private investors. In Kenya, Tiko, a non-profit outfit whose Kenyan operations Ms Mugambi now leads, has piloted what it calls the world's first development-impact bond focused on sexual health—a scheme in which investors are repaid only if agreed outcomes are met. A similar bond may be launched in South Africa this year.

Governments also need to find ways to save. Previously, notes Benoit Renard, a co-founder of Tiko, fragmented donor projects and parallel delivery systems meant “you could find, in one facility, three different NGO people, in three different hats, doing a single task.” Tackling such inefficiencies can cut costs and make health systems less vulnerable to shocks.

Take Mozambique. After last year's cuts, health officials in Tete, a province in the north-west, began integrating services for immunisation, malaria, family planning and other parts of primary care. Resources that had been split between several NGOs were pooled. The result, according to Jonny Barty of Acasus, a consultancy that advises the government, is health workers reaching “more people, in more places, in less time”. The health ministry is rolling the model out nationwide.

Africa also needs to expand medical production. Currently it imports 70-100% of finished pharmaceutical products, and about 99% of vaccines. Africa CDC, the continent's main public-health body, wants at least 60% of vaccines, diagnostics and therapeutics to be produced locally by 2040. Capacity for research and development (R&D) needs to grow so that medicines can be developed closer to home.

Promising manufacturing hubs include South Africa, which wants local drugmakers to make lenacapavir, an advanced HIV prevention medicine, and Nigeria, which is taking what Robert Matiru of Unitaid, a global health body, calls an “all-of-government” approach to boosting production. A factory in Kenya is also set to deliver its first batch of drugs soon.

R&D is making progress. Thanks to philanthropies and non-profit groups such as the Drugs for Neglected Diseases initiative, clinical trials for some of Africa's most gruesome scourges are now run across the continent. Congo's national bureau for biomedical research detected the latest Ebola outbreak too late. But it has led genomic sequencing for Bundibugyo, the strain that is causing it.

There are plenty of snags. Not every country can flourish as a pharmaceutical producer, so regional co-operation will be

crucial. Yet South Africa has so far refused to join the African Medicines Agency, a regulator to certify vaccines and medicines that started work in 2021. That will delay a continental market for pharmaceuticals.

The biggest problem remains money. In 2024, 28 African countries spent more on servicing external debt than on health. A recent analysis by Bajeti Hub, a Kenyan NGO, found that long-run health spending in Kenya falls by 1.2% when the debt-to-GDP ratio increases by one percentage point. Some 22 African countries have signed deals with the Trump administration to restore some lost American health funding—in exchange for national health data and, in some cases, mineral rights. The road to sovereignty is bumpy. ■

Vigilante politics

So much for Pan-Africanism

CAPE TOWN

What a surge of anti-migrant protests says about South Africa

AS ANTI-FOREIGNER protests have swept South Africa in recent months, their rallying cry has become increasingly audible: “Abahambe! Sekwanele!” The words, in isiZulu, the most commonly spoken language in South African homes, can be translated as, “They must go! Enough!”

In the first half of 2026 there have already been more anti-migrant protests than in any other year of the 2020s, according to ACLED, a violence-monitoring group. And there is growing concern in the government that on June 30th there will be a violent climax. March and March, a vigilante group, has given the date as a “deadline” for illegal immigrants to leave South Africa. It also pledges to take to the streets that day in what some analysts worry could be an echo of July 2021. Back then the arrest of Jacob Zuma, South Africa's former president, led to the worst civil unrest since the end of white rule in 1994.

March and March, founded in 2024 by Jacinta Ngobese-Zuma (seemingly no relation to Mr Zuma), a DJ, is the latest anti-foreigner group to emerge in recent years. Like Operation Dudula, an outfit formed in 2021, it holds protests against immigrants living in poor urban areas. It goes door to door, asking other Africans for proof of their immigration status. It has tried to block migrants' access to public hospitals. Many shops run by foreigners have been looted. There are “Idi Amin's Uganda vibes”, says Fred Khumalo, a writer, referring to the campaign that chased people of Asian descent from the east African country in the 1970s.



Making it on their own



► Vigilante groups reflect—and exacerbate—xenophobia among the public. In 2025, 42% of South Africans polled by the Human Sciences Research Council (HSRC), a state-funded research group, wanted “no immigrants”, the highest share since surveys began in 2003 (see chart). The share welcoming “all immigrants” fell from a peak of 43% in 2008 to 15% in 2024 and 2025. The International Social Survey Programme, another poll, found in 2023 that 70% and 74% of South Africans say, respectively, that immigrants take away jobs and cause crime (the highest shares among all 29 nationalities surveyed except for Russians, who were more likely to believe immigrants caused crime).

Economists generally argue that immigrants are not to blame for South Africa's high rates of joblessness and crime. Research by both the World Bank and the OECD has found that foreigners are net job creators because they set up firms and add to consumer demand. The Helen Suzman Foundation, an NGO, has pointed out that foreigners commit fewer crimes on average than South Africans. Census data suggest that only around 5% of the population is foreign-born, a share that has not changed much in recent years.

Why, then, has there been a spike in anti-migrant sentiment? It may be that research is out of date or does not capture illegal migration or the effects in the informal economy, where many African migrants toil as farm workers or nannies. The HSRC data suggest that anti-migrant views are hardening the most among the least affluent. The realities in poor areas could be escaping researchers.

Yet opinion is also being shaped by campaigns that scapegoat Africans. These include online movements as well as offline ones. Many of the same online activists involved in Operation Dudula have re-emerged in 2026, notes Kyle Findlay of Murmur Intelligence, a data-analytics firm.

Many politicians have taken a strong anti-migration stance, too. ActionSA, led by a former mayor of Johannesburg, has

welcomed a member of Operation Dudula as a candidate for local elections in November. Gayton McKenzie, the leader of the Patriotic Alliance and a cabinet minister in the coalition government, has a history of incendiary statements about immigrants. UMKhonto weSizwe (MK), the party founded in 2023 by Mr Zuma (the country's most famous Zulu), is backing March and March, whose protests include open displays of Zulu nationalism. The HSRC poll finds that of South Africa's nine provinces, opposition to immigrants is greatest in KwaZulu-Natal, the Zulu stronghold.

The security services seem better prepared for June 30th than they were in 2021. On June 7th President Cyril Ramaphosa said that protesters “deserved to be heard”

but could not take the law into their own hands. Police and the private security firms that proved crucial to responding in 2021 are anticipating hotspots. The police minister has cancelled all leave and has said the army will be called up if needed.

Yet even if less damage is done on the day, South Africa's brand is tarnished. Several countries have organised repatriation flights or bus journeys for their citizens who fear for their safety. Africans often root for other Africans at the World Cup. But there has been little pan-African backing for South Africa. After the country's loss in the opening match, a Kenyan posted on X: “I hope South Africa isn't blaming African migrants...for the 2-0 beating and two red cards.” ■

Defences in the Sahel

The return of the rampart

Shrivelling state authority has prompted a revival of medieval-style fortifications

NOT MUCH remains today of the walls, ramparts and moats that once surrounded Benin City in southern Nigeria. Yet for centuries these giant earthworks—second in length only to China's Great Wall among man-made structures—bespoke a mighty civilisation whose authority extended across much of west Africa. By the standards of pre-colonial Africa, the Benin state was exceptionally strong: erecting the wall in a single dry season might have required mobilising as many as 5,000 men, each working ten hours a day. But as the empire withered and eventually succumbed to British invaders in the late 19th century, most of the earthworks vanished. So did those of many other fortified towns across west Africa.

Now they are returning. Since the mid-2010s, as jihadist insurgencies have spread across northern Nigeria and the Sahel, defensive earthworks have risen up around towns and cities. Many are so extensive they are clearly visible in satellite images. According to new research by Olivier Walther and Steven Radil, geographers at the University of Florida, all urban centres in north-east Nigeria with populations of more than 10,000 are now secured by trenches. Most big towns there and in the Lake Chad basin are surrounded by sand berms up to three metres high.

Unlike those of the former Benin empire, the new walls and moats are a sign of state weakness. In Nigeria, the army began digging trenches with the

help of the American government shortly after the notorious “Chibok Girls” kidnapping by jihadists in 2014. The goal, says an American contractor involved in the initiative, was to protect civilians and “hold the line” against jihadist advances. Militants in the Sahel typically carry out attacks using light vehicles such as motorcycles or pickup trucks. Trenches and berms are a simple, low-tech way to slow them down.

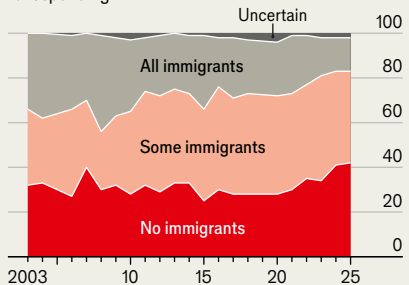
Governments in Burkina Faso, Cameroon, Chad, Mali and Niger have followed suit. In 2024 at least 133 civilians were killed in Burkina Faso by jihadists while they were digging a trench around the town of Barsalogo on the orders of the armed forces. It is not just cities and towns. Attracted to what they see as a quick fix for security threats, governments in the region are fortifying all sorts of assets under their control, from power stations and government offices to hospitals and schools. Togo has dug over 36km of trenches along parts of its borders with Burkina Faso and Benin to prevent militants crossing.

In the long run, the effect may be to cede vast swathes of the countryside to the insurgents. Mr Walther notes that large parts of the Sahel now resemble an “archipelago of secure islands surrounded by an insecure sea”. For much of its history Africa was characterised by weak states struggling to control scattered, mobile populations. The landscape of the Sahel today looks increasingly like a snapshot from its past.

Keep out

“I generally welcome to South Africa...”

% responding



No data for 2005 and 2019

Source: Human Sciences Research Council

Europe



War in Crimea

From resort to battle zone

FEODOSIA, SEVASTOPOL AND YALTA

Like Russia proper, Crimea is running short of petrol and patience

SIRENS DID not wail in the early hours of June 21st in Feodosia, a resort town in Russian-occupied Crimea. There was only the buzzing of drones and the sounds of explosions. The warning systems were turned off some time ago, not because the peninsula is safe but because otherwise “sirens would be sounding 22 hours a day and nobody would get any sleep,” an official recently told radio listeners.

Residents suspect the authorities are mostly concerned about the tourist season. Preserving the façade of normality grows harder by the day. Crimea is under constant attack: military units, railway stations and power plants are being hit, and some residential buildings as well. Yalta, a resort town with no military targets, is an exception; there is hardly an empty sun-lounger on the beach. But the rest of the peninsula has become a war zone.

On the night of June 20th Ukrainian drones hit power lines, blew up an oil ter-

minal in Kerch and damaged a ferry transporting cargo. Ferry crossings have now been shut down, severing one of the peninsula's arteries of supply. Lorries get in only via a highway through the Donetsk, Zaporizhia and Kherson regions. But the land corridor is increasingly vulnerable to Ukrainian drones, as burned-out vehicles along the road testify. Last week Mykhailo Fedorov, Ukraine's defence minister, said Crimea could “in effect become an island”.

Blackouts are frequent. On June 21st

Crimea's governor announced the most drastic step yet: a halt to sales at all petrol stations. Meanwhile, Ukrainian drones have brought the war deep into Russia itself. In the past few days they have hit refineries in Siberia and, with dramatic effect, in Moscow. Columns of smoke towered over residential neighbourhoods; a video of the lid of an oil tank blown high in the air went viral. Petrol is being rationed in more than half of Russia's regions.

The attacks on Crimea are especially significant. The peninsula serves as a bridgehead for Russia's army, though Ukrainian drones have chased its Black Sea fleet to more distant ports. It also has enormous political importance. Russia's conquest of the territory in 2014 was a triumphant moment of Vladimir Putin's imperial ambitions, and greatly boosted his popularity. Crimea was the jewel of the Russian and Soviet empires; Mr Putin called it “the spiritual source” of the Russian nation. He has been largely silent about Ukraine's recent attacks, as have state media.

Ukraine's constant strikes have shaken locals' confidence in the Russian state. Ordinary people feel suddenly vulnerable, explains a local mother. The conversation is punctuated by automatic-weapons fire outside: air-defence teams are trying to shoot down drones. Her 14-year-old son wants to leave, she says, but she tells him

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► that as Crimean Tatars (the territory's indigenous population), they must remain: "It is our only homeland."

Apart from the Tatars, only a minority of the peninsula's residents resisted annexation. Many, especially pensioners who remembered the Soviet era, welcomed it: Russia promised investment and more social benefits. A decade ago businessmen in Sevastopol wanted to reinvent the city, centred on a naval base, by developing winemaking, commercialising the port or mining crypto. Now, "people no longer see any prospects for the future," says Nikolai Chestiakov, a resident. Those with money are buying property in Russia and leaving.

While Sevastopol has emptied, Dzhankoi, a sleepy railway town, has boomed. Military personnel use it as a staging point for the Kherson region. Soldiers fill the hotels. "Thank God the army is here," says Mikhail, who owns a kebab shop. "The guys come, they relax—business is good." The drawback is that Dzhankoi has become a target. Five civilians died when a drone hit a residential building.

Yet there is no sign that the attacks have caused second thoughts among Crimeans who back Russia. Some see a Ukrainian return as a nightmare: Ukrainian officials have threatened retribution. They also fear being denounced for disloyalty by fellow residents. Counter-espionage propaganda is ubiquitous. "Ready to sell out your homeland? Waiting for instructions from your handler? Then we're coming for you," announces a Crimean radio programme titled "Comrade Major, Please Investigate!" A resident of Kerch who photographed a fuel tanker and forwarded it to a friend abroad was detained this month and could face charges of treason.

In Russia proper, too, frustration with the war is rising. Members of the ruling elite increasingly see it as a dead end. Promises that the near-stalemate would give way this spring to rapid Russian advances have proved empty, as in earlier years. Strikes in St Petersburg and Moscow have forced ordinary Russians to reckon with the war's consequences. Internet blockages engineered by security services interfere with citizens' daily lives.

Yet Mr Putin's main problem is not rising casualties or the mediocre economy, but the fact that he has nothing to show for his war. As Russian history shows, leaders perceived as failed or weak are rarely forgiven. For now, the main sentiment in Crimea and across Russia is exhaustion. Even those who support Russia cannot see the purpose of the war. "We don't need any grand ambitions, we don't need anyone trying to make things better. We just want the sun to rise in the morning, tourists to come in summer," says Tatyana, a tour guide in Feodosia. "We are so tired of everything else." ■

France's populist right

Provence of concept

BEAUCAIRE AND NICE

The south is testing out hard-right rule

THE BEACHFRONT in Nice is a spot for admiring the self as much as the sea. Biceps are big; dogs are tiny. Superyachts float in the bay. In short, not an obvious place to stir up populist indignation. Yet in March France's fifth-biggest city was captured by a party allied to Marine Le Pen's National Rally (RN), making it her biggest prize yet. Nice hints at how the nationalist right is conquering the south of France—and how it hopes, at the presidential election next April, to win the country.

At this year's local elections most big cities were won or held by the left or greens. Some in the south narrowly resisted the RN, including Marseille and Toulon. Nice was a crucial victory for the nationalist right because of both its size and location. The RN and its friends now run mid-sized town halls in a wide band of southern France that includes Beaucaire, Carpentras and Orange in Provence, Menton on the Riviera, and Carcassonne and Perpignan in the south-west.

Eric Ciotti, the new mayor of Nice, who quit the centre-right Republicans to found his own party and hooked it up to the RN, promised to tighten security, cut waste and turn the page. The outgoing centrist mayor, Christian Estrosi, had held office for 17 years. Mr Ciotti's tough law-and-order promises resonated. Away from the seafloor cocktail bars, flanked by the airport and the motorway, lie the sprawling

housing estates of the Moulins quarter. In May two bystanders were shot dead there by a man wielding an assault rifle. In the past two years drug-gang rivalry has led to 11 deaths. Mr Ciotti has opened a police outpost near the site of the shooting. He vows to increase by 50% the number of local police in the city by 2027.

Yet the capture of Nice reveals something else: a potent new RN electoral strategy, embraced by Jordan Bardella, the party leader, which marries its traditional working-class base with a liberal-minded bourgeois electorate. This works in Nice and other southern towns with a history of leaning to the right, as well as a high share of pensioners, who tilt right too. Mr Ciotti cut the *cordon sanitaire* that previously made the RN untouchable. But he studiously balanced these links with the recruitment of moderate local notables. "We are the real party of the right," he insists.

It is a liberal fiscal message—"France wins the world cup for taxes," he complains—with a populist twist. Mr Ciotti admits to admiring Argentina's Javier Milei and his chainsaw, and has begun a showy assault on town-hall profligacy. In Nice families who display extravagant wealth coexist with others on squeezed budgets. Mr Ciotti has blended headline anti-immigrant nativism with a clampdown on the town-hall car fleet and banquet budget.

The politics play out differently across the south, where the RN has a long history. In the 1980s Jean-Marie Le Pen, Ms Le Pen's father and the party's co-founder, began to score in double digits in the region, drawing on xenophobia prompted in part by an influx of immigrant labour recruited from north Africa. In greater Nice 17% of the population are immigrants, compared to 12% in France as a whole.

Identity politics is a common theme. Beaucaire is a Provençal town of shuttered façades dominated by a medieval fortress and a cement factory. Under RN rule for 12 years, the town hall flies only the French *tricolore* (not the European Union's flag), and has revived folk traditions with an unapologetic Christian nod. Last year the town was fined €120,000 (\$136,000) for refusing to dismantle a nativity scene, illegal in public buildings under France's secular rules. This month Beaucaire organised a "weekend of tradition", including a Provençal dance show, with free entry for all.

Residents seem to appreciate hyper-localism: potted flowers, cleaner pavements, children's summer activities. "Beaucaire has been transformed," says a seller of *tape-nade* at the weekday market. "It's clean, there are no drug dealers, there are events for everyone." Julien Sanchez, until recently its mayor, is now the RN's campaign director. A study of RN governance by Fred Paxton and Timothy Peace, at the University of Glasgow, calls this approach "popu- ►►



In the heat of the right

► list pragmatism”, part of a national strategy of mainstreaming the party.

RN city governance has brought plenty of controversy. Several new RN mayors, including in Carcassonne, have stopped flying the EU flag. Many have cut subsidies to human-rights and immigrant-aid groups. Carcassonne has introduced a rule against begging. In Carpentras a song associated with the Nazi-collaborationist Vichy regime was broadcast through municipal loudspeakers on May 8th; its RN mayor denied responsibility.

Nor are the RN and friends unstoppable in the south. Marseille was held by the left after the hard left stood down in the run-off against Ms Le Pen's candidate; it showed the value of left-wing unity, which eludes national leaders. But the region now has a concentration of hard-right laboratories, which will test how competent—and how toxic—their leaders are. ■

Germany's cities

Crumbling schools, empty coffers

INGOLSTADT

Town halls are going broke. That could be good news for the far right

IT DOES NOT take long for a visitor to Ingolstadt, a medium-sized city in Bavaria, to notice that this is Audi town. Spot the Audi health-insurance booth near the town hall, the carmaker's splendid museum or the ads for its summer concert season, which opens this weekend. Meet a local, and chances are they work at the Audi plant—the firm employs 40,000 people here—or are related to someone who does. The success of Audi drove Ingolstadt's growth from a sleepy town of 30,000 after the second world war to 145,000 today.

And it is the struggles of Audi, where profits and the workforce are shrinking, that help explain why the city has fallen into a deep hole. Business-tax revenues have halved in just a few years, thanks in large part to the woes of Audi and Volkswagen, its parent company. So tight is the squeeze that earlier this year the Upper Bavarian government rejected Ingolstadt's budget, banning the city from making new investments. Marquee projects, such as the renovation of the city's celebrated brutalist theatre building, are on ice. Pupils had to be removed from one dilapidated school because the city could not afford to repair it. “Audi is a blessing for our city, and also a risk,” says Michael Kern, the mayor.

The volatility of municipal financing has bound the fates of many German cities to local business. In good times some swam in money: one town with a Daimler

plant near Stuttgart paved its pedestrian crossings with Carrara marble. But when industry sneezes, cities fall ill—and the bug now extends beyond post-industrial spots like the Ruhr or Saarland. Towns in wealthy Baden-Württemberg have been walloped by the car sector's woes. The troubles of Ludwigshafen, one of Germany's most indebted cities, are those of BASF, the local chemicals giant.

But even towns with a broader economic base are staring into the abyss. Few have escaped the fiscal trap of revenues failing to keep pace with exploding spending. A new report from the Bertelsmann Foundation finds that last year the budget deficits of Germany's 10,000-odd municipalities amounted to a record €32bn (\$36bn), up by almost 30% in just one year (see chart). Almost all expect it to get worse.

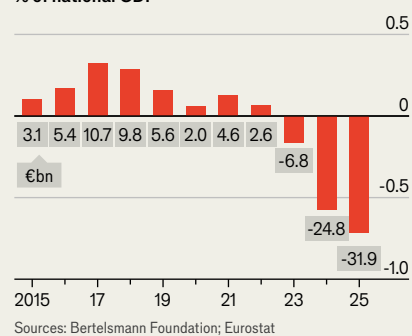
The crisis is the worst in the history of the federal republic, says René Geissler, a professor of public administration at the Technical University of Applied Sciences in Wildau. He cites three factors: surging inflation, especially in construction; personnel costs; and onerous laws passed by the Bundestag, especially on youth and disabled benefits, that municipalities must meet. Such mandatory expenditures now account for 80% of Ingolstadt's budget, forcing brutal cuts to discretionary outlays. In poorer towns the squeeze is tighter. Mr Geissler says he expects stressed municipalities to make “under-the-radar cuts” to help the numbers add up.

Consequences show up everywhere. Bayreuth has cancelled a party to mark the 150th anniversary of Richard Wagner's first festival. Berlin's skint boroughs are struggling to cope with an invasion of toxic hairy caterpillars. Like desperate workers pawning jewellery before payday, some cities have turned to *Kassenkredite*, short-term loans, to pay their bills. And across Germany schools, roads and sports halls are falling into disrepair as the brakes are pressed on capital spending. The municipal-investment backlog now stands at a record €231bn. Saying it was “three minutes to midnight”, on June 22nd mayors across Germany held public events to declare themselves at the limit. “We can manage this year, but beyond that many cities will have their backs against the wall,” says Martin Wilhelm, the treasurer of Offenbach, a satellite town of Frankfurt.

What's to be done? Municipalities want the federal government to stick to a rule of “who orders, pays”; ie, to provide funding for any new law whose costs they must bear. Those without rainy-day funds also want €30bn in short-term relief. The federal government, itself in a tight fiscal spot, has offered little. But so grim is the situation, says Henrik Scheller, a public-finance expert at the German Institute of Urban Affairs, that help is likely later this year. If

Sinking harder

German municipalities, budget balance, % of national GDP



not, “we could see a kind of revolution.”

That does not feel imminent in Ingolstadt. But Michelle and Christian, a local couple, say they notice small signs of decay, from libraries soliciting donations to a city festival that had its funding cut. Elsewhere, many fear a deteriorating public realm could help the far-right Alternative for Germany. “Every leaking school roof that is not repaired for three years boosts the right-wing populists,” says Marc Grandmontagne, Ingolstadt's education and culture chief. Perhaps that will concentrate minds in Berlin. ■

Hungarian democracy

Operation Cleansing Fire

BERLIN

How to burn away the remnants of Viktor Orban's regime

PETER MAGYAR, Hungary's prime minister, had two intertwined tasks when he came to power in early May. One was to restore Hungary as a full democracy and a reliable member of the European Union. The other was to unlock billions of euros in funds from the EU, which Hungary's ailing economy badly needs. These had been frozen because of numerous violations of the rule of law under the reign of Viktor Orban, his predecessor.

Mr Magyar is moving quickly on both fronts. In parliament on June 22nd he announced “Operation Cleansing Fire”: a constitutional amendment that would, among other things, remove Tamas Sulyok, the president (whose term runs until 2029), and Peter Polt, the head of the constitutional court, both loyalists of Mr Orban. He also unveiled plans for a new constitution. Mr Magyar pledged to “free our country from the economic and political mafia”.

The amendment reintroduces manda- ►►

► tory retirement at 70 for the constitutional court's 15 judges, which would mean the 70-year-old Mr Polt and three others must step down immediately. It would limit lawmakers' terms to 12 years. And it would create an agency for the recovery of national assets to prosecute the Orban regime's misuse of public funds. Mr Magyar claims corruption costs Hungarians 8-10% of GDP annually; economists say it is hard to estimate, but agree it is substantial.

"I am now more optimistic that Hungary can avoid a constitutional crisis," says Daniel Hegedus, a Hungarian who is deputy director of the Institute for European Politics in Berlin. When he took office Mr Magyar asked the president, the constitutional court's judges, the public prosecutor, the bosses of the competition and media watchdogs and the head of the audit office to resign by the end of May. All are allies of Mr Orban, and refused.

Mr Magyar thus found a more controversial way to purge Mr Orban's apparatus. Tisza, his party, enjoys a two-thirds supermajority in Parliament, enabling it to pass constitutional amendments. His proposed amendment states that the day after it comes into force, the president's term of office will end.

It will be Mr Magyar's second constitutional amendment. On June 15th parliament passed one to limit a prime minister to eight years in office. So Mr Orban, who ruled for a total of 20 years from 1998, cannot run again. The former leader is being hoist on his own petard: he often used his own party's supermajority to alter the constitution. Still, "from a rule-of-law perspective an amendment designed to force people out of office is not great," says Zselyke Csaky of the Centre for European Reform, a think-tank headquartered in London.

Supporters of Mr Magyar say using the toolbox left by Mr Orban's regime is necessary to restore democracy and regain access to EU funds. On May 29th Ursula von der Leyen, the head of the European Commission, approved the provisional release of €16.4bn (\$18.6bn) in EU grants and loans from the union's covid-recovery pot and its cohesion funds. The release depends on Hungary fully reaching 27 milestones, including revised public-procurement rules, new anti-corruption measures and strengthening academic freedom.

Removing the president and top judges from their jobs is not among the milestones. But it should signal that the Hungarian state's capture by Mr Orban is over. Next comes a referendum, following a period of national consultation, on a draft of a new constitution. The new one may make it harder to pass amendments, for example by requiring approval by two successive legislatures. That would stop the next party to win a big majority from changing the rules of the game. ■

Meloni and Trump

Declaration of independence

ROME

The dalliance between the populist-right leaders is off—or is it?

SEVERAL OF Giorgia Meloni's ministers are due at a reception at the American embassy in Rome on July 2nd to mark the United States' 250th birthday. The mood may not be festive. Last week a row between the prime minister and President Donald Trump led to her foreign minister scrapping a visit to Washington. Then on June 24th her defence minister tried to rebut statements by Mark Rutte, head of NATO, that Italy had (despite its denials) let American planes use its airbases in the war on Iran. It all illustrated Ms Meloni's impossible task: preserving relations with America while distancing herself from Mr Trump to appeal to Italians.

The ructions between the populist-right leaders began on June 19th, when Mr Trump accused Ms Meloni of begging him for a photo op at the recent G7 summit. She hit back with an impassioned video, telling him: "Neither I nor Italy ever beg." Mr Trump was already irritated by Ms Meloni's refusal publicly to endorse the attack on Iran. When he responded to her video by claiming she had lost popularity, she snapped back: "My popularity is none of your concern. I suggest you focus on yours."

The obvious explanation is exasperation. Italy's prime minister had clung more tenaciously than leaders of other

European states to a belief in "the West", and to an Atlantic alliance the president has repeatedly undermined. Mr Trump may also have touched a genuine sore point. Italy saw the summit as a chance to repair ties with America that had been frayed by the president's clash with Pope Leo. It is hard to imagine Ms Meloni begging. But it is possible an aide assigned to get a joint photo overdid it.

The cynical explanation is that Ms Meloni's beef is calculated. Her biggest headache is the rise of a new right-wing party, more extreme than her Brothers of Italy. Led by a retired general, Roberto Vannacci, Futuro Nazionale is rapidly gaining support—perhaps overtaking the League, one of the Brothers' coalition partners, which has fallen to around 6% in polls. To fend off Mr Vannacci, Ms Meloni might have to embrace hard-right positions that would, in turn, alienate mainstream conservatives who flocked to her in the 2022 election.

Her response to Mr Trump meant the general could not attack her for failing to defend the nation's honour, and won her the admiration of Italians who revile America's president. She may hope that will give her ministers room to quietly share an independence-day beer with the Americans. But Mr Rutte's tactlessness has restarted the fireworks.



Not going so steady anymore

CHARLEMAGNE

Ever slower union

The European Union takes too damn long



IN JULY 2025 America and the European Union agreed on a lopsided trade deal, signed at President Donald Trump's Turnberry golf course in Scotland. By September officialdom in Washington had completed the necessary paperwork. Getting it through the EU system in Brussels has proved more complicated. There were two legislative proposals from the European Commission, a report from the European Parliament, then some haggling between the bloc's 27 member states on how they should jointly negotiate with MEPs. Meetings ensued, press releases were issued. By May, some ten months later, a grand jamboree of negotiations between various bits of the EU known inelegantly as a "trilogue" had tentatively signed off on the deal. The parliament then decamped to its second home in Strasbourg (best not ask why) to vote on it. All that now remains is a few more ministerial signatures, then some more press releases, then translation of the whole thing into the EU's 24 official languages. Everyone involved seems delighted it has taken "only" a year or so to get the whole thing over the line.

Living the slow life is part of Europe's charm. Those long lunches and interminable summer holidays are as integral to the place as cathedrals and grumpy waiters. Alas, the idea of *la dolce vita* carries through to policymaking. In the case of the trade deal, forcing Mr Trump to twiddle his thumbs—he huffily threatened to scrap it if the EU didn't get a move on—may have been a feature, not a bug. But other European initiatives are into their second decade of gestation for no such good reason. Deliberation is a virtue. Europe often seems to be engaged in obstinate self-obstruction.

The EU does not so much kick policy proposals down the road as place them in bureaucratic black holes. What comes out—and when—is a matter of happenstance. A recently passed revision of air-passenger rights required 13 years of legislative contemplation. The idea of a "capital-markets union", to make citizens' savings flow more easily to European businesses, seems barely closer to reality today than when it was first proposed in 2015. More humdrum initiatives can drag on for decades. A scheme for companies to register at EU level rather than in each country was first presented in 1988. A fresh push to get it done was unveiled in Davos by the president of the commission, Ursula von der Leyen, in January

2025. And again, identically, in January 2026. Even if all goes well in the inevitable trilogues to come, it will be another two years before it comes into force. Some continents have drifted faster.

Delays blunt the usefulness of the EU's best-laid plans. In most countries economic stimulus schemes linked to covid-19 are a thing of the distant past: the money was out of the door within months, when the fiscal oomph was needed to combat recession. Not in the EU. There, covid stimulus cash is still being spent. This is unsurprising given that the plan involved a *triple-feuille* of administrative entities, from Brussels to national governments and regions. Anything to do with money slows Brussels to a crawl. The EU's budget is set once every seven years, and takes half that long to be agreed upon. Rarely has so little money—just 1% of GDP, give or take—been haggled over by so many for so long.

In part the policymaking torpor is hard-wired into the system. The EU machine in Brussels has many of the responsibilities of a national government, but the working methods of an international organisation. Vetoes abound, either formally or informally. Time-consuming "impact assessments" are required at every turn. Making sure everyone gets their say—often more than once—means that the average law takes nearly two years from its first draft to finally being adopted. In many cases whatever is passed at EU level then needs to be transposed into national law by parliaments in each of the member states, which typically takes another two years. Courts that rule on EU business are slow: antitrust cases take 43 months to get a first ruling, by which time the infraction allegedly committed has long been forgotten. Measures designed to counter abuses of trade rules take so long that triggering them often seems pointless.

Some see virtues in the EU's ponderous approach besides annoying Mr Trump. Yes, it is a crawling consensus machine. But it is better to legislate properly than rush into mistakes that later need to be corrected. (America's hasty enactment of the Turnberry deal was in part reversed by the courts, for example.) Yet that argument would be more convincing had the EU itself not recently been obliged to scrap a heap of red tape, crafted in a bout of over-regulatory zeal just a few years ago. And even this effort to simplify existing legislation is taking for ever, mired in procedure.

Summer is coming

The EU's slothful approach is a luxury it can no longer afford. Slow and steady worked in calmer geopolitical times, when America guaranteed Europe's security, China bought its exports and Russia was flailing. Now the world is changing faster than the EU can act. Its adversaries know this. Scott Bessent, America's treasury secretary, mocked "the dreaded European working group" that crops up as a response to all of the continent's woes—a gibe that stung. China is winning a trade war that Europe is only just now discussing, and promising to discuss again soon. Europe's sole appropriately speedy endeavour is its aid to Ukraine since Russia's invasion—and even there, it started slow in 2022.

Can Europe pick up the policymaking pace? That may seem an odd question to ask just as Eurocrats are leaving Brussels for the summer. The obvious solution is for gridlocked projects to advance among a subset of countries, an approach being tried with the capital-markets union. That seems more realistic than revisiting the EU's constitution, which would require a good decade of haggling before it could speed things up. Europe's deliberation once looked wise, or at least defensible. Now it looks foolish. ■

Britain



Andynamics

The yes prime minister

If Andy Burnham is to save Britain's economy he needs to be tougher, and not just a slicker version of Sir Keir Starmer

I'M MORE OF a spender than a saver. I would much prefer to say 'yes,'" confessed Andy Burnham in 2024. This attitude suited him well as mayor of Greater Manchester; it was his job to wring as much cash from Westminster as possible. But after Sir Keir Starmer resigned on June 22nd, the newly elected MP for Makerfield is now almost certain to become the next prime minister. He would be Britain's seventh since 2016. Being in charge of doling out the dosh (and responsible for the nuclear codes to boot) is a much tougher gig than Mr Burnham is used to. For a start, he will have to begin saying "no".

Mr Burnham has undeniable strengths. He is untainted with the government's failures and can talk human. His net favourability rating (-4 percentage points) might seem bad, but it is better than that of any British party leader (Sir Keir's is -45). Labour MPs look set to back him in force, convinced that his heart is in the right place, something the bloodless Sir Keir

never quite managed. Wes Streeting, a more centrist rival, has rowed in behind him. When nominations close on July 15th Mr Burnham is likely to be uncontested.

That gives Mr Burnham precious little time to prepare for his huge new job. He rails against the sins of politicians past. Deregulation, privatisation and deindustrialisation are just some of the abstract nouns in his sights. But the would-be prime minister is gnomish about what he would do in office. He talks vaguely of

"stronger public control" of essential services, without clarifying what he means. He flip-flops on key policies, recently re-committing to Sir Keir's fiscal rules having previously called for them to be relaxed to fund defence. And he is silent on some of the most pressing issues, like how to navigate the upheaval from AI.

Tackling the country's deeply entrenched problems will require a relentless focus on the future, not the 1980s. A main reason that Britons seethe at governments of all stripes is that their living standards have stagnated. Mean wages have grown by just 1% in real terms since Labour took office in 2024, and by a paltry 4.5% since April 2008 (a third as quickly as American wages). More trouble could lie ahead, if AI displaces the legions of white-collar knowledge workers who make up the British economy's backbone.

Low growth is also bad for Britain's ailing public finances. Public spending has risen from 39% of GDP in 2019-20 to 45% today. An ageing population and rising defence commitments could push it higher still. Tax revenues (37% of GDP) are struggling to cover this largesse. Nor will greater borrowing help. Debt interest this year will cost £109bn (\$143bn), more than the education budget. Bond investors charge Britain higher interest rates than any other big rich country, and jump at the slightest hint of profligacy. ▶▶

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— Jeffrey Donaldson's fall

None of these problems is insurmountable. But bold policies are needed to get Britain out of its rut. This includes tackling high health-care and welfare costs, pragmatism on “net zero” and cutting the red tape that stops things being built. Mr Streeter’s aborted campaign set out serious plans. If Mr Burnham appointed him chancellor, that would signal a willingness to embrace growth.

The problem is that these policies cut against Mr Burnham’s instincts, which are more aligned with those of Ed Miliband, another contender for chancellor, whose views are more statist (see Bagehot). Capitalising on AI will require creative destruction, letting unproductive firms fail and freeing workers to move to more AI-friendly jobs. Mr Burnham seems instinctively opposed to the deregulation needed to achieve this. His allies are calling for even stronger protections of workers’ rights than were introduced by Sir Keir.

Such wrongheadedness can be seen elsewhere. Mr Burnham has argued that “undue weight” is given to planning reform, preferring an expensive state house-building programme (it is unclear how he would fund this). He wants to reindustrialise the economy, a romantic idea which ignores the fact that Britain’s comparative advantage lies in services.

Mr Burnham’s fiscal instincts do not suggest a man willing to make tough choices. Whereas Sir Keir’s problems started when he accepted freebies for himself, Mr Burnham’s undoing might come from offering too many to other people. At a rally on June 19th he promised to bring down “water bills, energy bills, rail fares”. He claims savings would come from bringing these industries under greater public control. But costs are driven up by the need to improve dilapidated infrastructure and the net-zero push, not ownership structures. In reality lower bills can be funded only through reduced investment (contradicting his own pledges) or higher subsidies.

It is unclear how Mr Burnham could afford such subsidies while also increasing defence spending (another thing he has said yes to). The aspiring prime minister has committed to Sir Keir’s manifesto pledge not to raise employment taxes or VAT (yes!), but also wants to protect the triple lock, an expensive ratchet for increasing the state pension (yes! yes! yes!). He even endorses the least deserving causes, like the women protesting against pension-age increases that they were told about decades in advance (though he has since backtracked there, too). The only area he really says that he wants to save money from is working-age welfare, but he is vague on specific measures.

A few of Mr Burnham’s instincts are welcome. True, his partisanship for the north may become tiresome. And some of

his regional policies, like legally requiring Westminster to prove its decisions always reduce regional inequality, are barmy. But he is correct to identify over-centralisation as a drag on England’s regions, and to push for greater fiscal devolution. He is also right to want to reform Britain’s broken property-tax system and its shamefully patchy provision of social care (though again he shies away from outlining the painful measures this would require).

Might being prime minister stiffen Mr Burnham’s backbone? His reported appointment of James Purnell, a pragmatic veteran of Sir Tony Blair’s government, as his chief of staff augurs well. A much-touted speech on the economy next week will give him the chance to flesh out his vision. But if the man who likes to say yes fails to make the necessary trade-offs, he need only look at Sir Keir to see the fate that awaits him. ■

Mountains and molehills

Burnham’s bounce

New prime ministers tend to provide a polling boost. Those don’t always last

“ROME IS SAVED!” quipped an opposition MP as Andy Burnham entered the House of Commons on June 22nd. “Turn the water into wine,” said another, as the former Manchester mayor was sworn in as an MP. “He’s not the messiah,” another heckled. Mr Burnham, with his cheeky charm (and a quick-witted “Monty Python” reference), replied: “Naughty boy.”

He won his place in Parliament with a by-election triumph on June 18th. With Sir Keir Starmer stepping down, Labour MPs hope Mr Burnham will take over and work his magic on their own constituencies.

Switching horses is a common tactic. Since 1945, ten of Britain’s 18 prime minis-

ters have taken over between elections. They lasted for an average of two years and 324 days, ranging from Harold Macmillan (nearly seven years) to Liz Truss (49 days). Five went on to win a general election (albeit while losing her majority, in Theresa May’s case) and four to lose one (including Mr Burnham’s ex-boss, Gordon Brown). Ms Truss resigned before an election could be called, but after her the Tories were toast.

The Economist analysed opinion polls since 1955 and found that when the governing party changes prime minister it has got an average poll bounce of 3.8 percentage points (see left-hand chart). This is partly the reward for removing an unpopular incumbent—with an average 1.8-point uptick after the prime minister’s resignation. When Margaret Thatcher resigned in 1990, the Tories had a ten-point surge. When a fresh prime minister is installed, the ruling party tends to gain a further two points.

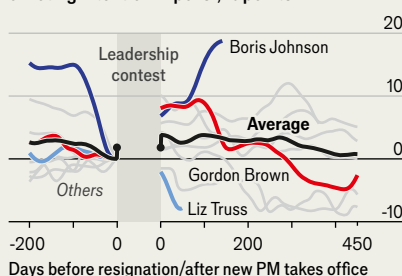
But these gains can be fleeting—on average all but disappearing 400 days into the new leader’s time in office. When Mr Brown succeeded Sir Tony Blair in 2007, Labour’s poll numbers rose by nine points. A year later they had collapsed to five points below when Sir Tony left office.

There are already warning signs for Mr Burnham. During the by-election campaign the share of Britons with an unfavourable view of him rose from 30% to 41% in YouGov polls, before ticking down to 38% after his victory (see right-hand chart). In power he will face the same problems as Sir Keir. His popularity could plunge.

Or he could follow the example of another charismatic former mayor: Boris Johnson. In spring 2019 Mrs May was still struggling to persuade MPs to vote for her unpopular Brexit deal. Mr Johnson was chosen to break the deadlock and reunite Tory voters. It worked. Between May 2019, when Mrs May announced her resignation, and the general election in December 2019, the Conservatives’ support rose from a little over 20% to 44%—enough to secure a landslide victory. If Mr Burnham can unite the left, he could repeat that trick. ■

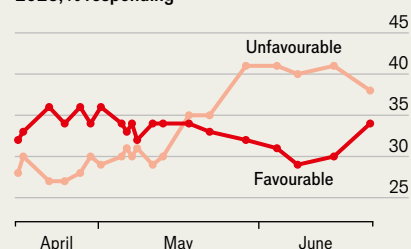
Can he do a Boris?

Change in British governing party’s share of voting intention in polls*, % points



Sources: Pollbase; YouGov; *The Economist*

Andy Burnham, favourability, 2026, % responding



*Between 1955-2022

Another northern town

Meet the Warrington model

WARRINGTON

Never mind Manchester. The really interesting town associated with Andy Burnham lies to its west

ANDY BURNHAM has sailed into Westminster largely on the strength of his performance as mayor of Greater Manchester. As his accent suggests, though, Mr Burnham is not from Manchester. He grew up a few miles to the west, on the edge of a very different northern English town, which deserves to be better known.

Warrington is not a glamorous place. It has no skyscrapers and is small, with about 215,000 inhabitants. Its people like it that way. “They don’t want us to be too big, and be a city,” says Hans Mundry, leader of the Labour-run borough council. Warrington Town Football Club plays in the seventh division of English football (its rugby-league team is far better). In 2015 the Royal Society of Arts ranked the town last out of 325 places for cultural amenities. Angry Warringtonians pointed out, not altogether convincingly, that Britain’s first IKEA furniture store had opened there.

For all that, it is successful. GDP per person is one-third higher than the British average, and has grown faster since 2010 (see chart). The place has prospered by pursuing policies distinct from what Mr Burnham calls “Manchesterism”, a vaguely articulated, leftist approach to economic development which gives the state a big role. Warrington is an alternative model for the next prime minister, and a warning.

Along with more than a dozen other urban areas in England, it is a post-war “new town”. The government in Westminster gave it special planning powers in 1968, assuming that it would build homes to mop up working-class people pushed out of Manchester by slum-clearance projects—much as the new towns around London did. But within a few years Warrington was defiantly going its own way.

Middle-class people converged on it, attracted by its location near three motorways: the M6, M56 and M62. Warrington’s planners pivoted in response. Rather than erecting large social-housing estates, as originally envisaged, they accepted that private homebuilders would supply most homes. They frowned on underpasses and overpasses, symbols of modernity adored by the architects of other new towns like Milton Keynes and Stevenage.

The instinct to work with market forces is still detectable among Warrington’s leaders. Mr Mundry argues that the town is quick to move on when old industries decline (it was once a wire-making power-



house). A few years ago it used money from the developer of a warehouse park not to create prettier public spaces—a common use of such funds—but to set up a bus route to take workers to the warehouses.

Ian Cox, who runs the Golden Square shopping centre, says that the town is “ridiculously easy to escape from” because of its transport links to Liverpool and Manchester. Competition from those cities, plus online shopping, mean that people need a reason to visit the town centre. His solution has been to put on gigs and events. Local councils can stymie such initiatives, but Mr Cox says that Warrington has been simple to deal with—more like a company than a public body.

Today engineering drives the economy. Birchwood Park, on the north-east edge of Warrington, churned out bombs during the second world war. It now crawls with PhDs, many working on nuclear energy. At



Amentum, an American firm, engineers show off a device that can trundle through sewers, looking for flaws. Other robots are intended for more dangerous environments such as nuclear-storage tanks. “We have 270 vacancies,” says Loren Jones, the head of Amentum’s energy and decommissioning business outside North America.

Unfortunately, in the past few years Warrington has been rather too keen on business for its own good. In the 2010s it went on an investment spree, buying Birchwood Park, investing in a challenger bank and pouring money into solar farms. By 2023 Warrington’s debts amounted to £1.8bn (then \$2.2bn), a colossal sum for a town of its size. It looked like a fine strategy when borrowing was cheap. But that is no longer the case. And last year one of the solar parks, near Cirencester in the Cotswolds, caught fire.

When outside experts scrutinised the council’s investments, they found that most had performed poorly. Earlier this year Warrington asked the central government for emergency help to stave off bankruptcy. It has raised council tax (levied on households) by 7.5%, more than local governments are usually permitted to without calling a referendum. Birchwood Park and other assets will be sold.

Other hazards loom. Warrington is surrounded by green-belt land, on which building is difficult. It has been able to take large bites out of the belt. But local tolerance for sprawl may be eroding. Stuart Mann, an independent councillor and self-described “pain in the arse”, argues that too little infrastructure is being built for the new residents and businesses, and that Warrington feels ever less distinct from Greater Manchester and Merseyside.

Lessons for Burnham

Over the next few years politics could spoil Warrington’s run. The first hurdle is the race to succeed Mr Burnham as mayor of Greater Manchester. If Reform UK, a right-wing populist party, wins, it could harm the town. Reform’s national leadership firmly opposes a scheme known as Northern Powerhouse Rail, which would improve Warrington’s railway links. Next year the town will join a combined authority with largely rural Cheshire, which may or may not work well. In 2028 local voters will get a chance to punish the council for raising their taxes.

Warrington’s most famous son can learn from the town’s successes and its troubles. Creating an environment in which business can thrive is wise (Mr Burnham should know this anyway: despite talk of statist “Manchesterism”, Manchester has succeeded partly because since the 1990s it has allowed property developers to build residential towers). But he needs to keep an eye on the debt. ■

BAGEHOT

Forget Andy. It's Ed's Britain

Ed Miliband is the Copernican centre of British politics



ON MONDAY MORNING, another bout of history happened. At a podium on Downing Street, a tearful Sir Keir Starmer became the latest prime minister to quit mid-term. On the West Coast Mainline, a train carrying Andy Burnham, Sir Keir Starmer's inevitable replacement, hurtled from Manchester to London, like Lenin on an Avanti Pendolino. But inside Labour, the changes at 10 Downing Street provoked less chatter than who would occupy 11 Downing Street as chancellor. There are two candidates: Ed Miliband and Not Ed Miliband.

British politics has undergone a Copernican revolution: it now revolves around Mr Miliband. Research spelling out how grumpy fund managers would be about Mr Miliband was pushed out by people who want Wes Streeting, a former health secretary from the party's right, or "anyone but Ed". During an interview at ARC, a right-wing conference, the Conservative leader, Kemi Badenoch, mentioned Mr Miliband more times than she did Mr Burnham, the future prime minister. "DON'T PUT ECONOMY AT MERCY OF 'RED ED'", screamed the *Daily Mail*. The past, present and, perhaps, the future of British politics orbit Mr Miliband.

If politics in Britain is but the actions of individuals, then Mr Miliband's decisions have shaped the country more than most. He stood against his older brother David, a former foreign secretary who was less left-wing and more photogenic, in the fight for the Labour leadership, beginning the party's leftward drift. His decision to switch the party into a one-man, one-vote organisation, rather than a complex intersection of MPs, unions and party members, paved the way for the leftist Jeremy Corbyn to succeed him.

Mr Miliband's tenure at the helm of the party was an electoral failure but, in the long term, a political success. Within a few years the interventionist, fussy agenda that Mr Miliband advocated turned from a manifesto rejected by voters into a cross-party consensus. Skim through Mr Miliband's Labour manifesto of 2015 and it is a premonition of Britain in 2026: a world of price caps on energy bills, strict regulation of landlords and a higher minimum wage. Bizarrely, it was the previous Conservative government that put many of these policies in place. In this period, a meme emerged: "Ed Miliband is the prime minister".

In many ways, he still is. Nowhere is this more true than on "net zero". It was Mr Miliband who committed Britain to binding carbon budgets, while energy secretary in 2008 under Gordon Brown's Labour government. Although now a *bête noire* of the Tory party, "net zero" was put into law by the Conservative government, nodded through as a sort of going-away present for Theresa May, who had agreed to resign as prime minister: a card, a few well-wishes and some consequential climate legislation.

And so now, as the cabinet minister responsible for the climate, Mr Miliband dominates the present. Such is his gravitational pull, "net zero", which should have been but one part of Labour's stint in office, took on an outsize importance. The soft checks and balances of the British system, namely other cabinet ministers, proved inadequate for the simple reason that Mr Miliband knew precisely what he was doing, unlike most of his peers. Complaints from businesses about high energy prices were not heard. Quickly, no Googling, who was the business secretary for Labour's first year in office? Next round: who is the current one?

The result of this lopsided government is that Britain is stuck with decisions Mr Miliband has made. For the moment, Britain has one of the strongest records of decarbonisation among big, rich countries and, at the same time, the highest industrial energy costs. If gas prices stay high and volatile, Mr Miliband's plan may look reasonable; if not, it could prove an expensive error. Britain is hurtling towards a largely carbon-free energy grid by the 2030s and Mr Miliband is destined to be either visionary or villain.

What to do with the man who sits at the centre of Britain's political solar system is the biggest looming decision Mr Burnham will have to make. A frantic operation is afoot to stop Mr Miliband becoming chancellor. His own colleagues hint darkly at bond-market tantrums should he be appointed. For all that Ms Badenoch likes to take aim at Mr Miliband, her feelings are nothing compared with the bile aimed at him from people within Labour. In one quote that still rattles around Planet Miliband, an anonymous Labour sort declared that Mr Miliband was "the only minister who really knows how to work the system". Before adding: "And Ed is the one minister we don't want to be a success if we want to win the next election."

I'm with the Miliband

Therein lies the dilemma for Mr Burnham on what to do with Mr Miliband. If being chancellor were a normal job, Mr Miliband would be the foremost candidate. He worked in the Treasury for eight years. Old hands, such as Nick Macpherson, the former permanent secretary of the Treasury during the austerity years, offer glowing references. Even phrenologists approve: he has "the biggest economics brain", claimed one ally, presumably snapping a pair of callipers together. Appointing Mr Miliband as chancellor would be bold. But it would turn the new prime minister into merely another celestial body in Mr Miliband's orbit.

Mr Burnham defines Manchesterism, his ruling creed, as business-friendly socialism. But which comes first? Installing an instinctive Blairite, such as Mr Streeting, may in the short term allay concerns from the more hysterical quarters of business (even if they would bleat about the realities of any chancellor from the centre-left). Other names are floated—Yvette Cooper anyone?—simply because they are not Mr Miliband. For now, British politics is stuck in Mr Miliband's gravitational pull. It is up to Mr Burnham to determine if he wants to break free. ■

International



Not adding up

Class struggle

Some college and university students test no better than ten-year-olds

IN RECENT weeks more than 1,800 maths and science lecturers at the University of California, one of America's largest and best public university systems, have signed an open letter detailing a tricky problem. First-year undergraduates, they say, are increasingly arriving without the basic skills they need to succeed. At the campus in Berkeley, they write, some 20-30% of students taking an early calculus course turn up displaying "severe preparation deficits". The challenge has become so great, they add, that instructors are having to re-teach middle-school mathematics.

The letter is the latest entry in a widening debate about post-secondary education, university admissions practices—and the acuity of America's young. It follows a gobsmacking report in November at the California system's San Diego campus. Academics there noted that the number of first-year students entering with maths skills below high-school level had in-

creased nearly thirtyfold in five years, to almost one in eight. Some 70% of the lagging students, they argued, were not performing at the level expected of a 14-year-old.

Worries about undergraduate maths skills pair with long-running dismay at falling levels of literacy. Lecturers warn of literature students who seem incapable of finishing books. It is not just on the west coast, or in public universities, where these problems are reported. At Harvard some humanities and social-science professors say they feel compelled to shorten texts, according to a report released to faculty in October. Students are arriving at America's most famous university "with less experience reading complex prose and less capacity for focus and sustained attention."

→ ALSO IN THIS SECTION

— The Telegram is away

They "struggle with readings that students completed with ease just ten years ago".

Professors have been griping about students for as long as there have been ivory towers. Backing up anecdotes with broad-based data can be hard. After secondary school, learners very rarely sit nationally or regionally standardised exams. Fact-finders thus rely a lot on spotty reports from faculty, such as those in California, who feel compelled to raise the alarm.

But observers seeking a sense of global trends—and America's place within them—need not fly completely blind. At least some insight may be wrung from a once-a-decade test carried out by analysts at the OECD, a club of mostly rich countries. Its "Survey of Adult Skills" aims to gauge how far citizens in dozens of countries have the literacy and numeracy necessary to thrive in the real world.

At their simplest, its tests find out how well people can make sense of the instructions on a pill bottle, or work out how much wallpaper they must buy to redecorate a room. At more advanced levels, they explore how well people can draw sound conclusions from complex analysis and charts. Test-takers are divided into five levels of acuity, in each discipline. Level 1 ought to be achievable by a rich-world pupil at the end of primary school, says Andreas Schleicher of the OECD.



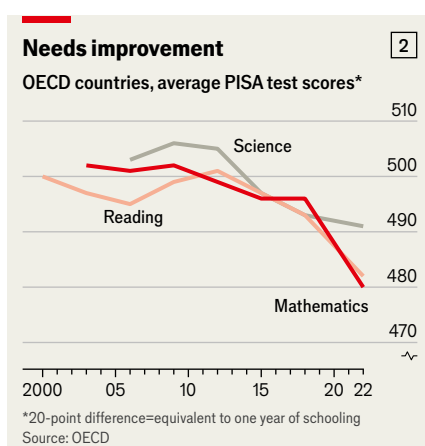
Some 160,000 people of all ages were tested in the last round (the results of which were published at the end of 2024). *The Economist* asked the OECD for data just of those under 35 who were enrolled in “tertiary” education at the time they took the tests. That includes students in all universities along with learners in most kinds of colleges (but only those taking courses that are in theory more advanced than are offered in high schools).

Many of them do very well—but a striking share perform abysmally (see chart 1). Across rich countries some 8% of students in tertiary education notch up a score in literacy no better than one might expect from a ten-year-old child. The share is about the same for numeracy. Worse, the share at or below this bar has risen since the tests were last run, a little over a decade ago. The share of very poor performers in literacy has more than doubled.

Scores at the individual-country level vary widely. In Estonia fewer than 2% of tertiary students score at or beneath the bottom bar. That rises to a fifth in Poland (for literacy) and a quarter in Chile (for maths). Brits can feel reasonably sanguine, despite their growing public disdain for academia; results for its students are better than average, and improving. America's scores, by contrast, are among the most disappointing. One in seven of its tertiary students scored at or below primary-school level in the literacy tests, up from about one in twenty a decade ago. The share at or below the bottom level for numeracy, meanwhile, was almost one in five.

Every child left behind

What is going on? In part, colleges and universities are inheriting problems that have originated in the world's schools. One cannot overestimate the impact of the pandemic. Countries enforced national school closures lasting 20 weeks on average. Rota systems for in-person learning, and quarantines for “close contacts”, then disrupted lessons more. In the years immediately after that disaster it was as if some stu-



dents “had not gone to high school”, says Jessica Hooten Wilson, a professor at Pepperdine University in California. “It was actually a very scary thing to see.”

Yet in many places schooling was already going backwards when the mega-lurgy hit. Scores in NAEP, America's national reference test, reached a peak in the early 2010s and have been edging down since. Scores from PISA, an international exam taken by 15-year-olds, are drifting the same way in a swathe of other countries (see chart 2). Places with unusually deep, long-running declines include France, Germany, the Netherlands and New Zealand.

The causes of these trends are hotly debated. Rising migration is relevant: newcomers tend to be poorer than native-born students, and more likely to speak a foreign language at home. Meanwhile traditionalists accuse school reformers of watering down testing and accountability schemes. And of replacing time-tested syllabi with faddish curriculums that downplay learning concrete facts in favour of nice-sounding but vapid “soft” skills.

Claims that social media have been “rewiring” children's brains bear strong echoes of past panics, such as over television and computer games. But there is not much doubt that screens of all kinds have displaced more nourishing hobbies: the share of nine-year-olds in America who say they read books for pleasure has fallen from nearly 60% in the 1990s to 37% now. Indeed, it is not just school pupils or university students that are seeing declines in literacy: the OECD's tests also find this trend among older populations, notes Mr Schleicher, perhaps because people are getting less practice than in the past reading long and complex texts.

Yet colleges and universities that claim to be merely passive observers of this are marking their own homework. In many countries they enjoy broad control of their own admissions policies. They have frequently failed to use these freedoms to hold standards high.

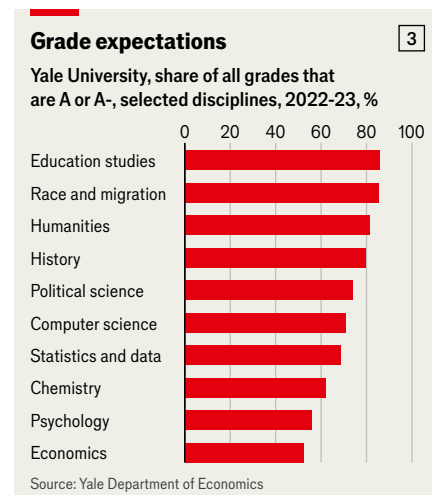
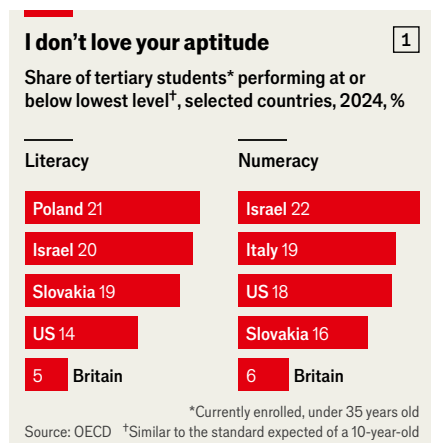
For decades critics have accused col-

lege and university officials of lowering entry criteria to capitalise on growing demand. Nowadays the dynamics are a bit different: in some rich countries the number of 18-year-olds is nearing or past its peak. Administrators may find it even harder to resist watering down standards when the alternative is to downsize. Indeed, comparing the OECD's data on students' skills with the changing number of students in tertiary systems throws up a correlation that deserves further study: shrinking systems are especially likely to have collected lots of students who score in the lowest levels of those tests.

Falling achievement in schools has mostly been driven by children who already ranked in the bottom half of their classes, not by clever clogs at the top. So the drip of unprepared students into some of America's best universities demands additional explanation. The irate academics in California, and in many other parts of the country, blame it on the scrapping of entry tests. Before the pandemic more than half of bachelors-granting universities in America required applicants to sit tests of numerical and verbal reasoning—usually the SAT or ACT (these tests help substitute for the standardised exams that exist in many other countries). Now it is as few as 10%.

In the depths of covid American universities argued that it would be impossible to sit these tests safely. But they were also inspired by claims that the exams are biased against black and Latino students, who have tended to do worse on them than average. Cynics say that canning them has made it easier for administrators to keep moulding the ethnic make-up of their campuses in whatever fashion they decide is just—despite a Supreme Court ruling in 2023 that outlawed race-based affirmative action. For institutions down the food chain, binning the tests probably simplified the more pressing task of simply making sure there are enough bums on seats.

All this has made America's admissions ►►



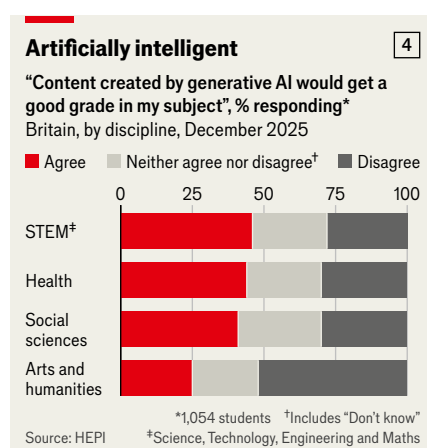
officers depend more on alternative signals that are growing only less reliable. Application essays were already often written by parents and teachers but are now worth next to nothing, given how easily they can be whipped up using AI, says Mina Aganagic, a maths professor at Berkeley and one author of the open letter. As for high-school grades, they have been inflating fast. In recent years many American states have lowered the thresholds high-schoolers must cross to earn graduation certificates. About a quarter of all the students whom professors in San Diego have been dispatching to their weakest remedial maths class had notched up perfect scores in maths in their last years at school. Admissions is becoming a “black box”, says Professor Aganagic. “I think most people will agree that selecting students at random does not serve anyone.”

A major question for colleges and universities is not just how they will respond to a growing pool of underprepared applicants, but whether they are themselves prepared to continue enforcing high expectations. The open letter in California warns that with weaker students has come “growing pressure to dilute quantitative rigour”. In Britain big national exams somewhat blunt grade inflation in secondary schools, but that is not the case in higher education. Though grades have fallen somewhat from a pandemic high, in 2025 some 30% of bachelors’ students in Britain got a first-class degree, up from 7% in 1995.

At Yale 79% of grades were an A or A- in 2022-23, compared with 67% in 2010-11. The rate was lowest in economics courses, at about 50%; it was above 80% for humanities, ethnic and education studies, among others (see chart 3 on previous page). In April senior academics at Yale said, in an essay exploring why faith in higher education has been declining, that “decades of inflation and compression have rendered the college grading system almost meaningless as an academic measure”.

A report issued last year by the dean of undergraduate studies at Harvard contains an admirably frank summary of the pressures driving grade inflation, drawn from conversations with faculty. Students who have never once received a mediocre mark during their school days have become more confident about appealing less-than-perfect grades at university. Academics at Harvard fear students will avoid courses run by tough-markers, and that this might hurt their own careers. They perceive that managers are taking more notice of what students write on end-of-course evaluation forms. That provides further incentive to try to keep the punters happy.

For a decade Harvard has been “exhorting faculty to remember that some students arrive less prepared for college than others, that some are struggling with diffi-



cult family situations or other challenges...and nearly all are suffering from stress”, according to the report. Academics who grade them badly have not always been certain the university would “have their back”. Changing fashions in teaching also play a role. Group projects are more difficult than exams to mark objectively. Some lecturers even cite interest in concepts like “ungrading” or “contract-based learning” in which students earn A-grades for completing all assigned work.

CheatGPT

On top of all this now lands AI, which appears to be enabling rampant cheating. Some 94% of undergraduates in Britain report using AI to help with assessed work, according to a survey released in March by HEPI, a think-tank. Some 12% admitted pasting AI-generated text directly into coursework, up from 3% in 2024. Nearly half of students in STEM (science, technology, engineering and mathematics)—and a quarter of humanities students—judged that AI-generated content would “get a

good grade” in their subject (see chart 4).

In America during the 2023-24 academic year some two-thirds of learners in public universities were using AI, and an estimated 9% of those were using it to cheat, according to research published in May. The rate of deception was highest—gallingly—for students of economics (17%) and journalism (16%). The numbers are surely much higher now, says Igor Chirikov of Berkeley, one of the study’s authors.

For now, the cheating pays off. In a second study published as a working paper last month Dr Chirikov analysed 500,000 grades awarded by a big (unnamed) university in Texas between 2018 to 2025. He found the number of top marks being handed out in courses that involve skills AI is good at, such as writing and coding, has soared since ChatGPT launched in late 2022. The share of “A” grades in these subjects has risen 13 percentage points, or about 30% from the baseline. He finds no similar inflation in subjects for which talking robots are probably not very useful.

Academics have little faith in tools that claim to detect AI-written papers, and lots to lose from hurling accusations of fraud. Says Dr Chirikov: “You can’t ban a tool that you are also expected to teach.” Many lecturers are reintroducing invigilated assessments (during covid, open-book tests that students may take home and complete within a 24-hour period became popular). But that can meet resistance from administrators who must find the space and staff for properly proctored exams.

For some, AI has induced a sense of resignation. It is growing common to hear academics shrug that perhaps students no longer need strong basic skills, because so much of the work they do in future will involve tweaking things made by AI. That is not pragmatism; it is surrender. ■



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Business



Artificial intelligence

Do not compute

NEW ALBANY, OHIO

America's data-centre backlash is putting the AI boom at risk

FROM THE top of the slide, which curves into a little backyard pool, the view as recently as April would have been of lush Ohio farmland, dense forest and pretty clapboard houses. Now it is of six giant weatherproof tents of the type used by the armed forces to house fighter jets or by aid organisers in disaster zones. These will soon contain perhaps \$30bn-worth of cutting-edge semiconductors. Along with a clutch of gas turbines to provide power, they occupy a site the size of an airport terminal. If Meta, the site's owner, stays on track to bring its "Prometheus" data centre online in 2026, it will dedicate an entire gigawatt (GW) of power—the amount needed to power as many as 1m homes, or roughly the output of a large nuclear reactor—to artificial intelligence.

The mammoth data centres of the future, capable of training frontier AI models in 2030, will not be in the urban clusters in

Virginia or California that house most of America's server farms today, but in the emerging "Silicon Heartland" of Michigan, Wisconsin and Ohio, or in southern states like Louisiana, Mississippi and Texas. Vast sums—as much as \$750bn by Amazon, Google, Meta, Microsoft and Oracle, plus billions more by various others—are being ploughed into investments in these places. An estimated \$3trn will go into AI data

centres globally between 2026 and 2030. Much of that is earmarked for America. The money will expand total AI computing capacity, measured in the amount of power consumed by major data centres, from just under 12GW in America currently to as much as five times that amount by the end of the decade (see map on next page). And across the country, people of all political stripes are furious about it.

There is plenty to dislike: the ugliness of the buildings; the roar of generators and cooling systems; the skeleton army of new transmission towers criss-crossing the landscape; the fear of contaminated water. Surveys suggest that Americans would sooner live next to a nuclear plant than a data centre. The issue has surged in salience. Gubernatorial candidates facing voters in November are routinely quizzed on where they stand.

Already local activists are claiming scalp. At least 20 data-centre projects worth \$42bn, which would have used 3.5GW of power, were cancelled in the first three months of 2026 after local pushback; \$85bn-worth have been cancelled over the past three years, including small centres proposed by Amazon and Meta. Residents of Cedar Rapids, Iowa, are resisting Google's plans to build one there. Several townships in Michigan have passed mora- ➤

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► toriums after OpenAI broke ground on a project in Saline despite local opposition.

Yet the resistance is more than simple NIMBYism. In April a survey by Pew Research, a pollster, found that Americans who have merely heard of data centres are just as opposed to them as those who live within five miles of one. Philosophers have long worried that a rogue AI—fixated on a singular goal, like maximising the production of paperclips—would doom humanity by hoovering up all its resources and paving Earth with servers. Sam Altman of OpenAI and Dario Amodei of Anthropic have spent years warning that AI will throw most people out of work or be turned to mass harm. Now the infrastructure they need is arriving on people's doorsteps, and it looks like something out of a war zone. Residents across America are standing up in council meetings begging for projects to be axed in the hope of slowing the technology's progress. Will they succeed?

That question is not just on the mind of the AI industry. "We need to stay a ways ahead of China," says Chris Wright, America's energy secretary, in an interview with *The Economist*. Ensuring America leads in artificial intelligence is "the overriding goal" of his tenure, he says. "We've got to enable these data centres to get permitted and built and [have] turned on power."

Perhaps 1-2GW of America's current data-centre capacity is dedicated to training frontier models across the major providers—Anthropic, OpenAI and Google—as well as those trying to keep up with them, including Meta and xAI. It follows that perhaps 10GW should be available for inference—allowing customers to use the models to ask questions, write code or perform other tasks. Yet after demand for AI tools soared in early 2026, the available "compute" proved woefully inadequate. Anthropic has throttled model usage, OpenAI has scrapped its compute-intensive video tool and Microsoft has repriced its coding assistant so steeply that some programmers are returning to the lost art of writing software themselves.

New data centres are supposed to relieve this pressure. Construction is already under way on big projects that should provide almost 30GW of additional capacity by the end of 2028. Yet the amount needed to make new models has been rising quickly. In a white paper published last year Anthropic argued that as much as 5GW would be required to train a single frontier model by 2028. According to Epoch AI, a research firm, that figure could rise to as much as 16GW by 2030. If true, much of the capacity that will come online in the next few years will be absorbed by training alone.

Some forces will curb the need for ever more capacity. Chips get more efficient over time, generating more computing oomph from the same amount of power.

Crypto-mining facilities are being repurposed for AI. And as Andrew Feldman, the boss of Cerebras, a chipmaker, points out, inference does not require the enormous facilities needed when training a model, meaning it is possible to use portions of existing data centres. Even so, this may not be enough. So far only coding has been truly disrupted by AI. Other industries it might revolutionise—like law, finance or media—are still nascent adopters.

Most of the data-centre projects already under way were approved and started before the backlash reached its present fever pitch. Some—like the OpenAI site in Saline, Michigan—scrapped through by the skin of their teeth: the project was voted down by the council and was able to proceed only because the county had no land zoned for industrial use, which violated "exclusionary" zoning laws.

Even areas that have long embraced data centres have turned against them. In March 2025 Loudoun County in Virginia—known as "data-centre alley"—eliminated rules that made data-centre development easy. New sites now need a "Special Exception" involving public hearings. In Texas, which hosts the second-largest number of data centres after Virginia, the city of San Marcos has passed a moratorium on them.

Permission denied

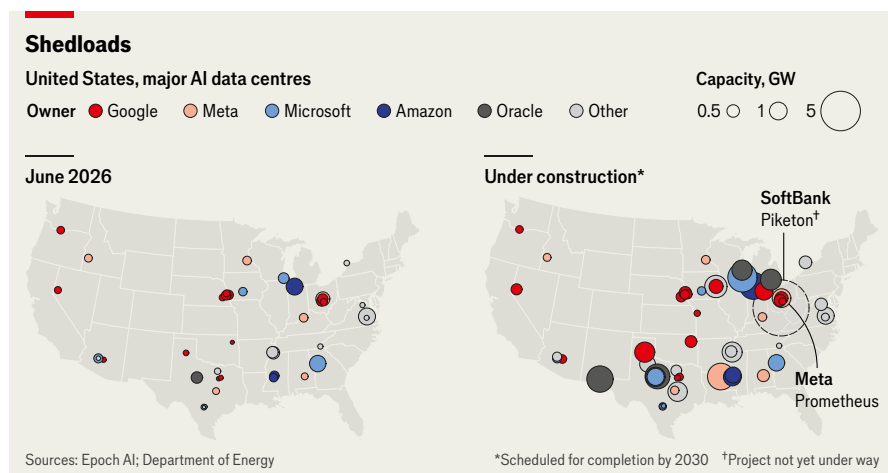
Americans are resisting data centres partly for understandable (if sometimes misguided) concerns about the impact on their neighbourhoods and the environment. For instance, the myth that AI data centres consume vast quantities of water was popularised in 2025 by a book that relied on a serious miscalculation. A mid-sized data centre uses about as much water annually as two golf courses, but far less if it incorporates water-recycling technology, as plenty now do.

Then there are worries over power. SemiAnalysis, another research firm, reckons there are around a terawatt (1,000GW) of large-load grid-connection requests

outstanding across American states, nearly all for data centres. That is equivalent to almost the entire capacity of the American electrical grid, which can produce a maximum of 1,250GW. America's electricity demand averages around 470GW through the year, but peak summer demand can reach 750GW and utilities like to maintain a 15-20% buffer above that. Moreover, only around 975GW of capacity is reliably available on demand.

That has led to concerns over a spike in electricity prices for consumers and other businesses. There is little credible evidence of that happening so far. Growing demand for electricity allows utility companies to spread the cost of upgrades over more users. And because data-centre operators always install a backup source in case their energy supply is interrupted, they can reduce their demand in the event of an extreme event, like a storm. "We want to be good grid citizens," says Alistair Speirs, who leads the data-centre build-out at Microsoft. He adds that the batteries typically hooked up to the hyperscaler's data centres allow it to "choose when to sip and when to slurp from the grid".

Still, the vast investments in data centres planned over the next few years will require America to produce a lot more power—with the resulting infrastructure sure to generate opposition of its own. The Department of Energy expects that by 2030 the country will need to add 50GW of generation to support AI and another 50GW for the manufacturing renaissance the administration expects. "We need to grow our capacity of dispatchable power," says Mr Wright, who is sceptical of wind and solar projects that produce only intermittent energy. In pursuit of this goal he has prevented coal-fired power stations from closing and favours restarting nuclear plants and building more gas-fired ones. Although over a third of data centres are expected to generate all their power on-site by 2030, that will make these projects even more of an eyesore, and the remain-



der will still rely on the grid.

Ohio, now home to the fourth-largest concentration of data centres in America, has been shrewder than most states in its approach. In July last year the state utility commission passed a requirement that each month data-centre operators above a certain size pay for at least 85% of the power capacity they have asked to be made available, even if they do not use it, in order to reassure Ohioans that they will not end up footing the bill for grid investments. This innovation became one of the ideas included in the “ratepayer protection pledge” that tech companies signed in the Oval Office in March. Ohio’s provision is better than the pledge, for it is binding.

Hillbilly uprising

Despite all this, the Buckeye State has struggled to soothe its residents. Three-quarters of Democrats and two-thirds of Republicans in Ohio still oppose local data-centre development. Opposition is so great that, despite Donald Trump winning the state by 11 points in the 2024 presidential election, Vivek Ramaswamy, an AI enthusiast who is running to be Ohio’s governor, is neck and neck in the polls with his Democratic opponent.

The Trump administration has means of bypassing local opposition. In March the Department of Energy announced an enormous 10GW project in Piketon, in rural Ohio, on federal land, thereby avoiding some of the usual permitting processes. The project will be funded by SoftBank, the Japanese conglomerate run by Masayoshi Son, which will build a gas-power plant to supply what would be the world’s largest data centre. “Imagine being in an Appalachian farmer’s field with mud on our shoes with Secretary Lutnick, Secretary Wright, Mr Son and me... and all the other hillbillies!” says Adam Holmes, an Ohio state representative, who attended the groundbreaking.

Piketon’s residents are hardly squeamish: the area was home to America’s nuclear-enrichment programme in the 1950s. Even so, Shane Wilkin, the state senator who represents the district and sits on the Ohio legislature’s data-centre committee, is not having an easy time of it. “We had the water utilities folks in here to testify and I asked him how many data centres in Ohio have a discharge permit? One. And then I asked him a question I didn’t know the answer to—which is always a risk—have they had any violations? And he said two: for late paperwork.” Mr Wilkin says he explains to his constituents that data centres won’t push up power prices, because they bring their own power, and they can’t contaminate the water supply because they don’t discharge their water. But they simply say: “Well, I just don’t want it.” ■



Chinese carmaking

A sucky business

SHANGHAI

Despite ferocious competition, new EV-makers keep appearing in China

NEXT YEAR Dreame, a Chinese business better known for its robo-vacuums, plans to start selling sleek electric vehicles (EVs). It is not the only firm branching out from small household appliances to machines that can carry an entire household. Rox Motor, which started selling electric SUVs in 2023, was also founded by a vacuum tycoon. Xiaomi, which made its name producing cheap smartphones (and also makes vacuums, among other things), began selling sports cars a year later.

Perhaps more surprising is that there are new entrants to the crowded Chinese EV market at all. No fewer than 143 EV brands sold at least one car last year. But 46 of them did not sell more than 1,000, according to AlixPartners, an advisory firm. Even so, 23 new EV brands were launched while just nine were halted.

Admittedly, lots of these are “sub-brands”, created by big companies to differentiate their high-end rides from mass-market models. Geely, China’s third-largest carmaker by sales, has a portfolio of more than ten brands, including Zeekr, Polestar and Lynk & Co.

Only ten Chinese companies managed to sell at least 1m cars apiece in 2025, accounting for 84% of the total. However, that share was down slightly from the year before, suggesting that the consolidation of China’s EV market that industry insiders have long predicted is

yet to take place. Only occasionally does an entire company collapse. Hozon Auto, which made the Neta EV brand, is in the process of doing so. It suddenly stopped paying many of its employees last year and attracted headlines when angry staff cornered the founder at his Shanghai office demanding their wages.

Still, things look rather bleak for the industry in 2026. Overall car sales in China have been falling swiftly. In April they were down by a fifth from a year earlier, marking the seventh straight month of decline. The central government has criticised EV-makers for engaging in a vicious price war since 2023. Prices started rising in May, suggesting that the industry may have started listening to the state. But BYD, the world’s largest EV-maker, attributes this instead to rising costs for components such as chips and batteries, which may mean margins remain under pressure.

As demand falls and prices rise at home, China’s carmakers will become even more reliant on foreign sales, which are advancing in leaps and bounds. In April they were up 80% year on year. Setting up foreign distribution networks is costly and time-consuming. But as pressure on the industry continues to intensify, foreign sales, which tend to have higher margins, will make all the difference. China’s EV-makers are eager to Hoover them up.

Venture capital

Unicorpse

PALO ALTO

A growing herd of zombie unicorns is haunting Silicon Valley

FANTASTIC BEASTS roam Silicon Valley. Locals are familiar with the unicorn, a startup valued at more than \$1bn, the decacorn (\$10bn) and even the hectocorn (\$100bn). Lately, however, another creature has been stalking the land: the “zombie” unicorn, a firm once worth \$1bn-plus but now a mere shadow of its former self. Their spread is keeping venture capitalists (VCs) awake at night.

By May 2026, 332 of the 1,900 unicorns in a database maintained by Ilya Strebulaev of Stanford University had raised money at a valuation at or below their peak (see chart). Of those, 212 were valued at under \$1bn. As many as 383 had disclosed no new funding in the previous three years; 41 of these had lost unicorn status. As Mr Strebulaev points out, data on startups are notoriously noisy. Some unicorns may have raised no money because they did not need it. But others may be struggling to justify earlier mythical valuations. A growing number of unicorns appear to have joined the ranks of the undead.

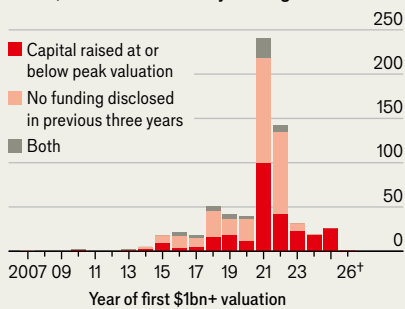
When unicorn-spotting was at its height, plenty of those startups had scant revenue and sketchy business plans. Low interest rates encouraged investors searching for better returns to hand fistfuls of money to VC funds. In 2022 VCs raised a total of \$223bn, according to Pitchbook, a data provider. Interest rates have since risen and fundraising has plunged, to a mere \$66bn last year.

Some businesses seem simply to have been wildly overrated. Cameo, a video site where celebrities give personal greetings for a fee, was valued at \$1bn in 2021 amid a pandemic-era surge in demand. Now it is reckoned to be worth \$82m.

Even for unicorns whose businesses are on a surer footing, the fundraising downturn has punctured their ability to raise more money at their old valuations. Startups typically aim to spend five to ten years growing before listing on the stock market or finding a buyer. Zombies and their backers may find it hard to do either. VC agreements often allow investors to veto an initial public offering (IPO) if they believe they would do better to wait. Later investors who bought in at high valuations are

Silver-blooded and undead

World, selected unicorns by funding status*



*Based on a database of around 1,900 unicorns †To May 1st
Source: Ilya Strebulaev, Stanford University

especially incentivised to block an IPO that might value a startup at a lower amount than they did, in the hope that its prospects may later improve.

In the past the easiest option for startups has been to raise more from VCs. But now these investors are more interested in today's buzzy AI firms than in taking a chance on ageing ones, says Peter Cohan, a tech investor. To persuade them to cough up, startups often have to accept a cut to their valuation, among other unenviable terms. Quora, an online question-and-answer service, was valued at \$1.7bn in 2017. By 2024, it was raising money at a \$500m valuation and seeking to pivot its business toward AI. “Our valuation is lower

than our previous peak, but we are happy to finally be marked to this new market,” its boss wrote in a blog post.

Nearly half of the firms in Mr Strebulaev's database that are now valued at less than their peak raised money in 2021. From late next year their investors will start needing to cash out, reckons PitchBook. It expects net cuts in valuations of between \$500bn and \$1trn as older startups reprice, scramble to find a buyer or go bust. That is quite a cull: PitchBook puts the total value of unicorns at \$8.6trn, or \$5trn once the ten most valuable are excluded.

Falling valuations spell horror for VCs. More recently launched funds have been returning markedly less money to investors than those of earlier vintages, according to research by the World Economic Forum. They have also underperformed the S&P 500 by a wide mark, particularly those that did not invest in a small club of artificial-intelligence superstars, says Mr Cohan. That will make backers like pension funds even more reluctant to fund VC firms outside the top ranks: already just 5% of them produce 90% of the industry's profits.

Spooked VCs are experimenting with novel approaches. Some have shifted from conventional funds with predetermined lifespans to continuous ones that mix stakes in private companies with liquid holdings in public ones, allowing them to pay investors periodically. Others have turned to secondary markets for shares in private firms, which allow them to sell partial stakes, though most buyers are looking instead to get their hands on shares in the most in-demand startups.

Some VCs hope that if this year's giga-IPOs of star AI companies are a success, public markets will become more welcoming to tech's lesser lights. It may take more than that to bring the zombie unicorns back to life. ■





Logistics

Delivery targets

UPS is losing ground to FedEx

FOR AMERICA'S biggest delivery companies, the pandemic brought a boom. To fulfil a torrent of orders from people stuck at home in lockdown, FedEx's white lorries and UPS's brown ones kept rumbling along America's highways and their planes crisscrossed its skies.

As demand has cooled, both have taken similar turns, away from low-margin deliveries. In January 2025 UPS said it would halve the business it did with its biggest customer, Amazon, which once brought in 13% of revenue. The tie-up, UPS said, was a drag on profit. FedEx has just spun off FedEx Freight, an underperforming division delivering bulky stuff on pallets, such as car parts. Both are angling for higher-margin, if more exacting, business-to-business work. On June 22nd, for instance, UPS said it would invest nearly \$50m in temperature-controlled facilities suitable for transporting medicines.

Investors are so far happier with what FedEx has delivered. In the past year its market capitalisation has risen by 38%, against just 5% for UPS (see chart). As well as unloading FedEx Freight, it has been merging its ground and air networks, which often duplicated each other: separate vehicles would pick up packages for road and air from the same address. FedEx expects this to save \$2bn by the end of next year. On June 23rd it reported increases in annual revenue and profit, though its operating margin fell slightly, thanks to higher wages and fuel costs.

UPS too has been slashing costs: last year it closed 93 buildings and cut 48,000 staff, reducing costs by \$3.5bn. Another \$3bn is due to be lopped off in 2026. This is partly owing to its extrication from Amazon; it says that by the end of this month it will have cut deliveries for the e-commerce giant by half.

Yet further savings will be hard won. UPS is the largest employer of members of America's Teamsters union. In 2023 it struck a wage deal with the truckers that runs until 2028, with a hefty increase still to come. FedEx's drivers, by contrast, are not unionised but hired as contractors (though its pilots are union members). By agreement with the Teamsters, UPS can shed drivers only if it buys out a maximum of 7,500 of them for \$150,000 each.

Prospects for revenue are questionable too. "It's really hard not to believe that UPS is going to be facing structural volume declines," says Brandon Oglenski of Barclays, a bank. As well as retreating from Amazon, UPS faces fierce competition from FedEx in the business-to-business market, where UPS's volumes fell by 5% year on year in the latest quarter. FedEx, by contrast, has said that its business-to-business activities are performing healthily. Its total revenue has now caught up with that of UPS.

Nor has Amazon gone away. It is a threat to both UPS and FedEx. It already delivers more parcels in America than anyone else. When in early May it declared that it would expand its own logistics business, making deliveries for companies of all sorts, the share prices of FedEx and UPS dipped by 10%. Both have since largely recovered, but the danger lurks.

The recent record may unduly flatter FedEx. Bruce Chan of Stifel Institutional, an investment bank, points out that the merger of its air and road networks, though a big undertaking, was "low-hanging fruit": UPS has no similar glaring redundancies. UPS promises that the second half of this year will be a turning point: revenue and operating profit will start growing again. But it has a long, hard road ahead. ■



EasyJet

Hard Landing

One of Europe's largest low-cost airlines may be heading for a break-up

A BUDGET AIRLINE'S purpose is to offer cheap air travel. But snapping one up for a bargain is another matter. On June 22nd, after weeks of rumoured interest, Castlake, an American investment firm, went public with its offer to buy easyJet, a European low-cost carrier (LCC), for £4.7bn (\$6.2bn), revealing that two lower offers had already been rebuffed. EasyJet accused Castlake of trying to buy the airline "on the cheap". It rejected the bid, and another, for £4.9bn, on June 25th. Yet it has said it will now start talks with its suitor.

The airline business is tough. Fierce competition means generally meagre profits, making unforeseen upheavals—such as the soaring cost of jet fuel resulting from America's conflict with Iran—especially painful. Since the start of the year IATA, a trade body, has cut in half its forecasts for the global industry's overall net profits and net margins, to \$23bn and around 2%. Bankruptcies are frequent, such as the demise of Spirit, an American LCC, which went out of business in May.

The past financial performance of easyJet makes for pleasant reading against such a disagreeable background. Over the two financial years to September 2025 its annual revenue increased by around a quarter, to just over £10bn, with operating profits rising from £476m to £703m. More recently, however, the war in the Gulf has hit the airline hard. It relies heavily on European leisure travellers, many of whom have delayed booking holidays, forcing the airline to keep ticket prices low. The result is that Barclays, a bank, reckons on operating profits of just £107m this year. That is why, prior to rumours of Castlake's interest, easyJet's share price was down by nearly a quarter from the start of the year.

Castlake may not be looking to hold on to easyJet in its current form. It led a consortium that rescued SAS, a Scandinavian carrier, from bankruptcy in 2023, but has agreed to sell out to Air France-KLM, a partner in the deal, suggesting it is not seeking long-term ownership of an airline. It has other means of making money from aviation, and may be more interested in the "value trapped within easyJet", says Alex Irving of Bernstein, a broker.

Castlake's leasing business manages hundreds of planes. The prospect of adding easyJet's owned fleet of around 200 Airbus short-haul aircraft to its total may be motivating its acquisition attempt. The ►►

▶ airline's order book of 290 additional Airbus planes for delivery over the next eight years would also add to the appeal. These are valuable assets, given Airbus's huge backlog of over 9,000 planes and perhaps a ten-year wait for new orders.

Castlelake may spy other opportunities too. Over the years easyJet has diverged from the ultra-low-cost strategies pursued by rivals such as Ryanair and Wizz Air by operating from more mainstream airports. As a result, it has amassed a portfolio of

valuable landing slots that would appeal to Europe's legacy carriers.

The airline has also built a thriving holiday business. Launched in 2019 to make use of a website that serves over 100m customers, some of whom may be interested in a package holiday, this business now accounts for over a third of pre-tax profits. Although it is hard to know what it would be worth if spun off, other European carriers with holiday businesses, such as British Airways or Jet2 may be interested.

EasyJet's investors may well be tempted to accept an improved offer from Castlelake. Rumours suggest that raising the current one of £6.50 a share to anything over £7 might be enough. But Castlelake could also walk away. As Mr Irving of Bernstein notes, a break-up that could take a couple of years would be a "full-on task". In a business as risky as aviation, where another big setback is as certain as it is unpredictable, dismantling easyJet for parts would be a brave manoeuvre. ■

BARTLEBY *Watch and learn*

Teaching AI how people work is fraught with problems

ARTIFICIAL INTELLIGENCE works best within companies when it knows the context in which it is operating. Some of that context is easy to codify upfront, in explicit rules and guidelines. Some of it can be captured by analysing raw data: Celonis, a German software firm, ingests information from enterprise systems to see how processes like invoices or procurement actually unfold within organisations.

But some information is harder to pass on, and none more so than tacit knowledge, the know-how born of experience and intuition. Michael Polanyi, a philosopher, famously distilled tacit knowledge thus: "We can know more than we can tell." How can AI learn about work when even humans cannot articulate why they do things a certain way?

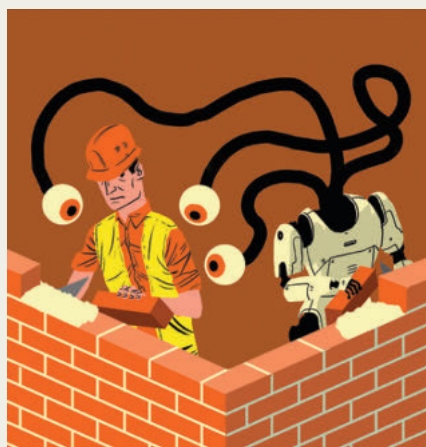
One answer is that they don't need to. Tacit knowledge is already embodied in the data that machines are trained on: a corpus of marketing copy for a specific brand, say, encapsulates organisational judgment and expertise that models can soak up. And one of AI's defining characteristics, says Enrique Ide of IESE, a Spanish business school, is its ability to identify patterns in data that humans cannot describe. From facial recognition to chess, AI has proved very handy at mastering activities that it was never explicitly taught.

Sometimes, however, the details of a process matter. A brick wall encapsulates bricklayers' tacit knowledge, for example, but staring at loads of them won't tell you how best to build one. Monumental AI, a Dutch startup that uses AI-powered robots to lay bricks, interviewed masons as part of its early research into how to design its machines. But their answers tended to be maddeningly vague things like: "That's the way I've

always done it." The bricklayers knew more than they could tell.

What they could not articulate for themselves, hours of video footage could reveal. Monumental saw, for instance, that masons were vibrating their hands a little bit when they pushed a brick into the mortar. This small movement helps push mortar into the pores of the bricks, creating a stronger adhesive bond, which is why Monumental's robots were then designed to replicate it.

An obvious answer, therefore, is to track how people work in ever finer detail. A few functions are already closely monitored. A customer-service call centre naturally throws off lots of relevant information: calls are usually recorded as a matter of course, and it is normal for agents' screens to be monitored as well. AI could be let loose on much more data. A survey of American employees by Danielle Li of the Massachusetts Institute of Technology and her co-authors found that workers believe they are sitting on large amounts of uncoded knowledge about their organisations.



But more intensive monitoring of the specific actions of individual workers is also sensitive terrain. Meta's employees have reacted noisily to the firm's Model Capability Initiative, a programme to track keystrokes and mouse clicks in order to train AI. Ms Li's survey found that workers think they have the ability to withhold a lot of valuable information from nosy employers. They may well choose to do so. In an experiment, the researchers offered to buy survey data from participants; those who had been shown a video on how data could be used to train AI were less willing to sell.

Even the most intrusive monitoring regime would struggle to capture what is going on in people's heads: what they know about clients' preferences or the thought processes behind certain judgments. So another option is to try and tease out experts' knowledge through evaluation processes. On some tasks, it is easy for an AI model to know how it is getting on: a piece of code either runs or it does not. Other tasks are much fuzzier to verify—judging whether a design is aesthetically appealing, say, or how well a research brief has been carried out. By getting human experts to rate how AI performs on these more subjective criteria, the models can progressively be refined to match their standards.

There's nothing wrong with trying to capture tacit knowledge. Organisations have long worried about losing expertise when old hands depart. But using AI to solve the problem raises a number of tricky questions. Who owns uncoded know-how? How much surveillance is acceptable? And as machines learn more and more, how will that affect the way that people acquire, practise and pass on the expertise that was previously gained through experience?

SCHUMPETER *Consulted*

Silicon Valley has much to learn from the spreadsheet jockeys it despises



THE MOST important question in financial markets can be put thus: will companies like JPMorgan Chase and Walmart find artificial-intelligence models useful enough to buy them at a price such that OpenAI and Anthropic can stay in business? If they do: rock 'n' roll. If not, investors will tire of funnelling capital into the AI model-makers, thereby destroying confidence in the vast construction of data centres, and the semiconductors and gas turbines that power them, and so on into economic oblivion.

Last year the answer was obvious, at least in Silicon Valley. The back-bending labour of office workers in the 21st century would soon go the way of the back-breaking tasks of the 20th. Which is to say it would disappear, but this time into computers rather than offshore: firms would buy tokens, not workers. The messianic confidence of tech bosses on this point has made them unpopular but not yet profitable. Partly it reflects their faith in the technology. But it also reflects a view of office work formed in the early 2020s, when tech firms hired vast numbers of paper-pushers whose emails, dashboards and coffee-chats their “technical” staff despised. Since then the “bullshit jobs” view of the world has simply given way to one about “bullshit tasks”.

Yet the war on white-collar work has progressed far more slowly than techies anticipated. The information that constitutes modern companies is not easily organised and fed into AI models (see *Bartleby*). The answers the models spit out are not easily acted on, either. How, for example, should a firm decide which tasks to automate? When should it use cheap open-source models rather than pricey cutting-edge ones? How should it calculate the returns on its investments? Can it hedge the cost of tokens? Answering these questions is arduous and possibly very lucrative work. Hence the idea—simple, elegant and brilliantly ironic—that model-makers should become much more involved in how their inventions are used. The bullish case works like this: companies get more out of AI; model-makers solve their existential cashflow problems; the world economy marches happily on.

To destroy the consultant, then, become the consultant. Each of the AI labs has announced partnerships with big professional-services firms. OpenAI recently bought a small British tech con-

sultancy. In May Anthropic and OpenAI both set up joint ventures with big private-equity houses, presumably hoping that AI models could turn around many software firms languishing in their funds. That private equity cannot sell what it owns is an existential concern for the industry; if things are as bleak even after the attention of the AI labs, it will be a concern for everyone.

Bodies must be thrown at the problem, but how many and by whom? When big tech firms built their cloud-computing businesses in the 2010s, much of the grunt work was outsourced to Accenture, a multinational consultancy which grew enormous implementing the innovations of giants like Microsoft and Google in its client companies the world over. It has warm bodies in abundance, which it describes in the impenetrable language of the corporate central planner: 30,000 workers to be trained on Claude in “a major investment in talent, solutions, and go-to-market muscle”, it declared in December (just a small detachment of Accenture’s total corps of nearly 800,000 souls).

The market does not believe Accenture will play the role of a cash-generating middleman this time. Its shares have lost more than half of their value this year. Perhaps investors are right. That said, snobbery is an underappreciated aspect of what firms choose to bring in house and what they leave in the marketplace: some at the top of the AI labs think a corporate culture that prizes individual genius is incompatible with hiring many thousands of unremarkable workers to plug their products into the economy. (Banks manage to combine star traders with huge back offices, though AI types are likely to find this analogy insulting, too.)

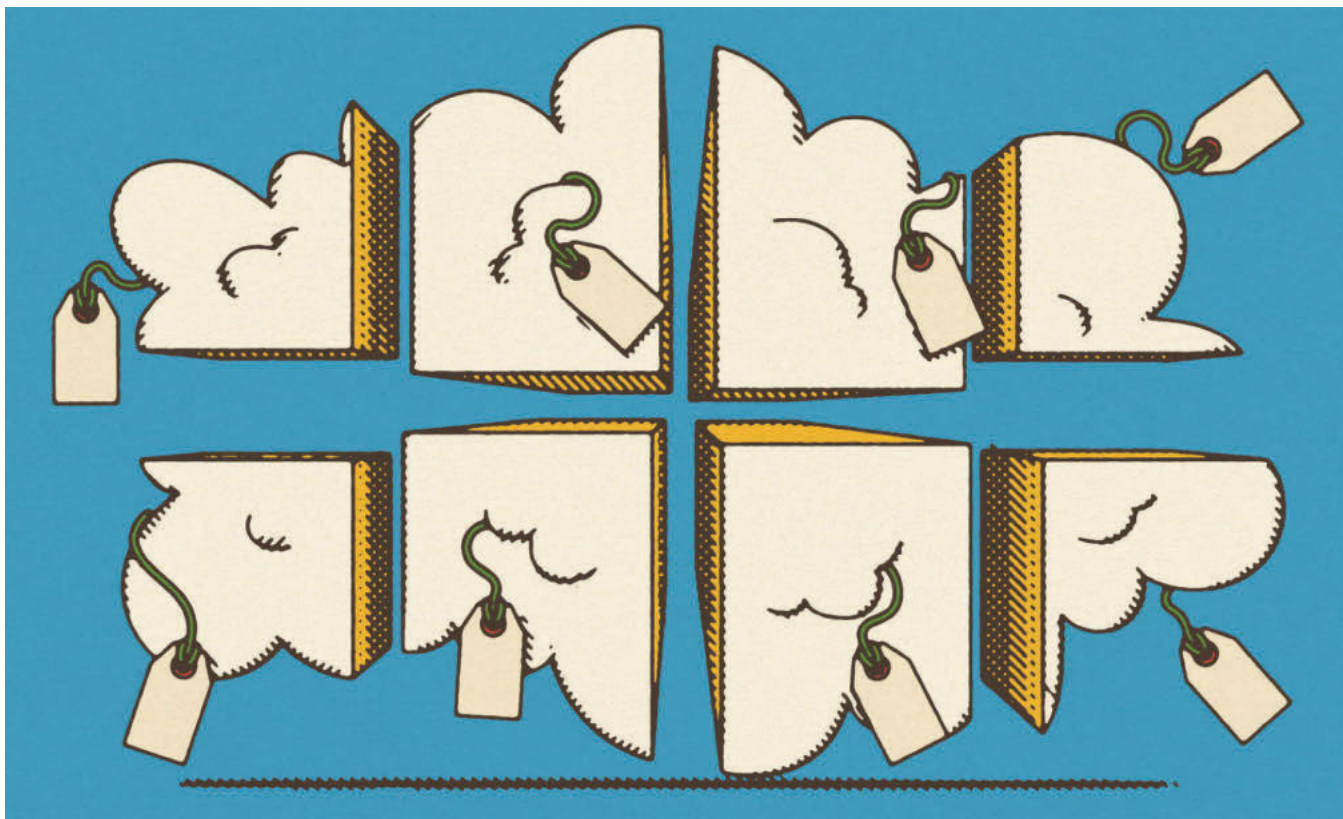
Palantir is closer to what OpenAI and Anthropic might eventually build—and that firm’s share price has also collapsed recently at the thought. The crux of Palantir’s business is sending consultants into firms, sorting their data out and displaying it in a way that makes a middle manager feel as if he works at GCHQ. It is an impressive company, but also an impressive linguist. In its hands a firm’s data become its “ontology”. Consultants are elevated to the status of “forward-deployed engineers”. What distinguishes them from Accenture’s new “reinvention-deployed engineers” or OpenAI’s “forward-deployed experts”? Who knows, but pity any consultant who remains backward and unemployed.

Which firms strike it rich telling companies how to use AI models depends to a large extent on whether model-makers develop a better understanding of the mind-numbing nuances of modern business than incumbent consultancies do of AI. Yet so far nothing the AI firms have done indicates the slightest understanding of how ordinary firms operate. Perhaps that is not surprising for an industry whose staff pride themselves on being incompatible with the bureaucratic corporate life.

Model students

Whether or not Silicon Valley’s AI labs become white-collar advisers, they have a great deal to learn from studying them. In investment banks, they will find instructive case studies on being hated and dealing with systemic risks: worryingly, popular mistrust is hard to dislodge and regulation generally only follows disaster. In elite law firms, they will observe the extraordinary longevity of small and focused companies. And in consultancies, there are lessons on expanding too much, as top firms like McKinsey have done in recent years, but also on the perils of making bold predictions about the future of capitalism. That is not a warning the masters of AI are likely to heed. ■

Finance & economics



Finance and compute

The futures of AI

NEW YORK

How to turn processing power into financial assets

IN 2014, AS cloud computing was taking off, a group of German technologists and financial-platform operators had an idea. They launched the Deutsche Börse Cloud Exchange, on which firms could buy and sell spare access to computing power.

The effort to turn processing power into a liquid asset did not scale well, according to Randolph Roth, its chief executive and veteran of the exchange business. Standardising central processing units (CPUs), the general-purpose chips that predominated at the time, proved too difficult and integrating different types of cloud capacity too expensive. Hopes that futures and options would follow in the wake of the original launch were dashed. The venture shut down in 2016.

But Mr Roth and his partners were early to an idea whose time really may have come. Businesses around the world spent \$129bn on cloud services in the first quarter of 2026, estimates Synergy, a

research firm, more than would have been spent in a whole year a decade ago. Immense investment in data centres is under way. America's five cloud "hyperscale" giants are poised to devote around \$700bn to capital expenditure this year.

The hottest part of the market is not for CPUs but for the graphics-processing units (GPUs) used to train and run AI models. Some businesses now spend more on artificial intelligence, and so on number-crunching, than on wages for employees.

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"Compute", once a piece of techno-jargon for processing power, has entered common parlance. Those who sell it want to turn their assets into cashflows. Those who buy it want to hedge against sharp movements in the prices of a costly, critical input. And middlemen are racing to create financial instruments to match the two groups.

Companies that own and operate data centres certainly want to wring more out of their hardware. Some "neocloud" firms, which focus on high-end GPUs for AI, have used the value of the chips as collateral for loans. In 2024 Macquarie, an Australian investment bank, helped organise a \$500m loan to Lambda, a neocloud, with the loan backed by the borrower's valuable chips from Nvidia, which furnishes most of the world's AI silicon. The same bank approved a GPU-backed loan to Fluidstack, another neocloud, in April last year for an undisclosed amount.

Debt backed solely by GPUs remains rare, even for neoclouds with masses of chips. Some of what looks like GPU-backed lending is actually more pedestrian. Core-Weave, the largest neocloud, runs 49 data centres. Its compute-related power capacity is one gigawatt, around the same as a medium-sized nuclear reactor. Since 2023 the company has used chips as part of the collateral for a series of loans from alternative-asset managers such as Blackstone ►►

► and Magnetar Capital. These are backed by chips but also guarantees from CoreWeave's clients with multi-year contracts for cloud capacity.

The guarantees let lenders take comfort that CoreWeave can pay back its debts while sparing them from having to think too hard about how fast its GPUs will lose their value. This funding structure is also reassuringly familiar, even if the technology is not: it is common in toll roads and other old-economy infrastructure finance.

Although such instruments serve big compute-sellers well, big buyers need something different. Locking in a price with a multi-year neocloud contract insures a buyer against compute getting more expensive but not against it getting much cheaper. So as businesses around the world spend ever more on compute, they want to be able to hedge against price volatility just as they insure against changes in energy tariffs, interest rates or foreign-exchange movements—ideally in deep and liquid derivatives markets.

Two startups want to help companies do this, by turning nascent indices tracking compute costs into a futures market. Silicon Data, founded in 2024 and backed by DRW, a trading firm, has paired up with CME Group, which operates large derivatives exchanges. Ornn, created by recent graduates of the Massachusetts Institute of Technology and run from a flat rather than an office just a few months ago, has paired up with Intercontinental Exchange, the parent company of the New York Stock Exchange, to do the same. Both aim to launch compute futures later this year, to be traded on their partner exchanges.

Each firm has its own indices tracking the hourly rental price of specific advanced chips such as the H100, Nvidia's workhorse GPU. Silicon Data's series tracks both transactions and quoted prices, which results in a smoother index. Ornn includes only real transactions, which makes its index harder to manipulate but noisier. Silicon Data's price for the H100 is now around 10% higher than Ornn's. Two months ago it

was 40% higher. Neither is likely to reflect more than a fragment of the wider market, most of which is private.

Broadening the appeal of compute derivatives requires standardisation. As Mr Roth discovered a decade ago, this is finicky. Different types of chip are imperfect substitutes. Even the same chips offered by different cloud providers may vary wildly in performance and price. The way in which hundreds or thousands of chips are connected to one another, the environment in which they are housed, and data centres' location all weigh on demand.

Such problems lead some industry insiders to suggest that a well-functioning derivatives market may be a decade away. The index-providers and designers of derivatives retort, reasonably enough, that no existing market was perfectly standardised at inception. Even in deep and liquid ones, some products will stray far from benchmarks—just look at oil prices. In March oil futures in Oman and Dubai, mostly for delivery to Asian buyers, climbed to over \$150 per barrel. At the same time, the price of West Texas Intermediate, the American benchmark, sat below \$100.

There are reasons to be bullish about compute standardisation. Chips from Nvidia's Hopper range, which includes the H100, have held much of their rental value even as next-generation Blackwell models entered the market. This flies in the face of a remark in March 2025 by Jensen Huang, Nvidia's boss, that "when Blackwell starts shipping in volume, you couldn't give Hoppers away". It also suggests that compute derivatives' underlying assets will not suddenly become worthless, the risk of which would slow adoption.

Other middlemen are getting in on the action. Architect Financial Technologies, a software firm, has launched an exchange for compute futures in Bermuda. It is also now offering its own futures, in competition with the big exchanges, and planning a platform licensed for trading in America called the American Innovation Exchange. Architect's boss, Brett Harrison used to run FTX US, the American arm of a cryptocurrency exchange which imploded in 2022. He says the AI Exchange (in unsubtle shorthand) aims to make a market not just for compute, but for raw materials, like copper and silver, and sources of energy needed for AI chips and data centres.

Meanwhile, Roundhill Investments and ProShares, two purveyors of exchange-traded funds, have filed paperwork with securities regulators to create ETFs for compute futures once these exist. Other ETF providers have filed to launch related "2x funds", which magnify the returns (or losses) with leverage. As compute becomes the currency of the AI era, all these efforts will intensify. As Mr Harrison sums up, "All of a sudden, the race is on." ■

American balance-sheets

A firm mattress

The household saving rate has plunged—but don't panic

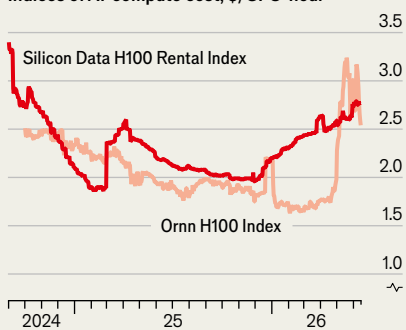
THE LATEST fashionable worry about America's economy is that consumers are spending beyond their means. Personal consumption—its engine, accounting for two-thirds of GDP—has grown by a respectable 2% or so over the past year. But the personal-saving rate fell to 2.6% in April. It has been lower only once since early 2008, when Bear Stearns, a bank, became an early casualty of the global financial crisis. It is only a matter of time, the argument goes, before consumption cracks.

From 1970 to 2000 the personal-saving rate, which measures how much income Americans have left after taxes and expenses, averaged around 10%. Even a year ago it was double April's figure. Annual inflation is now outpacing wage growth for the first time in three years. Larger tax refunds from Donald Trump's One Big Beautiful Bill Act—worth about \$350 more per household than last year—have mostly been disbursed. On the surface, then, the low saving rate looks like proof that Americans are running short of cash.

There are, though, some benign explanations. The first is demographic. The saving rate at the end of the 20th century was supported by a much higher ratio of workers, who save for retirement, to pensioners, who draw down their savings. Today America has more pensioners than ever: the number collecting Social Security benefits reached 54.5m in May (see chart 1). For the first time, more than half of those outside the labour force are 65 or older. Pensioners typically have less current income than workers, but keep spending by tapping lifetime savings. That pushes down the saving rate in a financially innocuous way. ►►

Add it to the hopper

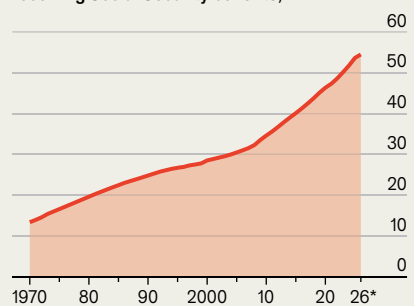
Indices of AI-compute cost, \$/GPU-hour



Source: Bloomberg

Great-great-Uncle Sam

United States, retired workers receiving Social Security benefits, m



Source: Social Security Administration

*To May

Pocket changes

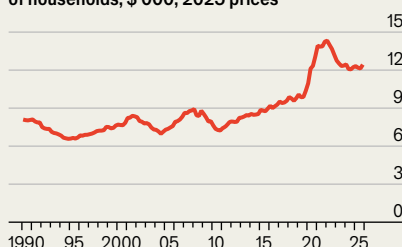
United States

Liquid assets, % of personal disposable income



Sources: Federal Reserve; FRED

Average liquid assets* held by the bottom 50% of households, \$'000, 2025 prices



*Deposits and money-market funds

▶ Adjust for demography and the picture looks less alarming. Research by Federal Reserve economists suggests that a typical pensioner spends \$15,000-22,000 more than their income each year. The gap is much larger for richer households with more assets. Across all oldies, such “dis-saving” knocks just over five percentage points off the headline saving rate. Some of today’s low figure, then, reflects older Americans spending down what they saved in the past, and doing so more freely as high asset values boost their wealth, not younger households running out of money.

A look at household balance-sheets offers more reassurance. The saving rate measures the money left over from each month’s income, not the stock of cash that households already have on hand. And that stock of cash looks better than the monthly flow. Data from the Fed show that liquid assets—cash, bank deposits and money-market funds—are equivalent to about 84% of annual disposable income, up from less than 70% in the three decades before the pandemic (see chart 2, left panel). The cushion is not confined to the rich—and is thicker than the saving rate might suggest. The bottom half of households by wealth average liquid balances of around \$12,800, more than at any point before the pandemic in real terms (see chart 2, right panel).

There is certainly no sign of an imminent consumption crunch in the latest private-sector data. PNC, a bank, finds that card spending (excluding petrol) rose by nearly 5% year on year in May, close to its fastest pace in four years. Spending on retail as well as travel, entertainment and other discretionary services looks especially perky, according to Bank of America. Walmart, the supermarket bellwether for middle America, reported that transactions in the first quarter this year rose at their fastest pace since 2024.

Together, tax refunds, high-rolling pensioners and fat cash mattresses have helped keep total consumption going. Yet consumers are not invulnerable. A big fall in the stock market could reduce willing-

ness to spend even among the old. Oil prices are likely to stay high for months despite America’s tentative deal with Iran to reopen the Strait of Hormuz. Tariff disruption may intensify as the Trump administration’s investigations—the legal route the president is using to justify punitive duties—move forward. The future of the USMCA, America’s trade pact with Canada and Mexico, is uncertain. American consumers can still withstand a lot. That does not mean they can withstand anything. ■

Russia

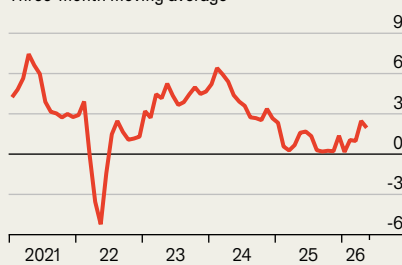
All quiet on the home front

Vladimir Putin’s war economy has problems—but is not about to crash

IS THIS IT, finally? After four years of sanctions-busting growth, a chorus of think-tank economists is calling time on Russia’s war economy. A new report from the Kiel Institute for the World Economy argues that Russia faces “structural exhaustion”. Charles Hecker of the Royal United Services Institute reckons it “is proba-

A thaw

Russia, current-activity indicator*, % change on previous month, annualised
Three-month moving average



*High-frequency measure of economic activity
Source: Goldman Sachs

bly already in recession”. Nigel Gould-Davies of the International Institute for Strategic Studies talks of “the coming crisis in Russia’s political economy”. Even Russian official figures point to GDP shrinking by 0.2% in the first quarter, year on year.

Since invading Ukraine in 2022 Vladimir Putin’s Russia has thumbed its nose at those who have predicted economic doom. It has defied Western sanctions by trading more with places like China and India, and spent ample fiscal reserves on the armed forces, infrastructure and social benefits. Between 2022 and 2025 GDP per person, adjusting for inflation, rose by 12%—weak by the standards of other emerging markets, and flattered by defence production that does not benefit households, but not bad when compared with the dire predictions. Despite fresh strains, the economy is not about to collapse.

Start with the weak official statistics. These are, largely, a statistical mirage. A rise in value-added tax (VAT) in January, from 20% to 22%, encouraged Russians to make lots of purchases in late 2025, boosting that quarter’s growth at the expense of the next. Early 2026 also had fewer working days than a year earlier, and—even by Russian standards—grim weather. A potentially cleaner gauge of economic activity, produced by Goldman Sachs, is consistent with sluggish growth but no slump (see chart 1). Data from VEB, another bank, points to a pick-up in GDP in March and April, thanks in part to a surge in oil prices. Russia is almost certainly not in recession.

Elsewhere, the picture is mixed. Consumer confidence has slipped, according to a measure tracked by the Levada Centre, an independent pollster. But that was from close to an all-time high (see chart 2 on next page). Jobs may be a bit scarcer, yet unemployment remains close to a record low of around 2%. Russia is finding it more difficult to export fossil fuels as Ukraine ramps up attacks on its energy infrastructure, and oil prices are down from the highs attained during the Iran war. Even so, overall goods exports in April were slightly higher than a year earlier.

In some ways the economy is improving. Inflation is half its recent peak of over 10%. Real wages, already 25% higher than in 2019, keep rising. Many firms are doing just fine. From January to May Aeroflot, the flag-carrier, flew passengers a total 40bn kilometres, nearly 10% more than the same period last year. Oligarchs are doing even better. Sales of luxury cars smuggled from the West are up; those of Lamborghinis, by 80% from 2025.

This resilience owes a lot to heavy fiscal stimulus. Last year the government spent 7-8% of GDP on the armed forces. These outlays, those who foresee a crisis argue, suck manpower from the rest of the economy and drain the exchequer. Perhaps. Yet ▶▶



Red Square still in the black

Sanctions

Cancelled penalty

Letting Iran sell its oil is a huge economic concession by America

IRAN'S FOOTBALLERS are having an unexpectedly good World Cup. On June 21st they held Belgium, ranked ninth in the world, to a draw in Los Angeles, leaving them within reach of the knockout stages. Over in Switzerland, Iranian diplomats are scoring even bigger successes. On June 22nd the Treasury Department up-ended four decades of American policy by issuing a sanctions waiver permitting the production, sale and delivery of Iranian petroleum for 60 days. The move will bring instant relief to the Iranian regime, and it could over time make Iran rich again.

America banned its firms from buying Iranian oil in 1980 in response to the hostage crisis in its embassy in Tehran the previous year. That embargo was supplemented in the early 2010s by "secondary" sanctions exposing other buyers to American penalties. Those were suspended under Barack Obama's nuclear deal of 2015, then reimposed, in harsher form, when Donald Trump tore up the pact three years later.

The latest waiver goes much further than any previous relief. An earlier one, issued by Mr Trump while war between the countries raged, covered only Iranian oil already loaded on ships. Licences granted to third countries under Mr Obama required them to cut purchases, which drove exports down from 2.5m barrels per day (b/d) in 2011 to 1.5m in 2012. Even Mr Obama's nuclear deal lifted only secondary sanctions. Mr Trump's new licence imposes no such constraints. American refiners may now buy Iranian petroleum directly, pay for it in dollars and receive it from blacklisted tankers—rolling back, temporarily, the original embargo from 1980.

Why be so generous when talks have so far yielded no Iranian concessions? One transparent motive is to keep negotiations alive—and the Strait of Hormuz open—despite rising tensions over Israel's continued attacks in Lebanon. Beyond that, says Michelle Brouhard of Kpler, a data firm, who also advises America's energy department, the administration hopes the move will push oil prices down, stop China getting cheap Iranian crude and deter Iran from closing the strait. In reality it will make a marginal difference at best.

One reason is that Iranian crude already flowed more freely after America lifted its blockade of Iran's ports in mid-June. Exports of oil went from nearly nothing in May to 1.5m b/d, notes David Wech

of Vortexa, another data firm (see chart). Loadings from Kharg island, Iran's main export terminal, have risen, too. Iran still has some way to go before it reaches the monthly average of 2m b/d recorded before the war. But the price of Brent crude, the global benchmark, which has barely moved since Mr Trump's concession was announced, suggests that markets were pricing in a surge in Iranian shipments even before the waiver was announced.

For exports to rise much more, and prices to come down further, Iran must find new buyers for its oil. In recent years nearly all of its barrels have gone to small "teapot" refineries in north-eastern China. They are "quite excited" about the prospect of purchases with less need for costly efforts to conceal them, says Tom Reed of Argus Media, a price-reporting agency. However, the teapots cannot easily increase their purchases; Iranian crude is now priced on a par with Omani and Emirati oil, limiting the incentive to binge.

For other buyers to come forward, their bankers, insurers and compliance officers must first have confidence that they can do business with Iran for longer than 60 days—and that Mr Trump will not suddenly revoke his waiver. They still face European and British sanctions, which remain in place. And they must consider the reputational risk of putting money directly into the Iranian regime's pocket, says Amrita Sen of Energy Aspects, a consultancy.

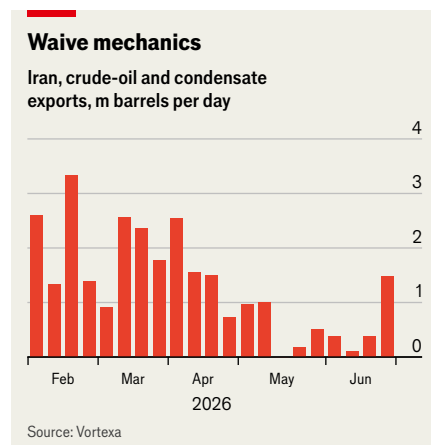
These obstacles will put off many potential customers. India, which once bought lots of oil from Iran, might take some. Japan and South Korea—regular buyers as recently as the late 2010s—may start coming back if the current arrangement lasts a few weeks, says Nader Itayim of Argus. Western purchases probably will not revive before a permanent deal is sealed.

As for keeping Hormuz open, the sanctions relief looks unlikely to achieve the clarity America hopes for. Days after Mr Trump signed the preliminary deal with his Iranian counterpart on June 17th, Iran

the military splurge represents an increase of 3-4% of GDP from the pre-war norm—not nothing but not enough to have huge knock-on effects. The civilian economy is treading water, not contracting.

Russia's fiscal problems, meanwhile, are not yet acute. To pay for the war the government can raise taxes, as with VAT. It can fund any shortfall—currently some 3% of GDP—by dipping into its rainy-day funds. It can borrow from the captive domestic market. At a pinch, Mr Putin's financiers can raid the rouble deposits held by corporations and households. This would be a last resort with real knock-on consequences. But who would stop them?

All things considered, Russia can expect GDP growth of 1% or so this year: about as good as France or Canada. Tougher sanctions, like those unveiled by Britain on June 16th, may cut this growth a bit. So will oil prices, if they continue to fall, and more Ukrainian attacks on Russian oil infrastructure. But it would take something much more radical to keep Mr Putin's war economy from trundling along. ■



declared the strait closed again. Non-Iranian traffic—which had begun rising after the signing—immediately paused, even as Iranian shipments increased. It now appears to be rising again, but so are tensions between America and Iran. In the longer run fears linger that Iran will seek to impose a toll on Hormuz crossings, which would constrict traffic. On June 22nd it said it would “administer” the waterway and set up a “telephone hotline” to co-ordinate the passage of ships.

In other words, the sanctions waiver so far looks ineffective from America’s perspective. For Iran, it is a boon: it hastens the recovery of exports and, by freeing up space in its nearly full storage, enables curbed production to restart. By reducing friction in logistics and payments, it lets Iran’s oil firms—and the regime—earn a little more on every barrel they sell.

If the licence is renewed indefinitely, as some experts expect, Iran would attract a bigger and more diverse group of buyers.

Add in billions of dollars a year in transit fees, the return of unfrozen assets and Mr Trump’s promised \$300bn reparation fund, and Iran could become one of the Gulf’s richest states within a decade, without having conceded much on its nuclear programme or its support for troublesome proxies, says a big trader familiar with the region. Mr Trump may face domestic opposition to what would be near-total capitulation. But the odds of such an outcome are steadily rising. ■

BUTTONWOOD *Man v machine*

For now, humans are still better macro traders than AI

HOW MUCH would you have paid ten years ago, as votes were counted for Britain’s Brexit referendum, to glimpse the next morning’s headlines and trade ahead of them? If you were betting on the pound, it would have helped a lot. The night of the poll £1 bought \$1.50; a fortnight later, less than \$1.30. But if you were trading stocks, a forewarning might have done more harm than good. Britain’s domestically focused FTSE 250 share index dropped at first, but only for two trading days. Then it began a bull market that lasted for a couple of years. Even the most fervent Brexiteer might not have predicted that.

Macro trading, meaning betting on how asset prices will move in response to political and economic trends, is enticing and glamorous. It is also hard, and a new study by Jerry Bell, Victor Haghani and James White of Elm Wealth, an investment firm, shows just how hard. They have designed a simulation in which both humans and leading artificial-intelligence models get access to the next day’s news in advance, and can place their bets before it breaks. In other words, they get to trade ahead of the rest of the market. Yet even with this advantage, it turns out to be difficult—for man and machine alike—to avoid ruin and turn a profit.

Mr Bell and his colleagues are updating an experiment they first ran in 2023. Then, they recruited 118 volunteers, most of whom were studying graduate-level finance at leading universities. Each was given \$50 and a chance to grow it by placing bets on America’s S&P 500 share index and 30-year Treasury bonds.

They could do this once per asset per day, at the market close before each of 15 trading days, selected by the authors from between 2008 and 2022. Before

trading, participants were shown the front page of the *Wall Street Journal* pertaining to the following day—so at Monday’s close they were shown Wednesday’s front page (with any actual price moves on Tuesday redacted). They could go long or short each asset and could leverage their bets by up to 50 times. This would translate a 2% price move, for example, to a double-or-nothing wager. Each trade was terminated at the following close.

Most participants did not make themselves proud. Roughly half lost money and one in six went bust; the average finishing pot was just \$51.62, or a gain of 3.2%. Since then Elm has hosted a similar game on its website, with an imaginary starting stake of \$1m. The 60,000-odd people who have played it have fared “substantially worse” than the original, paid cohort.

Perhaps the greater surprise is that, in the experiment’s latest iteration, several of the leading AI models did not truly excel, either. The Elm team gave each of ChatGPT, Claude, Gemini and Grok ten runs at the game, also starting with an imaginary \$1m. They were told to play as a

middle-aged American investor managing 100% of their financial wealth.

Only ChatGPT and Claude made money, with average finishing pots of \$1.5m and \$2.6m respectively. Grok’s was \$970,000 and Gemini’s just \$490,000. So what makes macro trading so hard, even for those with a crystal ball?

The biggest lesson is that both humans and AI are bad at sizing bets. The authors note that even ChatGPT and Claude were lucky as well as skilful in this respect. None of the models correctly predicted the direction of stocks and bonds more than about 60% of the time, yet their average leverage applied to their bets ranged between seven and 12 times. Given that American share prices moved by more than 5% on 23 days since 2000, and by more than 9% on seven days, the models were therefore running “too much risk of a catastrophic loss of capital”, including the possibility of a complete wipe-out.

The humans were even worse. In the original experiment, players in aggregate bet no more heavily when the news made price moves easy to predict. And like the AI models, they took too much risk overall. On 30% of days they used leverage above 20 times, which could easily have sent them bust.

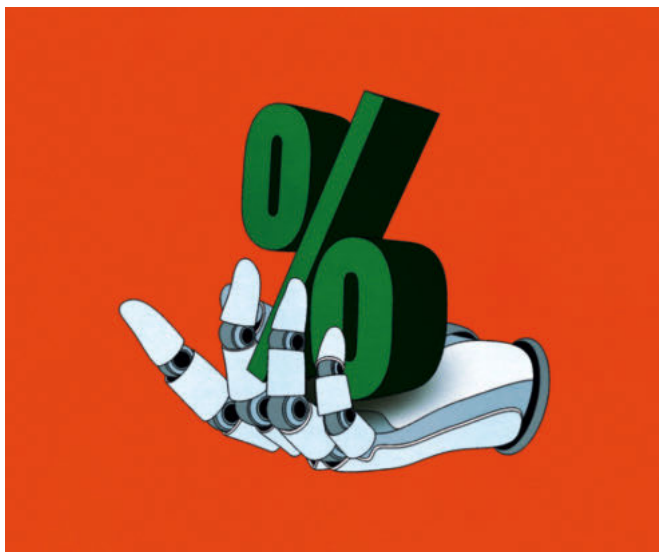
Some humans, of course, are much better. The Elm team also invited five expert macro traders to play their original game. All five professionals finished in the black, with an average return of 130%. They did a bit better than AI at predicting directions (a hit rate of 63%). But, crucially, they varied their position sizes a lot, betting more when they felt confident and not at all when they did not. Even with excellent foresight, knowing which assets to buy is tough. Deciding how much, it seems, is tougher.



FREE EXCHANGE

Of robots and r-star

Kevin Warsh thinks AI calls for lower interest rates. Is he right?



AS CHAIRMAN OF the Federal Reserve in the 1990s, the jazz-loving Alan Greenspan earned a reputation as the maestro of monetary policy (see Obituary). Many of his colleagues believed that unemployment had fallen so low that it was certain to stoke inflation, and called for higher interest rates. But Mr Greenspan heard a different tune. From anecdotes about retailers tracking inventories by satellite or airlines using computers to adjust fares to match demand, he inferred that an IT-driven productivity boom was taking hold before it appeared in official statistics. The economy could thus grow faster without stoking inflation. He held rates steady for much of 1996 and 1997, trusting his inference over economic models. For a time, it worked.

Kevin Warsh, the Fed's new chairman, has cast himself as Mr Greenspan's heir. In his telling, artificial intelligence will expand supply and lower costs, allowing the economy to grow faster with less inflationary pressure. He, too, thinks the productivity statistics will arrive late. In an interview last December Mr Warsh praised Mr Greenspan for forming his views "from anecdotes and rather esoteric data", arguing that when the data and the anecdotes diverge, central bankers should trust the latter. He sees the "excitement" in the eyes of chief executives as evidence that a productivity revolution is afoot. This forms part of Mr Warsh's broader case, disputed by many of his colleagues: that AI is a disinflationary force, and that it calls for lower interest rates. Has Mr Warsh drawn the right lesson from his hero?

In the near term the effect of AI on rates comes down to a race between supply and demand. For now, demand is winning. Debt-fuelled spending on data centres, chips and power plants has surged. A stock-market bull run makes the share-owning public feel rich, boosting consumption even as income growth slows. As Austan Goolsbee of the Chicago Fed has warned, "the bigger the hype, the more rates may need to rise to prevent overheating."

The Fed's new conductor can hear the loud brass of demand. His bet is that the quieter strings of supply will soon join in. As firms reorganise work around AI, the same workers and machines should produce more, cutting unit costs and expanding the economy's capacity. Yet even a supply-side transformation may not en-

gender the lower rates Mr Warsh expects. That is because higher productivity raises real incomes, stirring fresh demand of its own and leaving the near-term effect on rates far from certain.

A related, longer-term force also points towards higher rates. If AI causes faster economic growth and higher future incomes, it is also likely to lift the "neutral rate" (*r*-star for short), the real interest rate at which monetary policy neither heats nor cools the economy. The intuition is familiar: a student expecting a fat salary after graduation is willing to take on debt today. Likewise, households borrow against higher future incomes and firms invest more as expected returns rise. Both raise demand for capital, increasing the interest rate needed to balance saving and investment. Recent work by Lukasz Rachel of University College London estimates that AI could lift *r*-star by around one percentage point. If he is right, a return to the pre-pandemic world of ultra-low interest rates is unlikely, even once supply catches up with demand.

Would the same conclusion hold in a world of benign superintelligence, where hyperproductive AI allows humans to pursue a life of leisure? In principle, yes. Explosive growth would lead people to expect booming future income—and so spend more today. If instead AI starts looking like an existential threat to humanity, why save for a future that may not arrive? Either way, rates must rise steeply to coax people into saving.

Some forces could pull *r*-star down. If superintelligence commoditises knowledge work and stiffens competition, corporate mark-ups, profits and expected returns could shrink, weakening demand for investment. Faced with upheaval in the labour market, households may build precautionary savings rather than borrow against a brighter future. Widespread displacement of jobs could push the neutral rate lower still. If less income flows to workers and more to owners of capital, aggregate spending may weaken, since the rich save more of each additional dollar they receive. Redistribution could offset some of this by directing income back towards households more likely to spend it. Without that, the result might well be a savings glut even as productive capacity soars—secular stagnation born of abundance at the top.

That all these outcomes are at least conceivable suggests that Mr Warsh would be wise to approach the question of AI and rates with humility. Even if the AI revolution proves as revolutionary as the technology's boosters insist, it could push rates in either direction, depending on how precisely its gains filter through the economy and who captures them. The range of scenarios also points to the limits of historical analogy. AI may be anything but an ordinary technology.

Play it again, Alan

And even if history of the 1990s does repeat itself, Mr Greenspan's record contains a second movement which Mr Warsh seems not have heard. The late Fed chair noticed the productivity boom before others, but then came to fear the demand it unleashed. In 1996 he warned of "irrational exuberance" in asset prices. Once the boom was well under way, Mr Greenspan was warning that faster productivity would create "even greater increases in aggregate demand than in potential aggregate supply". In response the Fed raised rates six times between June 1999 and May 2000. Even so, it was already behind the beat. By early 2000 inflation had reached its highest level in almost a decade; the asset bubble burst soon afterwards. Mr Warsh may have learned the first part of the maestro's score. He should listen carefully to the coda. ■

Science & technology



Artificial intelligence

Computo, ergo sum

Why big AI labs are hiring philosophers to shape their products

TEN YEARS ago, as the AI revolution was gathering pace, arts and humanities students were told that, if they wanted to make themselves employable, they should “learn to code”. That may have been bad advice. These days, it is programmers who are nervous about AI taking their jobs.

They might consider learning to philosophise. Earlier this year the Federal Reserve Bank of New York published figures showing that American philosophy graduates are more likely to have jobs than their peers who studied computer science. In 2024, the most recent year for which numbers are available, 7% of those who had studied computer science were unemployed, against just 5.1% of philosophers.

Many are being snapped up by AI firms themselves. Students get job offers before they have graduated, says Luciano Floridi, a philosopher at Yale University. Academics are moving, too. Dr Floridi describes the scale of departures from philosophy

departments as a “haemorrhaging”.

Some of the lessons that philosophy can offer AI researchers are ancient. The Socratic method—as described by Plato, an ancient Greek philosopher—uses feigned ignorance and sequential questioning to clarify meanings, spot contradictions and reveal ramifications. Many current AI systems tend towards sycophancy. Models trained in the Socratic method, says Jörg Noller, an expert on philosophy and AI at Ludwig Maximilian University of Munich, are less keen on people-pleasing and more willing to pursue the truth.

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Then there is the idea of “Socratic ignorance”. In the “Apology”, Plato has Socrates claim that his wisdom consists mostly of being aware of how much he does not know. Implanting that humility into a model can help limit overconfidence, a common flaw that Dr Noller describes as “AI immaturity”. Iason Gabriel, a senior philosopher at Google DeepMind, an AI lab based in London, attributes an industry-wide decline in hallucinations to such efforts. More broadly, he says, philosophy lessons are “a powerful mechanism” for improving long AI reasoning processes known as “chains of thought”.

Philosophical training can also affect a model’s outlook in more specific ways. Feed an AI legal assistant the writings of John Locke, says Thomas Powers, a philosopher of technology at the University of Delaware, and it will favour robust property rights as an underpinning of political liberty. And if you don’t like those principles, the model-makers have others. The “Granite” series of models from IBM, an American computing giant, come with dials that let business customers better align outputs with their own corporate philosophies. Francesca Rossi, IBM’s head of responsible AI, says these can let users choose where to strike the balance between philosophical trade-offs, such individual agency versus social harmony. ▶▶

► Philosophy can help with safety, too. Researchers have documented all sorts of ominous behaviour in AI models, including attempts to evade oversight and even blackmail their users. One way model-makers try to discourage this sort of misbehaviour is called AI constitutionalism. This involves building a model around a scaffolding of rules and principles culled from philosophical writings with legal or moral authority.

Anthropic, an AI lab based in San Francisco, is one proponent. Constitutions for its Claude models have incorporated material from sources as diverse as Immanuel Kant, Apple's terms of service and the Universal Declaration of Human Rights. The latest iteration, led by Anthropic's top philosopher, Amanda Askell, was published on January 21st. Some staff at Anthropic have nicknamed the 78-page constitution Claude's "soul doc".

The biggest question, though, is what sorts of rules should be put in those constitutions in the first place. Philosophers have zeroed in on two main ethical frameworks. One is deontology. Popular with Kant, among others, this imposes strict rules that prohibit things like lying, coercion and treating people as a means rather than an end, even if it is for a greater good. Anthropic's constitution incorporates many deontological strictures. These can make AI behaviour more consistent, says Dr Powers—a plus for deploying robots in homes and public spaces.

Models with a deontological take on the world have other benefits. One is greater honesty, a trait widely noted in Claude. Models that are more truthful, says Nick Bostrom, a philosopher at the University of Oxford, are less likely to mislead their users. Inflection AI, another Silicon Valley lab, imposes deontological constraints on its Pi chatbot, which is designed to provide emotional support. Sean White, its boss, says Pi is good at spotting users at risk of harming themselves or others. Deontological constitutions also help with legal compliance, says Dr Floridi.

The other approach to ethics of interest to philosophers of AI is called consequentialism. It weighs costs against benefits to decide what to do. Models more sympathetic to consequentialism include OpenAI's ChatGPT and Google's Gemini. Google's AI models are designed to produce "likely overall benefits [that] substantially outweigh the foreseeable risks", a classic consequentialist goal.

Consequentialist algorithms are also crucial in software for autonomous vehicles: if an accident is unavoidable, a decision must be made on the least tragic way to crash. Chris Gerdes, a senior engineer at Waymo, which makes self-driving cars, says the trend is to make driving software more consequentialist. Consequentialism

is also central to AI weapon systems. Military objectives must be weighed against possible civilian deaths, says Jack Shanahan, a former head of the Joint Artificial Intelligence Centre, which studies AI for America's armed forces.

Thorny problems abound—a philosopher's favourite sort. Are there cases when deontological rules should be overridden? How do you make decisions when the consequences are unclear? Should AI systems take into account animal welfare, or the state of the environment? Would it be morally acceptable, asks Stefan Heck, a philosopher and the boss of Nauto, which makes AI-powered safety systems for lorries and other commercial vehicles, to prioritise young pedestrians over old ones? He predicts ethically fraught lawsuits: consequentialist algorithms, after all, explicitly permit one harm as long as it is designed to avert a worse one.

Critics fret about "moral deskilling": if computers increasingly make ethical calls, might people become less willing to make their own judgments? Roman Yampolskiy, an AI theoretician at the University of Louisville, argues that morality "is historically unstable, culturally variable, strategically manipulable, and often only retrospectively legible". Unemployed coders take note: there seems to be no shortage of work for philosophers of AI. ■

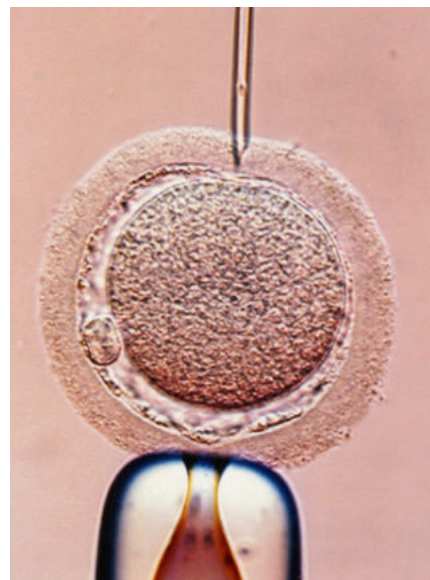
Fertility

Misconceptions

A review of popular "IVF add-on" procedures suggest most do little to help

TRYING TO CONCEIVE through in vitro fertilisation (IVF) leads to disappointment more often than not. About 60% of IVF attempts fail. Fertility clinics offer a long list of tests and procedures purported to boost their customers' chances. But a review of the evidence, published on June 23rd in the *Lancet Obstetrics, Gynaecology & Women's Health*, found no convincing evidence that most of these "IVF add-ons" are helpful. Worryingly, lots of the published evidence was dubious.

The review examined 157 randomised trials of various IVF add-ons. The researchers ran each through a checklist designed to spot signs that the data may have been manipulated. That list was developed in 2023 by fertility researchers (including some of the review's authors) who had noticed a growing number of fraudulent studies. They looked for things such as implausible study timelines, strange participant data or holes in a trial's paper trail.



40% of the time, it works every time

All told, nearly half of the trials did not pass muster. The remaining 85 covered ten commonly used add-ons. Of those, only three procedures showed evidence of abenefit, although the evidence was not particularly strong. The three procedures in question were endometrial scratching (which involves deliberately disturbing the lining of the uterus), EmbryoGlue (in which an embryo is dipped in a solution of hyaluronic acid, which is naturally found in the reproductive tract), and physiological intracytoplasmic sperm injection or PICSI (which tries to select high-quality individual sperm cells).

The data on three other IVF add-ons, though also limited, suggested they had no effect on the chances of a successful pregnancy. These three were corticosteroids (a class of anti-inflammatory drugs), genetic testing of the embryo for abnormal chromosomes and biopsy of the uterine lining to assess genetic expression.

Data for the remaining four procedures were too scant to make a judgment either way. These were acupuncture, intravenous infusion of fats derived from eggs or soybeans (in the hope this might calm an immune reaction to the embryo), and injections of platelet-rich plasma into either the uterus or the ovaries (platelets being rich in tissue-rejuvenating proteins).

The trials were not just few in number but mostly small in scale too. The median trial had only about 160 patients, which would limit its ability to detect small or subtle effects. The reviewers could not rule out the idea that some interventions might work for a subset of patients. But being able to say who, if anyone, might benefit from any of them would require more and better research—as well as eagle-eyed journal editors weeding out the fishy sort before it is published. ■

3D printing

Additive artistry

Electronics can now be 3D printed onto living tissues

IT BEGAN IN the 1980s as a way to quickly produce simple prototype models in plastic. These days 3D printing—or “additive manufacturing,” to give it its posher name—is used to make kitchen gadgets, jet-engine parts, dental implants and even some buildings.

In a paper published in *Science Advances*, a team of researchers led by Yong Lin Kong and his colleagues at Rice University in Houston, Texas, propose another application. They have come up with a way to print electronic circuits onto delicate materials, including living tissue. The researchers hope that could eventually lead to medical implants that wirelessly monitor the condition of their recipients, high-tech pills and more.

3D-printing circuitry is not a new idea. Nano Dimension, based in Massachusetts, for instance, prints electronics into communications equipment and medical sensors. Printers made by Optomec, based in New Mexico, build antennas directly into smartphones cases, as well as printing circuits onto touch screens and solar panels.

The process uses electronic inks that contain tiny particles of conductive material, such as copper, silver or gold. The particles are then fused together to form a conductive pathway, usually by a process known as annealing, which involves either heating the entire product in a furnace, or zapping the inks with a laser.

But that approach has limits, says Dr Kong. The heat needed to fuse the particles can damage sensitive materials, including some polymers and living tissue. It is this issue that Dr Kong and his colleagues now think they have overcome. They have developed a way to anneal electronic inks using microwaves focused into a beam smaller than a human hair. The precision of the beam ensures that only the ink particles heat up, avoiding any damage to the surrounding substances.

The researchers produced a special resonator to amplify the microwave energy and a tapered tip to focus the beam into a dot less than 200 millionths of a metre across. The kit is compact enough to be fitted to a desktop 3D printer. By adjusting the power of the beam, it is possible to change the microstructure of the ink. This can be used to create regions of higher or lower conductivity along the circuit, allowing some components, such as resistors, to be created within the wiring.

Dr Kong and his colleagues have plenty of applications in mind. 3D printing has already been used to make biological tissues for transplant, including heart valves and tracheas, without the need for human or animal donors. Adding circuitry could turn such grafts into wireless sensors, allowing them to monitor a patient's progress in real time, or even stimulate the activity of the organ they are printed on.

As a proof of concept, the researchers have printed a strain gauge onto both a cow's femur and the sorts of polymers used in replacement knees and hips for human patients. Another idea is printing circuits onto “ingestible diagnostic devices”. Essentially computerised pills, these are designed to relay information from inside a patient or carefully control the release of drugs. The researchers have even managed to print a humidity sensor onto a living leaf. The natural world, it seems, is the next canvas for additive artists. ■

Heatwaves and climate change

Another sweltering summer

Why Europe is the world's fastest-warming continent

ON THE MORNING of June 24th the Shaw Library at the London School of Economics was meant to be full of people discussing the impacts of climate change as part of London Climate Action Week. It was not to be. Britain's Met Office had, for only the second time, issued a “red” warning about temperatures high enough to pose risks even to the fit and healthy. The

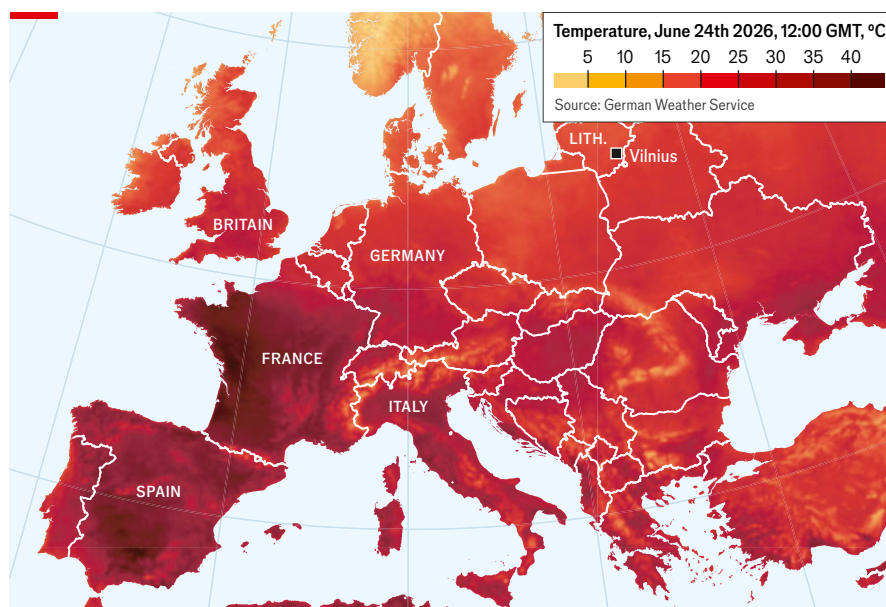
organisers decided they could not proceed with their meeting—the subject of which was “Extreme heat: Improving governance and strengthening action”.

Qualitatively, the European heatwave that started on June 18th is not unusual. A horseshoe of lower pressure—a so-called omega block—is sustaining an area of high pressure over the western part of the continent. The air near the surface in this high-pressure region came from the south-east, rather than off the Atlantic, and so started off fairly warm. High pressure above stopped it from losing its heat through convection, a process that produces clouds. A lack of clouds let the Sun stoke things further. Temperatures rose.

Quantitatively, it has been something else (see map). Temperatures over 40°C (104°F) were recorded in much of France and Spain; they were not ruled out for Britain, which has seen them only once before. France saw its hottest day ever on June 24th, with 58 administrative departments under a red alert. The same applied to 16 Italian cities. Germany's heat record could be broken by June 28th. Across the continent schools are shut, trains are cancelled and power grids are groaning.

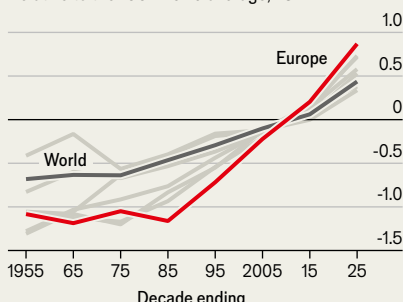
What makes an unexceptional weather pattern lead to exceptional heat? The obvious answer—climate change driven by greenhouse gases—is the correct one. A quick assessment offered by ClimaMeter, a consortium of scientists based at France's Institut Pierre-Simon Laplace, suggests that climate change has made the heatwave 2°C–4°C worse than it would have been under the same conditions in the second half of the 20th century.

Discussions of global warming tend to stress the effects on poor and middle-income countries at low latitudes. That is proper. The extremes experienced are ►►



The Great European Bake Off

Average temperature anomalies by continent
Relative to the 1991-2020 average, °C



Source: Copernicus ERA5

▶ worse, and people who live there have fewer—or none—of the resources needed for adaptation. Recent heatwaves in Asia have been horrific. But higher latitudes are warming faster. Europe's temperature is rising by 0.56°C per decade, twice the world average and faster than any other continent (see chart).

This is mainly because of a basic principle of climate change. Other things being equal there will be a greater warming at the poles than in the tropics, in part because the poles lose ice, which reflects sunlight, as the planet warms. The geographical centre of Europe, which sits near Vilnius at roughly 55°N, is closer to a pole than that of any other inhabited continent. Also relevant is Europe's clean air, which allows more sunshine to warm the surface. Air pollution is falling in lots of places, but the European decline has gone on longer and progressed further than most.

Climate change may also be influencing the frequency of omega blocks. But Friederike Otto, a climate scientist at Imperial College London, stresses that it is the outcomes such patterns can now lead to, rather than changes in how often they occur, that matter most. "Heatwaves are becoming more frequent, more severe...and longer-lasting," says Will Lang of the Met Office.

The health impacts will be dire. In the heatwaves of 2003 there were over 70,000 heat-related deaths across 16 European countries. In 2022—at the time the hottest European summer on record—there were more than 60,000 across 35 countries. In 2023, not quite so hot, estimates were a little under 50,000. Europe has been getting better at protecting itself: a study by Elisa Gallo of the Barcelona Institute for Global Health and colleagues, published in 2024, estimates that without such adaptations the death toll in 2023 would have been around 90,000, and the death rate among the over-80s twice what it was. But if Europe is going to lower the risks of the worsening climate, rather than just keeping up with them, it needs to do more. ■

Well Informed

How good is infant formula?

Good. Though perhaps less than its marketers say

BABY MILK has come a long way since a patented "soup for infants" made with cow's milk, potassium bicarbonate, and wheat and malt flour was introduced in Prussia in 1865. Today the Food and Drug Administration (FDA), an American regulator, requires baby formulas to have minimum levels of 30 nutrients including iron, iodine and vitamin D.

Formula-makers these days are just as keen to tell people about the extras they add to their products, too. In 2023 the *BMJ*, a journal, published a study of 757 formula products in 15 rich and poor countries. Many trumpeted additives such as prebiotics, probiotics and long-chain polyunsaturated fatty acids (LCPUFAs; some of which must be included in formula sold in Britain and the EU). The benefits claimed included supporting "growth and development" and "a healthy immune system". The most common claim, appearing on just over half the formulas that claimed benefits, was that they helped develop the brain, eyes or nervous system.

Do they? Infant formulas are usually classified as food rather than medicines, which leaves them freer to make claims about benefits without the backing of rigorous trials. Three-quarters of the products evaluated did not cite scientific references. Of the references that were cited, only half were clinical trials. Of

those trials, fewer than a third were registered. Among the small group of registered trials that used a randomised comparison, nine in ten claims relied on evidence the study's authors deemed to have a high risk of bias. Industry funding and affiliations were common.

The risk of bias might give some parents pause for thought. But it does not, in itself, prove that their claims are wrong. Only the evidence can do that, and as best as scientists can tell, it is equivocal. A review of 15 randomised trials involving 1,889 infants, last updated in 2017, found that those given formulas supplemented with LCPUFAs had no better brain development or visual acuity than those fed less-fortified formulations. A study published in 2021 examined 1,607 British adolescents and found that those who had been fed supplemented formula as babies had not gained an academic edge.

On the other hand, a meta-analysis of nine randomised controlled trials involving 1,039 babies was published in the journal *Nutrients* in 2025. It found a slight, but statistically significant, improvement in cognitive development among infants fed formulas supplemented with two LCPUFAs, ARA and DHA. The FDA notes that other studies have suggested that babies born prematurely may especially benefit.

Artificial formulas have generally proved safe in trials. Experts worry, though, that breast-feeding is undermined by slick, emotive marketing of formulas with an aura of scientific superiority.

The benefits of breast-feeding vis-à-vis any type of formula are well established. Many studies link breast-feeding with lower rates of infection for infants; several recent ones have linked it to lower rates of asthma and allergies. It is also much cheaper than buying formula milk, even accounting for the extra calories that a breast-feeding mother must consume. The FDA, and many other rich-country health bodies, advise that breast-feeding is optimal. But mothers who cannot breast-feed, or who find that they must top up with formula, should not worry. Today's infant formulas are safe and nutritious, even if some of the marketing runs ahead of the science.



Culture



Is America becoming a gerontocracy?

When time's not on your side

A new book spots a real problem and offers terrible solutions

Gerontocracy in America. By Samuel Moyn. Farrar, Straus and Giroux; 288 pages; \$30

TALK TO YOUNG people in any rich country and similar complaints pop up. Housing is “not affordable at all”, says Elijah Edwards, a 20-year-old attending university in Washington, DC. Prices are “ballooning up, and it seems like nothing [ever makes them] go down”. The young want homes near where the good jobs are. But it is so hard for builders to build that legions of 20-somethings are stuck seething in their parent’s basements. Could the culprit be...old people?

“Show up to any planning commission meeting, and you can clearly see the generational divide,” says Joh Gehlbach, a YIMBY organiser in Virginia. Silver-haired NIMBYs oppose any new construction that might spoil their view or make parking slightly trickier. The young typically don’t show up

at all. “If you have a public hearing on Tuesday at four o’clock, it’s a little easier for a retiree to [get there] than it is for, you know, a 35-year-old working professional who’s got a kid at home.”

America is ruled by the old, argues Samuel Moyn, a Yale professor, in “Gerontocracy in America”. It is not just that Donald Trump is 80 or that his predecessor left office at 82 and was palpably impaired. Mr Moyn sees a society that privileges the elderly, blocks the young and “is more set

on preservation than on renovation”.

When it comes to politics, he has a point. American lawmakers grow mightier with seniority, and there is no good mechanism for getting rid of them when they can no longer do their jobs. Kay Granger, a member of Congress from Texas, served despite living in a retirement home and suffering from dementia. Dianne Feinstein, a senator from California who died in office at 90, often failed to understand what was going on around her. In most lines of work, people past a certain age gradually cede responsibilities—and give them up entirely when they are incapable of fulfilling them. Not in politics. The role of “president pro tempore” of the Senate, who is third in line for the American presidency, is traditionally given to its longest-serving member. Today that is Senator Chuck Grassley of Iowa, who is 92.

Even younger politicians must bow to elderly constituents, who are much likelier to vote than the young. The median age of voters in America is 52. More importantly, the median age of primary voters—who pick the roughly 90% of House members whose seats cannot plausibly be won by the other party—is 65. “The more boring an election is”, the older the electorate tends to be, notes Mr Moyn. As a character in “Parks and Recreation” put it: “Senior citizens are basically the only people who ▶▶

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▶ vote in local elections. So if you want to win, you gotta get the grey vote.”

Throughout the rich world, politicians tend to favour the interests of the old over the young. From Brazil to France, this makes it fiendishly hard to reform pension systems. In America, any leader who tries is portrayed in attack ads throwing grand-ma off a cliff.

The old have less of a stake in the future, so they prefer lower taxes to good schools or investments in infrastructure, Mr Moyn argues. They are often untroubled by soaring public debts, since the burden of repaying them will fall on others. In 2016 Mr Trump cut short a briefing on how America's fiscal trajectory will eventually lead to a crisis after only five minutes, shrugging: “Yeah, but I’ll be gone.” Mr Trump has added more to the national debt than any president in history.

In foreign policy, polls find that young people think the most important issue is a very long-term one, climate change; old folks worry about terrorism. America now has a government that treats climate change as a hoax and talks about “terrorism” a lot, even slapping the label on suspected drug-smugglers before sometimes killing them. This looks tough on the televisions that old people watch but has no discernible effect on the supply of drugs.

Young voters are growing disenchanted: a study by Alonso Amarales of Bocconi University in Milan finds that when political leaders are much older than the population they represent, confidence in democracy itself tends to suffer. So Mr Moyn is right to lament the disproportionate political sway of the old.

He is on dodgier ground when describing their power in the workplace. They dominate corporations, he says, “monopolising” opportunity and keeping the young in lower-paying jobs. He calls this “Gerontocracy, Inc”. Yet the most striking examples he cites are from an atypical industry: academia. Tenure makes many professors unsackable, so they cling to their comfortable chairs long after they have ceased to perform adequately. Other workplaces are not like this. As employees age and grow less productive, their wages tend to fall. Pay typically peaks in the late 40s and early 50s, when workers are experienced but still vigorous, and then gradually dwindles as they wind down towards retirement.

Mr Moyn favours mandatory retirement ages. That might make sense for judges, who wield immense power and are almost impossible to sack, or airline pilots, who could endanger lives as they grow less sharp. But it makes no sense for most jobs, where employers should be free to keep staff on for as long as both parties benefit.

Mandatory retirement would free up jobs for the young, claims Mr Moyn, falling for the “lump of labour” fallacy. It would

also reduce inequality, he enthuses, because the old are richer than the young—though this is largely because they have been saving for longer.

He is strangely untroubled by the fiscal consequences of forcing people to stop working while they are still able and willing to do so. The only plausible way to keep public pensions solvent, for example, is for retirement ages to rise as life expectancy does. Fiddlesticks, says Mr Moyn: America can somehow preserve entitlement programmes while shortening people's working lives. (The Social Security trust fund is expected to run dry by 2032.)

What if old people enjoy working? Tough luck. Those who are “addicted to their vocations” or “the drug of importance” should receive help “reimagining the meanings of their lives”. Forced retirement can be made attractive if pensioners are helped to find rewarding hobbies or voluntary work. I’m from the government,

and I’m here to teach you pickleball.

As well as kicking old people out of the workplace, Mr Moyn would restrict their voting rights, since they have a smaller stake in the outcome. One idea is to reduce the value of a person's vote after they pass retirement age, so that by the time they are 83, they have only a tenth of a vote. Another is for 18- to 25-year-olds to have two votes each. “Why should everyone have an equal vote?” asks Mr Moyn. Because democratic decency and the constitution say so, judges would reply. Judges say the darnedest things; maybe it's because they are old.

In short, Mr Moyn spots a real problem but offers outlandish solutions. Congress may be a gerontocracy, but the workplace is not. Stopping silver-haired taxpayers from earning is no way to make society fairer. As the debate about AI illustrates, work gives people purpose as well as a paycheque; forcing them to quit early is a recipe for a miserable old age. ■

The first great Depression

Boom and gloom

A new book offers some lessons for the AI craze

1873. By Liaquat Ahamed. Penguin; 368 pages; \$32. Hutchinson Heinemann; £25

LIAQUAT AHAMED, an American author, has excellent timing. His first book, “Lords of Finance”, charted how blundering central bankers plunged the world into a depression after the Wall Street crash of 1929. It was published in 2009, when crises were on policymakers' minds and lips.

His new book also catches the moment. Rocketing share prices have made some investors feel torn between giddy enthusiasm about AI and fear that the boom will turn to bust. Perhaps in part because of this, Andrew Ross Sorkin's “1929”, an account of the Wall Street crash that was published last year, spent more than six months on the *New York Times* bestseller list. In “1873” Mr Ahamed tells the story of how the global railway boom of the 19th century ran out of steam, setting off what was then called “the great Depression”.

The AI boom is often compared to the craze for railways; this book shows just how much the comparison flatters today's tech bros (though the author does not mention them). In the two decades to 1870, the output of the four dominant economies (Britain, America, France and Germany) almost doubled, and global trade volumes expanded five-fold. “Powering this

surge”, writes Mr Ahamed, “was a jump in capital investment...led by railroad construction.” In 1869 the first transcontinental railway across America collapsed the time taken to travel from New York to San Francisco from six months to six days.

How did it all go wrong? As in so many booms, investors' justified optimism about a revolutionary new technology eventually ▶▶



Mad money

► became divorced from reality. Central Europe's bull market was turbocharged by reparations from France to Germany after the Franco-Prussian war, "akin to shooting a jolt of cocaine directly into the bloodstream". Share prices of the biggest firms listed in Berlin doubled over the two years to 1872. In Vienna, share prices quadrupled over the three years to 1873. Fraud ran rampant on both sides of the Atlantic.

In 1873 a clutch of senior Austrian railway officials was arrested for fraud, and demand for stocks faltered at last, even as plenty of new ones were issued. And so within months, a huge crash began. Bank stocks in Vienna fell by 45% in a single day. American railway companies, also riddled with fraud, defaulted en masse. Global lending dried up and output with it.

Across the world the crash triggered "a massive deflation in prices that would reverberate for the next quarter-century". In America industrial production fell by 6% and the prices of wholesale goods by 35%. This was great news for owners of government bonds, since these became much more valuable as the purchasing power of money rose. But for others, the relentless rise in the real value of debt was a disaster. "The losers...came from every stratum of society and could be found in every country on every continent," writes Mr Ahamed.

In 2026, with the AI craze gaining a distinctly manic edge, a lot of this sounds familiar and frightening. In fact, Mr Ahamed's skilful and deeply researched narrative should help doomsters cheer up a bit. For the crash did not have to produce decades of upheaval. The lesson from 1873—and indeed from "1873"—is that many things need to go wrong at once for a massive, sustained shock. As in the 1930s, this one metastasised owing to blunders by flawed policymakers.

Mr Ahamed mostly blames the "totally unnecessary reordering of the global currency system" that followed railway mania. In the space of just two years America and much of continental Europe (notably France and Germany) abandoned silver as a monetary metal, shifting to a pure gold standard. The decisions to do so choked off the supply of money just when it should have been expanded to fight the slump, transforming a financial crash into a quarter-century of deflation.

It is unthinkable that central bankers today—primed to fear deflation above all and to bail out economies at the first sign of trouble—would make similar missteps. And had their 19th-century counterparts acted differently, the crisis of 1873 might have been remembered as a minor setback: just another stock-market panic among many. The bail-outs of the 21st century raise plenty of their own problems. But should AI mania evaporate any time soon, do not fear another depression. ■

The Economist reads

Tome will tell

The best new novels to read this summer

"THESE ARE all my politics: I enjoy music and painting; a good book is an event in my life." A character in "The Red and the Black" by Stendhal, a French writer, proclaimed those words in 1830, but plenty of people today will recognise the sentiment. If you are weary of politics and seek an escape in art, pick up one of our recommended novels. Perhaps it will even become an event in your life.

The Calamity Club. *By Kathryn Stockett.* A rollicking work of historical fiction set in Oxford, Mississippi, in 1933. Meg endures misery in an orphanage. Birdie comes to town to borrow money from her married sister. Inevitably the characters' paths cross; soon they find themselves joining forces with other downtrodden women intent on improving their lives. This novel has intricately woven storylines and a vibrant cast.

The Kindness of Strangers.

By Emma Garman.

Set in London in 1953, this accomplished debut revolves around a ragtag assortment of residents of a Victorian boarding house. When a young man shows up and threatens the stability of the household, it becomes clear all the occupants have something to hide. A skilfully constructed mystery full of sly twists, dark wit and rich period detail.



Some like it hot

Land.

By Maggie O'Farrell. In 1865 Tomás, a cartographer, and his son, Liam, travel to a remote western peninsula of Ireland to map their country for their British masters. After discovering a mysterious, mystical site, Tomás vows to carry out an act of "honour and resistance". It is a decision that will transform his life and that of his family. A bold, epic novel about the relationship between people and place.

Other People's Children.

By Ben Faccini. Tom has a lot to contend with. He is plagued by memories of his late father. He struggles to bond with his partner's young sons. Adding to his woes is the increasingly strange behaviour of his Italian grandmother, Alma. When he delves into her wartime past, he unearths a devastating secret that alters his outlook. This book is elegantly written and astute as it traces family ties and the long shadow of history.

The Things We Never Say.

By Elizabeth Strout.

Artie Dam is a happily married man and a popular history teacher. But deep down he feels trapped in a "state of dismalness" and is contemplating suicide. A shocking revelation, however, makes him appreciate all he holds dear. This bittersweet novel about the effects of trauma, anxiety and loneliness examines how well people really know each other.

Transcription.

By Ben Lerner.

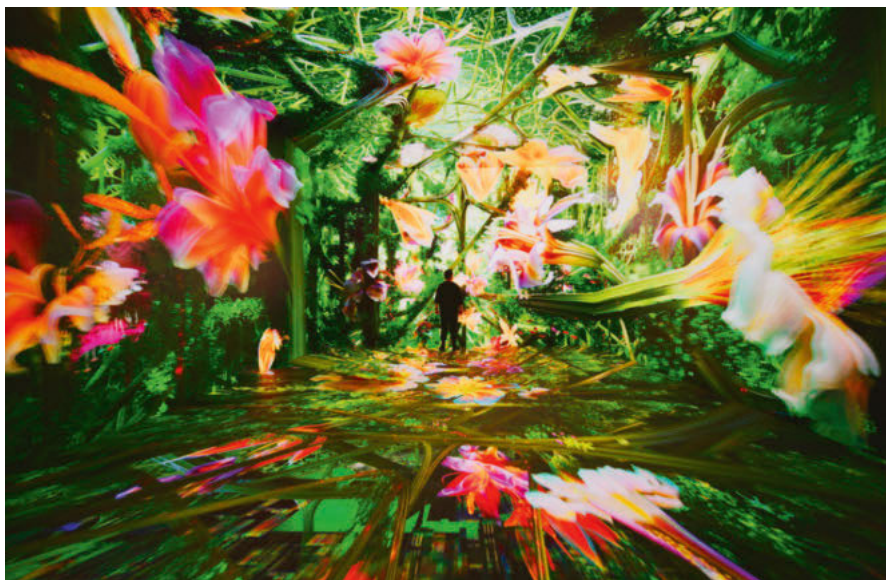
In this short, clever novel, an unnamed narrator conducts a magazine interview with his 90-year-old college mentor—but fails to record it. It is the first of several exchanges with different individuals, all of which offer shrewd insight into art and technology, friendship and fatherhood, and the reliability of memory.

Uprising.

By Tahmima Anam. Kusum Khan is tricked and sold into slavery on an island. In this prison, children look on while their mothers engage in sex work to pay off crippling debts to the cruel madam. But Kusum finds a way to unite her fellow prisoners into resisting and revolting. A fierce and unflinching novel inspired by the real women of Banishanta, Bangladesh.

Whistler.

By Ann Patchett. Daphne hasn't seen her stepfather, Eddie, since they were involved in a near-fatal car accident 44 years ago. A chance meeting in New York allows the pair to reconnect. It also gives Daphne the chance to finally learn the truth about why Eddie's marriage to her mother fell apart. A tale of secrets and lasting love, enriched by sharply drawn characters. ■



A new AI art museum

Lights, camera, (too much) action

LOS ANGELES

Dataland takes your senses by storm

VISITORS TO DATALAND, a new museum for AI arts in Los Angeles, step onto an escalator and descend into a tempest. Rain blows sideways across mirrored walls. Next, psychedelic pink amoebas and green squiggly lines gyrate across the walls and floor. It is hard to walk straight. Is this what it is like to be trapped inside a lava lamp? Soon guests are enveloped in the sounds of the rainforest, and fireflies flit across the screens. "It's a living sculpture," says Refik Anadol, a digital artist and the museum's co-creator. He points to lines emanating from people's feet. "That's your data."

Dataland is the creation of Mr Anadol and his wife, Efsun Erkilic, two Turkish artists. Mr Anadol is probably the most prominent and polarising AI artist working today—regarded by some as avant-garde and by others as spectacle-heavy and too commercial. (His works have appeared in the Museum of Modern Art in New York and the Sphere, a concert venue in Las Vegas.)

Dataland opened in a downtown development that Frank Gehry, its architect, always hoped would become part of an arts district. Mr Anadol's ambition is for the new museum to convince visitors that there is more to AI art than hype. He starts with the premise that there is ethical and unethical data. The former is about "asking permission" and "getting licences" for art works' data inputs, not training models on data irrespective of consent. "If a human and machine collaborate, it's a new

form of art," says Mr Anadol.

Dataland's more radical premise is the idea that its AI creations might help visitors get in touch with their humanity. This is done, in large part, by assaulting the senses. When guests arrive, they are given medical-grade devices—a watch and a heavy collar—that record heart rate, body temperature and emotional responses. As visitors move around, their measurements and location are fed into a system that changes the art to react to each person. Your correspondent's collar occasionally emits a fresh scent evoking summer rain and laundry detergent.

The question of how to use tech to enhance visitors' experiences is one with which every museum is grappling. Many are opting for interactive experiences in the virtual-reality vein. Some are designing whole exhibitions around screens, which can leave you feeling both over-stimulated and isolated. Are the days of asking the person next to you what they think of a painting over? What about being sure that you are seeing the same thing?

At Dataland, the experience is transfixing and overwhelming. It fails in its more grandiose ambitions—though it does inadvertently make you appreciate nature. By the end of the visit many guests will find themselves desperate for sunshine. After being immersed in a kaleidoscopic digital forest, they will crave the silence and stillness of a real one. ■

"Toy Story 5"

When it's not all fun and games

The film's real villain is unexpected

COWGIRLS ARE an intrepid bunch. Intent on figuring out why the neighbours' twin children refuse to play with eight-year-old Bonnie, Jessie (Bonnie's doll) saddles up, crosses the street and peers in the window. Inside, the twins are clenching their screens and swiping. Outside in the garden, discarded playthings are griping. "The age of toys is over," wails a small robot.

Pixar has turned the world of junky plastic into gold. "Toy Story" (1995)—the first full-length film made with computer-generated imagery—and the next three films earned more than \$4.5bn at the box office (in today's money). "Toy Story 5", released on June 19th, could become the highest-grossing film of the year. But unlike many mindless blockbusters, this film is soulful and splendid.

That is because it deals amusingly with urgent themes, such as how technology has hijacked childhood. Hoping to help their shy daughter make friends, Bonnie's parents buy Lilypad, a tablet; you then watch Bonnie's mood change with every ping from her group chats. The discarded toys are only the most vocal victims. The biggest loser is Bonnie, who misses out on the thrill of imaginative play and feels even greater social alienation online.

Previous "Toy Story" flicks have also reflected their era's crazes and concerns. The second film came out during the dotcom boom; in it, Woody, a cowboy, has become a collectible, raising questions about whether value stems more from relationships or money. "Toy Story 5" reprises and updates the first film's premise—displacement by something new and snazzy.

Unexpectedly, the film's biggest villain is not Lilypad, but Bonnie's unwitting parents. Passive and mindlessly gentle, they are tech-addicted themselves, fail to enforce the screen-time rules they set for their child and are too distracted to notice her descent into greater gloom.

Young audiences will walk out of this film perceiving old-fashioned toys and screens a little bit differently. Parents will leave wondering whether they are really as blind as the film makes them out to be.

Resurrection on the page

J.D. Vance's second coming

The vice-president fails to see his own vices

Communion. By J.D. Vance. Harper; 304 pages; \$35. William Collins; £20

J.D. VANCE IS a very godly man, as his very godly new book makes very clear. He goes to church; he is polite to priests; and he has read the Bible (twice, even the dull bits). He tries to be a very godly man in his life, too. He prays to God in intense italics to make him a “good dad”; starts out naughty words like “p***y”; and he ends his book with the word “Amen”. For him, actions matter because, as he says, quoting the Gospel of Matthew in yet more italics, “*By their fruits ye shall know them.*”

However if ye shall read “Communion”, the vice-president’s religious memoir, ye shall probably also feel a little confused. For which of the fruits of Mr Vance shall ye know him by? Shall ye know him as the man who prissily stars out words like p***y? Or as deputy for Donald Trump, a man who once boasted that he could “grab [women] by the pussy”? Shall ye know him as the person who predicted Mr Trump might be “America’s Hitler”? Or as the man who—as he does here—calls Mr Trump’s words “moral”? It is hard to tell.

Mr Vance used to be far simpler. This is his second literary coming; his first, published in 2016, was “Hillbilly Elegy”. Its title nodded at romantic poetry, but there was little romance here. For page after page, he chronicled the lives of the left-behind in America’s rustbelt: people without jobs and with opioid habits. Its heroine, for him, was his grandmother, who he called “Mamaw”—but do not imagine her as a granny in the apron-and-apple-pie vein. When her husband irritated her, she poured gasoline over him, “lit a match, and dropped it on his chest”.

For Americans, the book’s hero was Mr Vance. Just as medieval readers were inspired by stories of saints who did battle with demons and then won eternal life, so modern ones were heartened by how Mr Vance had done battle with his own demons and drug-addicted relatives to ascend to secular sainthood—or, at any rate, a place at Yale Law School. After Mr Trump was elected in 2016, “Hillbilly Elegy” became a sensation—selling over 5m copies worldwide—and it made his name.

Mr Vance’s second coming was always going to be harder. This is not wholly his fault. Virtue, as Milton and Dante disco-

vered, is much less fun than vice: everyone reads “Paradise Lost”; no one bothers with “Paradise Regained”. Everyone knows Dante’s “Inferno”; his “Paradiso” is an also-ran. St Augustine managed a good conversion memoir, “Confessions”, in the fourth century, but he seasoned it with ample sin, having prayed for God to make him good “but not yet”.

God made J.D. Vance good a bit too soon for “Communion” to be much fun. “Hillbilly Elegy” is filled with family tragedy, violence and uncles who do invigorating things like attacking their enemies with electric saws. By its end, however, Mr Vance is ensconced among the elite where his greatest discomfort is attending drinks events with lawyers. At these, he is first baffled by butter knives (surely that’s what index fingers are for?) then starts to doubt the meaning of life. (Lawyers can do that.)

For a time, this is fine: he fills his days with work, ambition, atheism and friends. Then, as he marries and settles down, the God-shaped hole (or, arguably, given Mr Vance’s ambition, the political poll) attracts him to religion. God was not alien—his relatives had bashed Bibles as well as each other—but he had never been devout.

Initially he is unsure on differences between denominations, except that “Catholics kneeled more and the Pentecostals had better music.” At first his faith feels so anodyne it is almost Anglican: he calls Chris-

tianity “my new interest”, as if it were chess or yoga; reads a lot of C.S. Lewis; and is inspired by that great theological thinker Aslan the lion. St Augustine this is not. In the end it is Catholicism, with its offer of bells, smells and Satan, that most appeals. Soon he is going to confession (“like therapy, but with less whining and more guilt”), befriending priests and reading Thomas Aquinas. He was baptised in 2019; today he evangelises with the zeal of the convert.

He quotes the archaic King James version at length, deploring “false idols” and champions traditional motherhood. Most startlingly, he wants to weaken America’s traditional separation of church and state.

But for all the fervour, his faith feels very erratic. At one point he states that Christianity demands “a constant evaluation of trade-offs”, as if Jesus were an ecclesiastical economist rather than, as Catholics believe, the son of God. When covid-19 breaks out, he realises that food might run short, so he buys “enormous bags of rice and flour” and “one thousand rounds of ammunition”. It is almost like reading about the loaves and the fishes.

Mr Vance is a good writer, but “Communion” is a confusing book. Which Mr Vance is he? The one who bangs on about God’s grace? Or the one who treats people like Ukraine’s president, Volodymyr Zelenskyy—a man in need if ever there was one—with a deplorable lack of grace? He keeps repeating, “*By their fruits ye shall know them.*” Yet by the end of this book, ye shall probably not be much the wiser as to who J.D. Vance was, or is.

Or, most worrying of all, who he shall next become. Though even if he does ascend to the presidency, he may find that fruit is bitter, too. For, as his beloved Matthew says, “What is a man profited, if he shall gain the whole world, and lose his own soul?” ■



Can't hold a candle to the old Vance?

Economic & financial indicators

	Gross domestic product				Consumer prices			Unemployment rate		Current-account balance		Budget balance		Interest rates		Currency units	
	% change on year ago				% change on year ago			%		% of GDP, 2026 [†]		% of GDP, 2026 [†]		10-yr gov't bonds		per \$	
	latest	Q1	quarter*	2026 [†]	latest	May	2026 [†]		May					latest, %	change on year ago, bp	Jun 24th	% change on year ago
United States	2.6	Q1	1.6	2.2	4.2	May	3.8	4.3	May	-3.4	-6.5	4.4	\$	11.0	-	-	
China	5.0	Q1	5.3	4.9	1.2	May	1.6	5.1	May ^{†\$}	3.2	-5.8	1.4		-5.0	6.79	5.7	
Japan	0.4	Q1	1.8	0.7	1.5	May	1.8	2.5	Apr	3.6	-1.8	2.7		125	162	-9.6	
Britain	1.1	Q1	2.5	1.0	2.8	May	3.3	4.9	Mar ^{††}	-3.4	-5.1	4.8		26.0	0.76	-2.6	
Canada	-0.1	Q1	-0.1	0.8	3.2	May	2.6	6.6	May	-0.4	-2.1	3.4		16.0	1.42	-3.5	
Euro area	0.3	Q1	-0.9	0.8	3.2	May	2.8	6.3	Apr	2.3	-3.4	2.9		33.0	0.88	-1.1	
Austria	0.9	Q1	0.8 [*]	0.8	3.7	May	3.0	5.7	Apr	1.1	-4.1	3.1		18.0	0.88	-1.1	
Belgium	0.8	Q1	1.0	0.7	4.0	May	3.5	6.2	Apr	-2.7	-5.2	3.5		34.0	0.88	-1.1	
France	0.9	Q1	-0.4	0.8	2.8	May	2.3	8.2	Apr	-0.6	-5.3	3.6		39.0	0.88	-1.1	
Germany	0.3	Q1	1.4	0.6	2.7	May	2.7	3.8	Apr	4.2	-3.7	2.9		33.0	0.88	-1.1	
Greece	2.0	Q1	1.0	2.0	4.9	May	3.9	9.5	Apr	-5.5	0.8	3.5		23.0	0.88	-1.1	
Italy	0.8	Q1	1.1	0.8	3.2	May	2.7	5.1	Apr	0.8	-3.0	3.6		11.0	0.88	-1.1	
Netherlands	1.2	Q1	0.2	1.0	3.4	May	3.1	3.9	May	9.2	-2.6	3.0		22.0	0.88	-1.1	
Spain	2.7	Q1	2.5	2.2	3.6	May	3.2	10.3	Apr	2.3	-2.4	3.4		17.0	0.88	-1.1	
Czech Republic	2.2	Q1	0.7	2.0	2.1	May	2.0	3.0	Q1 [*]	0.3	-2.6	4.7		42.0	21.3	0.8	
Denmark	5.8	Q1	7.8	3.3	1.9	May	2.0	3.1	Apr	12.1	1.3	2.7		17.0	6.57	-1.8	
Norway	1.7	Q1	1.4	1.3	3.1	May	3.2	4.6	Apr ^{††}	14.8	9.0	4.3		32.0	9.79	3.1	
Poland	3.5	Q1	2.4	3.4	3.1	May	3.0	5.9	May ^{\$}	-1.3	-7.0	5.4		-9.0	3.76	-1.9	
Russia	-0.2	Q1	-2.5	0.7	5.3	May	5.5	2.2	Apr ^{\$}	2.2	-3.4	15.9		78.0	74.6	4.9	
Sweden	2.2	Q1	-0.6	2.1	0.8	May	1.9	9.4	May ^{\$}	4.8	-2.0	2.7		41.0	9.74	-1.4	
Switzerland	0.5	Q1	2.6	1.0	0.6	May	0.6	3.1	May	5.3	0.2	0.4		-2.0	0.81	nil	
Turkey	2.5	Q1	0.5	2.5	32.6	May	31.7	8.0	Apr ^{\$}	-3.0	-3.6	30.9		45.0	46.5	-14.8	
Australia	2.5	Q1	1.1	1.6	4.0	May	4.3	4.5	Apr	-1.9	-1.8	4.8		52.0	1.45	6.9	
Hong Kong	5.9	Q1	12.2	5.0	2.0	May	2.1	3.7	May ^{††}	4.3	-2.5	3.4		32.0	7.84	0.1	
India	7.8	Q1	6.4	6.5	3.9	May	4.8	6.9	May	-1.5	-4.5	6.8		52.0	94.8	-9.0	
Indonesia	5.6	Q1	6.0	5.0	3.1	May	3.5	4.7	Feb ^{\$}	-1.7	-3.5	7.2		42.0	17,912	-8.1	
Malaysia	5.4	Q1	1.7	4.5	2.0	May	2.5	3.0	Apr ^{\$}	3.4	-4.1	3.6		6.0	4.14	2.7	
Pakistan	4.8	2026 ^{**}	na	3.0	11.7	May	7.8	6.9	2025	-1.1	-4.7	12.1	†††	-33.0	278	1.9	
Philippines	2.8	Q1	3.6	2.9	6.8	May	5.3	4.7	Q2 ^{\$}	-4.1	-6.3	7.0		67.0	61.2	-5.8	
Singapore	6.0	Q1	3.9	2.9	1.8	May	2.8	2.0	Q1	15.3	0.8	2.1		-18.0	1.30	-0.8	
South Korea	3.8	Q1	7.5	2.9	3.1	May	2.9	2.9	May ^{\$}	8.2	-2.5	4.2		134	1,533	-10.6	
Taiwan	14.5	Q1	6.9	10.5	2.2	May	1.6	3.3	May	27.4	1.8	1.7		23.0	31.7	-6.5	
Thailand	2.8	Q1	2.7	1.8	2.8	May	2.8	1.0	May ^{\$}	0.5	-5.5	2.1		39.0	33.2	-1.4	
Argentina	2.3	Q1	3.0	3.1	33.2	May	32.6	7.8	Q1 ^{\$}	0.4	0.1	na		na	1,471	-20.3	
Brazil	1.8	Q1	4.5	2.0	4.7	May	4.7	5.8	Apr ^{†††}	-2.6	-7.3	14.6		80.0	5.20	5.8	
Chile	-0.5	Q1	-1.1	1.4	3.9	May	3.6	9.1	Apr ^{†††}	-2.0	-2.4	5.5		-23.0	914	3.7	
Colombia	2.2	Q1	2.4	2.5	5.8	May	5.7	8.8	Apr ^{\$}	-2.5	-6.6	11.8		-43.0	3,426	19.1	
Mexico	0.2	Q1	-2.4	0.9	3.9	May	4.0	2.6	Apr	-0.4	-3.8	9.0		-36.0	17.6	8.9	
Peru	3.5	Q1	3.2	2.5	3.9	May	4.1	4.5	May ^{\$}	1.8	-2.4	6.1		-24.0	3.40	5.6	
Egypt	5.0	Q1	-24.0	4.5	14.6	May	13.8	6.0	Q1 ^{\$}	-5.0	-5.2	25.3		8.0	49.7	1.9	
Israel	1.7	Q1	-3.8	3.4	1.9	May	1.9	2.8	May	1.3	-4.0	3.7		-60.0	3.00	15.3	
Saudi Arabia	4.6	2025	na	-2.0	1.8	May	2.3	3.5	Q4	-2.2	-6.1	na		na	3.75	nil	
South Africa	1.9	Q1	2.2	1.6	4.4	May	4.3	32.7	Q1 ^{\$}	-1.0	-4.4	8.4		-150	16.5	8.0	

Source: Haver Analytics *% change on previous quarter, annual rate †The Economist Intelligence Unit estimate/forecast \$Not seasonally adjusted †New series **Year ending June ††Latest 3 months †††3-month moving average \$95-year yield †††Dollar-denominated bonds Note: Euro-area consumer prices are harmonised

Markets

In local currency	Index	% change on:	
		one week	Dec 31st 2025
United States S&P 500	7,358.2	-0.8	7.5
United States NAS Comp	25,476.6	-2.1	9.6
China Shanghai Comp	4,110.8	0.1	3.6
China Shenzhen Comp	2,855.6	0.6	12.8
Japan Nikkei 225	69,175.0	-1.0	37.4
Japan Topix	3,963.8	-1.2	16.3
Britain FTSE 100	10,461.6	-0.4	5.3
Canada S&P TSX	34,736.1	-1.1	9.5
Euro area EURO STOXX 50	6,214.7	-1.4	7.3
France CAC 40	8,385.5	-0.5	2.9
Germany DAX*	24,740.4	-0.8	1.0
Italy FTSE/MIB	51,638.9	-1.8	14.9
Netherlands AEX	1,065.4	-1.6	12.0
Spain IBEX 35	19,389.5	-0.2	12.0
Poland WIG	134,688.1	-4.0	14.9
Russia RTS, \$ terms	945.2	-12.2	-14.7
Switzerland SMI	14,117.8	2.2	6.4
Turkey BIST	14,331.2	-0.6	27.3
Australia All Ord.	9,012.6	-1.9	-0.1
Hong Kong Hang Seng	23,412.2	-3.7	-8.7
India BSE	76,991.3	-0.2	-9.7
Indonesia IDX	5,883.9	-5.4	-32.0
Malaysia KLSE	1,682.1	-1.6	0.1

	Index	% change on:	
		one week	Dec 31st 2025
Pakistan KSE	179,571.3	-0.5	3.2
Singapore STI	5,216.0	0.8	12.3
South Korea KOSPI	8,471.0	-4.4	101.0
Taiwan TWI	46,043.6	0.4	59.0
Thailand SET	1,548.2	-2.4	22.9
Argentina MERV	3,110,490.2	-5.5	1.9
Brazil BVSP*	170,506.7	1.2	5.8
Mexico IPC	66,278.0	-3.0	3.1
Egypt EGX 30	51,710.9	-1.7	23.6
Israel TA-125	4,074.1	nil	11.2
Saudi Arabia Tadawul	11,006.8	-1.0	4.9
South Africa JSE AS	109,828.9	-5.3	-5.2
World, dev'd MSCI	4,752.9	-1.1	7.3
Emerging markets MSCI	1,733.7	-2.7	23.4
US corporate bonds, spread over Treasuries			
		Dec 31st 2025	
Basis points		latest	2025
Investment grade		88	93
High-yield		320	354

Sources: LSEG Workspace; Moscow Exchange; Standard & Poor's Global Fixed Income Research *Total return index

Commodities

The Economist commodity-price index				% change on	
2020=100					
				month	year
Dollar Index					
All items	151.4	148.5	-4.0	12.5	
Food	145.4	145.4	-1.3	-1.1	
Industrials					
All	156.3	151.1	-6.0	26.4	
Non-food agriculturals	155.7	154.9	3.4	28.2	
Metals	156.5	150.1	-8.2	25.9	
Sterling Index					
All items	144.9	144.5	-2.2	16.1	
Euro Index					
All items	149.0	149.1	-1.9	14.8	
Gold					
\$ per oz	4,325.6	4,127.9	-8.5	25.0	
Brent					
\$ per barrel	79.0	77.2	-22.5	14.7	

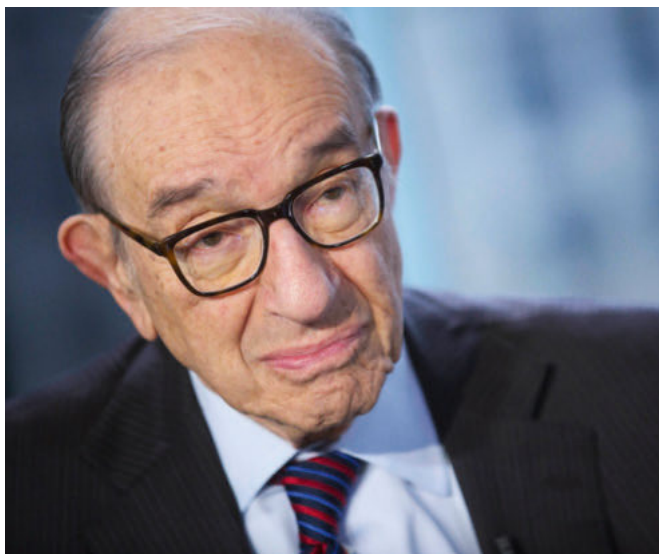
Sources: CME Group; LME; LSEG Workspace; NOREXCO; NZ Wool Services; S&P Global Commodity Insights; Thompson Lloyd & Ewart; USDA *Provisional

For historical indicators data, visit economist.com/economic-and-financial-indicators

OBITUARY

Alan Greenspan

The master-chairman of the Federal Reserve died on June 22nd, aged 100



THE NUMBERS were a guide, like the notes on sheet music. They were a guide in the 1930s, growing up in Washington Heights, at the northern tip of Manhattan, where Alan Greenspan tracked the statistics of his beloved New York Yankees. They were a guide in the 1990s, when as chairman of the Federal Reserve he presided over the then-longest American boom on record.

It was the numbers that led him to economics: the numbers he read in books on finance between sets, playing in a touring jazz band during the second world war. While the other musicians would sit and smoke a little reefer, he read up on American industry and J.P. Morgan, then picked up his saxophone and played that big-band music in Henry Jerome's orchestra. He had taken classes at Juilliard, the conservatoire in New York City, and played alongside geniuses like Stan Getz. They played by feel. He trusted the notes on the page, as he trusted his numbers.

By 1945 he was studying at New York University in the era of John Maynard Keynes, whose analysis of the Depression changed economics. He admired Keynes's analytical wizardry, but had no time for his sweeping vision. He kept to his data. As a consultant, his first job, he impressed clients with his knack for spotting turns in the business cycle—as in 1957, when growing steel inventories led him to intuit that a recession must be looming. It was all right there in the numbers.

There was more to life than data, though. He learned that from his friend Ayn Rand, a zealous, gut-level believer in individual freedom and the evils of state intervention. As the 1960s drew to a close, it was easy to see in America's slowing growth rates and rising inflation the inevitable side-effects of government bloat. He was not a rabid partisan by nature. But as a jazz-age conservative in Vietnam-era America, infused with the libertarian spirit of Rand and her circle, he gravitated towards the Republican Party.

He had a way with people in high places. In 1975 he began squiring Barbara Walters, and he spent the last four decades of his life with Andrea Mitchell, also a television journalist. He fell in with Richard Nixon, too, who brought him onto his team as a campaign adviser. Nixon's partisan rage turned him off, but he served happily under Gerald Ford as chairman of the Council of Econom-

ic Advisers: leading the band, now, stepping out from behind the music-stand. When Ronald Reagan ran for the presidency in 1980, he joined the campaign team.

It was Reagan who chose him for the job that would define his career. As chairman of the Fed, he would become the world's most powerful economic figure, coaxing the massive, complicated machinery of the American economy to perform its best, to play its sweetest. His tenure began on a sour note. He was just two months on the job on October 19th 1987, "Black Monday", when America's stock markets lost more than 20% of their value in a single day. It was the worst one-day drop in the country's history. In calm response, he jawboned banks into maintaining credit and pushed the administration to call for cuts to the budget deficit, which he reckoned would boost confidence.

He was fierce in defending the independence of the Fed; when the administration of President George H.W. Bush leaned on the central-banking system to do more to beat back the recession of the early 1990s, he had none of it. But independence did not mean that the chairman had to mind his own business. In the murky language known as FedSpeak, of which he became the most famous and fluent speaker, he would render his judgments on the plans of the president and Congress, never using one syllable when six would do. The American press proved a rapt audience.

The maestro's control never seemed more complete than in the 1990s. In 1994 he joined Robert Rubin, Bill Clinton's secretary of the treasury, and Larry Summers, a deputy secretary, to make up a trio of crisis-fighters: managing financial havoc in East Asia and Russia—and in American financial markets, when Long-Term Capital Management, a massive hedge fund, required a Fed-orchestrated bail-out. His Fed kept the American economy humming, speeding through an unprecedented boom. "If you want a simple model for predicting the unemployment rate in the United States over the next few years, here it is," wrote Paul Krugman in 1997. "It will be what Greenspan wants it to be, plus or minus a random error reflecting the fact that he is not quite God."

Finally, Mr Greenspan was playing by feel. He believed new technology was changing the way the economy worked. When Fed members worried that the economy was running too hot, he told them to be cool: the numbers had to be wrong. He let the boom run, counting on a productivity miracle to keep inflation in check. When rocketing stock prices threatened to create disharmony, he warned gently of "irrational exuberance", while still believing the economy was more resilient than ever.

But then, somehow, it began to get away from him. By the turn of the millennium he feared that productivity growth, and the stock market boom it had brought about, was inflationary after all. When stocks crashed and recession struck, the economy did not jump at his utterances as it had before. A jobless recovery gave way to a dangerous-looking housing boom, which his rate rises failed to dent. Even so, he left the Fed to a chorus of plaudits, hailed as the man who could make the American economy sing.

He barely had time to publish his memoir before the global financial crisis hit. It shook his faith in deregulation. He defended his Fed: the low interest rates that perked up growth, its light regulatory touch, and the bail-outs it had constructed in the years before the crash. He blamed the Obama administration for holding back growth. Sour notes sounded, however. Had he really been such a maestro? Had he been good, or merely lucky, and then unlucky? If only the numbers could say, one way or another. ■

The world is fracturing

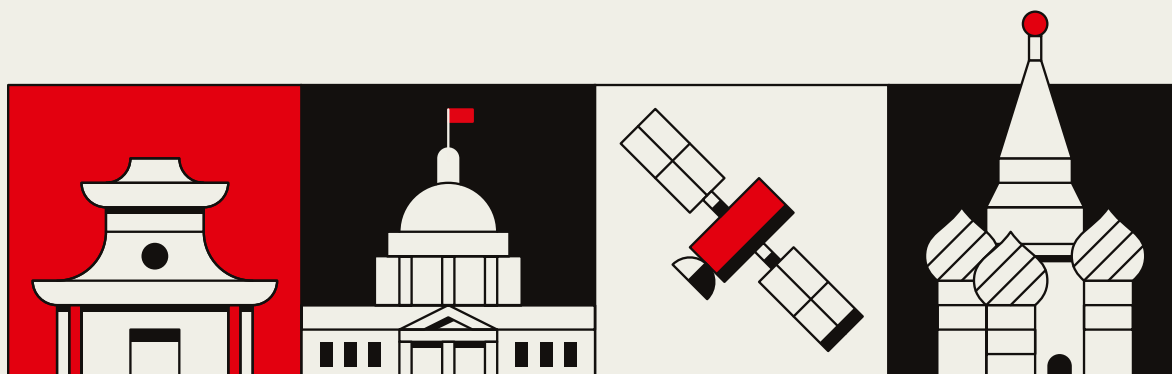
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Cartier